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Kottas et al.

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(54) **HETEROLEPTIC IRIIDIUM COMPLEXES AS DOPANTS**

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(51) **Int. Cl.**
H05B 33/14 (2006.01)
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(52) **U.S. Cl.**

CPC **H05B 33/14** (2013.01); **C07F 15/0033**
(2013.01); **C09K 11/025** (2013.01);
(Continued)

(58) **Field of Classification Search**

CPC H10K 85/342; H10K 85/622; H10K
85/626; H10K 85/654; H10K 85/6572;
(Continued)

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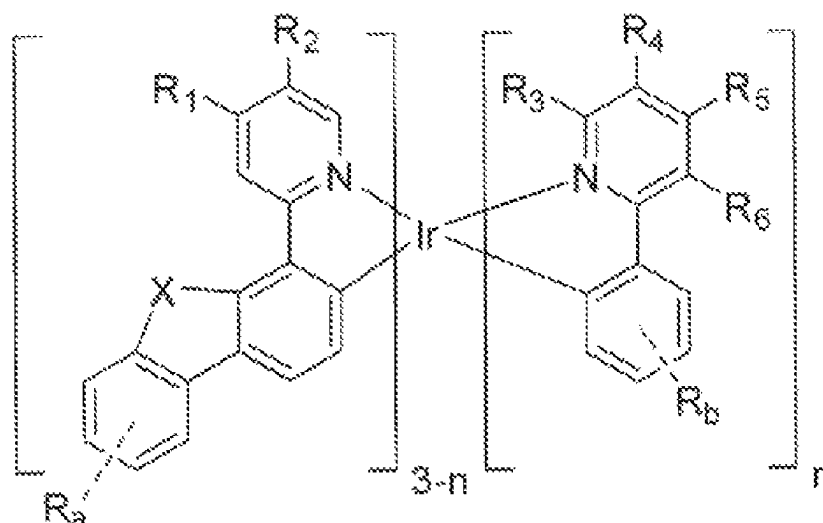
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(57) **ABSTRACT**

Novel phosphorescent heteroleptic iridium complexes with phenylpyridine and dibenzo-containing ligands are provided. Alkyl substitution at specific positions on the ligands gives rise to compounds with improved OLED properties, including saturated green emission.

20 Claims, 3 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/385,637, filed on Apr. 16, 2019, now Pat. No. 11,380,854, which is a continuation of application No. 14/875,147, filed on Oct. 5, 2015, now Pat. No. 10,312,458, which is a continuation of application No. 14/156,808, filed on Jan. 16, 2014, now Pat. No. 9,184,397, which is a continuation of application No. 13/193,221, filed on Jul. 28, 2011, now Pat. No. 8,709,615, said application No. 16/385,637 is a continuation-in-part of application No. 16/037,164, filed on Jul. 17, 2018, now Pat. No. 11,637,251, which is a continuation of application No. 14/225,591, filed on Mar. 26, 2014, now Pat. No. 10,056,566, which is a continuation of application No. 12/727,615, filed on Mar. 19, 2010, now Pat. No. 8,772,205.

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C09K 11/02 (2006.01)

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H10K 101/10 (2023.01)

(52) **U.S. Cl.**

CPC **C09K 11/06** (2013.01); **H10K 85/342** (2023.02); **H10K 85/622** (2023.02); **H10K 85/626** (2023.02); **H10K 85/654** (2023.02); **H10K 85/6572** (2023.02); **H10K 85/6576** (2023.02); **C09K 2211/1007** (2013.01); **C09K 2211/1029** (2013.01); **C09K 2211/1088** (2013.01); **C09K 2211/185** (2013.01); **H10K 50/11** (2023.02); **H10K 2101/10** (2023.02)

(58) **Field of Classification Search**

CPC **H10K 85/6576**; **H10K 50/11**; **H10K 2101/10**; **H10K 50/12**; **H10K 85/655**; **C07F 15/0033**; **C09K 11/025**; **C09K 11/06**; **C09K 2211/1007**; **C09K 2211/1029**; **C09K 2211/1088**; **C09K 2211/185**; **H05B 33/14**; **Y02E 10/549**

See application file for complete search history.

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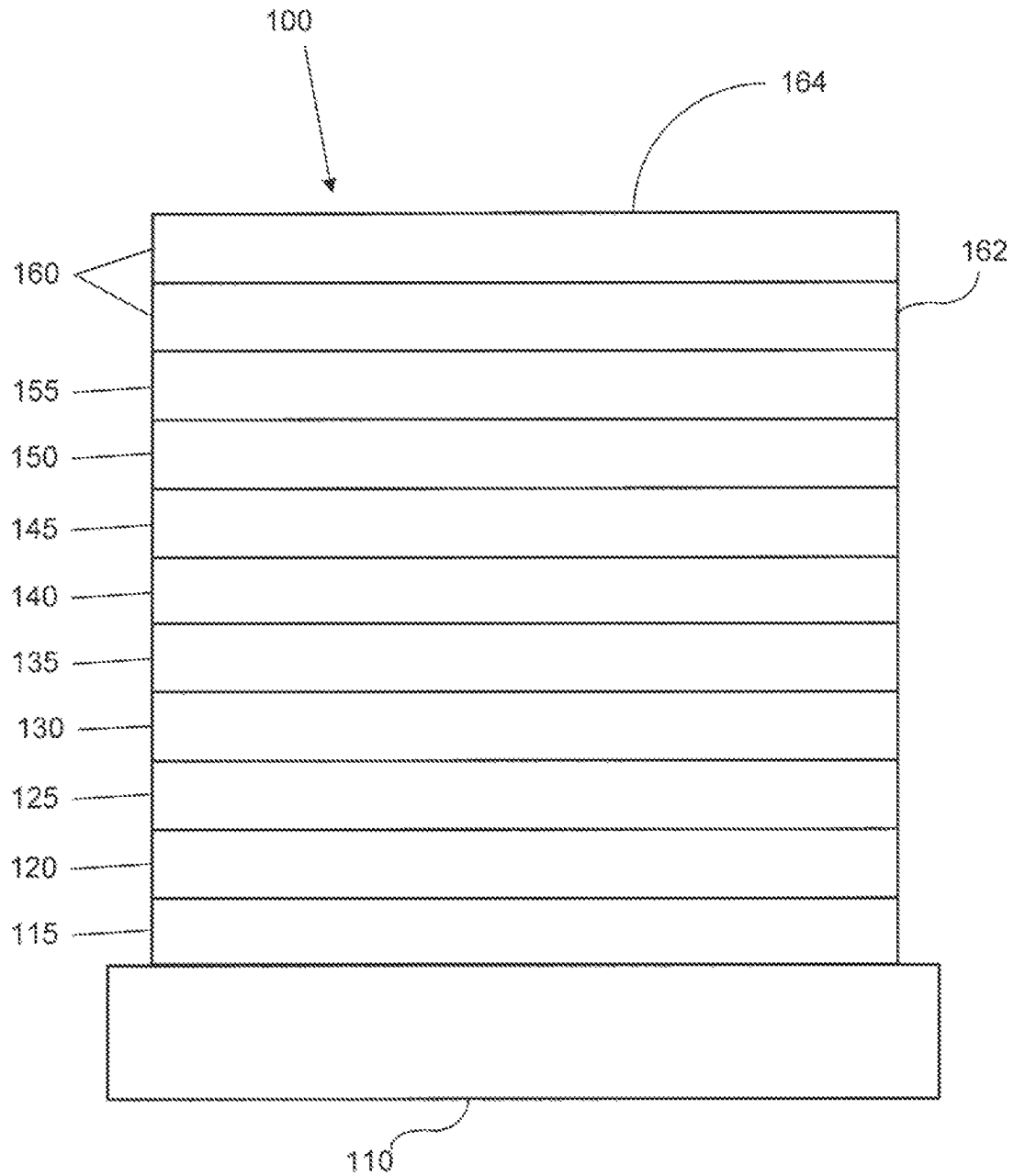


FIGURE 1

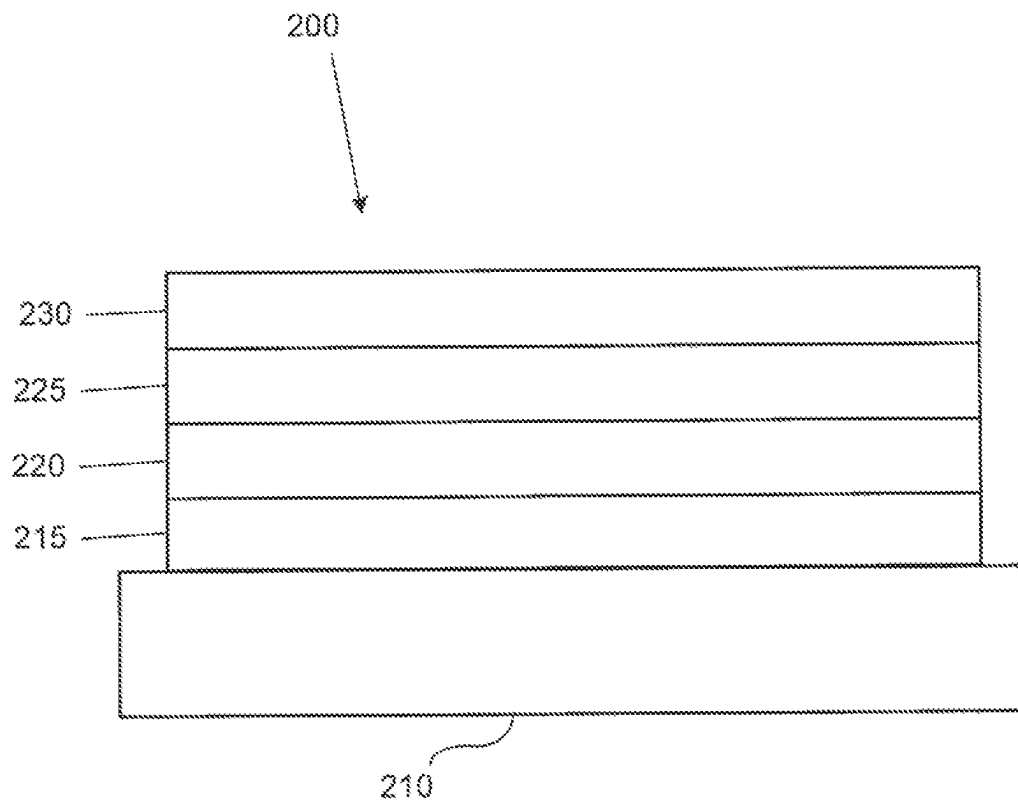
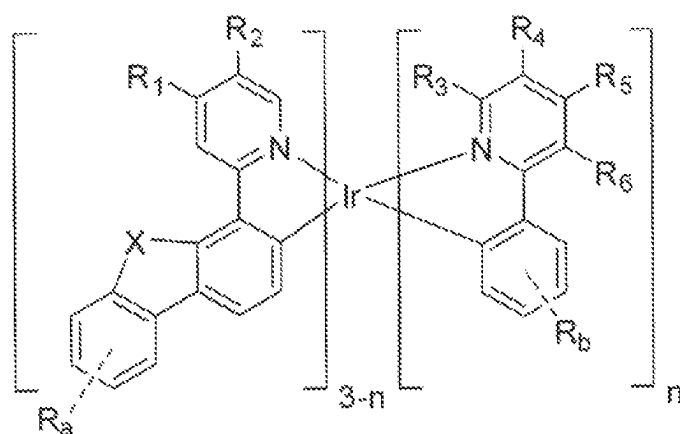


FIGURE 2



Formula I

FIGURE 3

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HETEROLEPTIC IRIIDIUM COMPLEXES AS DOPANTS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/746,351, filed May 17, 2022, which is a continuation of U.S. patent application Ser. No. 16/385,637, filed Apr. 16, 2019, now U.S. Pat. No. 11,380,854, which is a continuation of U.S. patent application Ser. No. 14/875,147, filed Oct. 5, 2015, now U.S. Pat. No. 10,312,458, which is a continuation of U.S. patent application Ser. No. 14/156,808 filed Jan. 16, 2014, now U.S. Pat. No. 9,184,397, which is a continuation of U.S. patent application Ser. No. 13/193,221 filed Jul. 28, 2011, now U.S. Pat. No. 8,709,615, and

U.S. patent application Ser. No. 16/385,637 is also a continuation-in-part of U.S. patent application Ser. No. 16/037,164, filed Jul. 7, 2018, now U.S. Pat. No. 11,637,251, which is a continuation of U.S. patent application Ser. No. 14/225,591, filed Mar. 26, 2014, now U.S. Pat. No. 10,056,566, which is a continuation of U.S. patent application Ser. No. 12/727,615, filed Mar. 19, 2010, now U.S. Pat. No. 8,722,205, which claims priority to U.S. Provisional Application No. 61/162,476, filed Mar. 23, 2009, the entire contents of which are incorporated herein by reference.

PARTIES TO A JOINT RESEARCH AGREEMENT

The claimed invention was made by, on behalf of, and/or in connection with one or more of the following parties to a joint university corporation research agreement: Regents of the University of Michigan, Princeton University, The University of Southern California, and the Universal Display Corporation. The agreement was in effect on and before the date the claimed invention was made, and the claimed invention was made as a result of activities undertaken within the scope of the agreement.

FIELD OF THE INVENTION

The present invention relates to heteroleptic iridium complexes suitable for inclusion in OLED devices.

BACKGROUND

Opto-electronic devices that make use of organic materials are becoming increasingly desirable for a number of reasons. Many of the materials used to make such devices are relatively inexpensive, so organic opto-electronic devices have the potential for cost advantages over inorganic devices. In addition, the inherent properties of organic materials, such as their flexibility, may make them well suited for particular applications such as fabrication on a flexible substrate. Examples of organic opto-electronic devices include organic light emitting devices (OLEDs), organic phototransistors, organic photovoltaic cells, and organic photodetectors. For OLEDs, the organic materials may have performance advantages over conventional materials. For example, the wavelength at which an organic emissive layer emits light may generally be readily tuned with appropriate dopants.

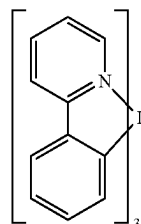
OLEDs make use of thin organic films that emit light when voltage is applied across the device. OLEDs are becoming an increasingly interesting technology for use in applications such as flat panel displays, illumination, and

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backlighting. Several OLED materials and configurations are described in U.S. Pat. Nos. 5,844,363, 6,303,238, and 5,707,745, which are incorporated herein by reference in their entirety.

One application for phosphorescent emissive molecules is a full color display. Industry standards for such a display call for pixels adapted to emit particular colors, referred to as “saturated” colors. In particular, these standards call for saturated red, green, and blue pixels. Color may be measured using CIE coordinates, which are well known to the art.

One example of a green emissive molecule is tris(2-phenylpyridine) iridium, denoted Ir(ppy)₃, which has the following structure:



In this, and later figures herein, we depict the dative bond from nitrogen to metal (here, Ir) as a straight line.

As used herein, the term “organic” includes polymeric materials as well as small molecule organic materials that may be used to fabricate organic opto-electronic devices. “Small molecule” refers to any organic material that is not a polymer, and “small molecules” may actually be quite large. Small molecules may include repeat units in some circumstances. For example, using a long chain alkyl group as a substituent does not remove a molecule from the “small molecule” class. Small molecules may also be incorporated into polymers, for example as a pendent group on a polymer backbone or as a part of the backbone. Small molecules may also serve as the core moiety of a dendrimer, which consists of a series of chemical shells built on the core moiety. The core moiety of a dendrimer may be a fluorescent or phosphorescent small molecule emitter. A dendrimer may be a “small molecule,” and it is believed that all dendrimers currently used in the field of OLEDs are small molecules.

As used herein, “top” means furthest away from the substrate, while “bottom” means closest to the substrate. Where a first layer is described as “disposed over” a second layer, the first layer is disposed further away from substrate. There may be other layers between the first and second layer, unless it is specified that the first layer is “in contact with” the second layer. For example, a cathode may be described as “disposed over” an anode, even though there are various organic layers in between.

As used herein, “solution processible” means capable of being dissolved, dispersed, or transported in and/or deposited from a liquid medium, either in solution or suspension form.

A ligand may be referred to as “photoactive” when it is believed that the ligand directly contributes to the photoactive properties of an emissive material. A ligand may be referred to as “ancillary” when it is believed that the ligand does not contribute to the photoactive properties of an emissive material, although an ancillary ligand may alter the properties of a photoactive ligand.

As used herein, and as would be generally understood by one skilled in the art, a first “Highest Occupied Molecular Orbital” (HOMO) or “Lowest Unoccupied Molecular

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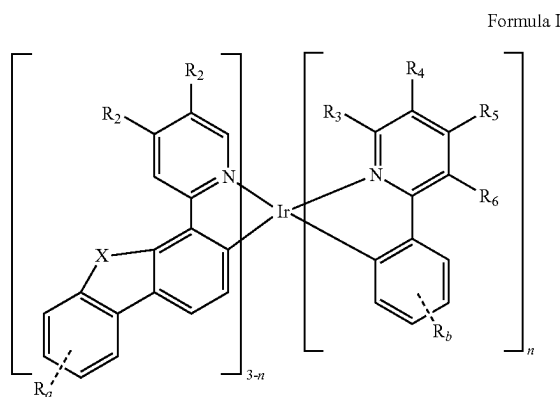
Orbital” (LUMO) energy level is “greater than” or “higher than” a second HOMO or LUMO energy level if the first energy level is closer to the vacuum energy level. Since ionization potentials (IP) are measured as a negative energy relative to a vacuum level, a higher HOMO energy level corresponds to an IP having a smaller absolute value (an IP that is less negative). Similarly, a higher LUMO energy level corresponds to an electron affinity (EA) having a smaller absolute value (an EA that is less negative). On a conventional energy level diagram, with the vacuum level at the top, the LUMO energy level of a material is higher than the HOMO energy level of the same material. A “higher” HOMO or LUMO energy level appears closer to the top of such a diagram than a “lower” HOMO or LUMO energy level.

As used herein, and as would be generally understood by one skilled in the art, a first work function is “greater than” or “higher than” a second work function if the first work function has a higher absolute value. Because work functions are generally measured as negative numbers relative to vacuum level, this means that a “higher” work function is more negative. On a conventional energy level diagram, with the vacuum level at the top, a “higher” work function is illustrated as further away from the vacuum level in the downward direction. Thus, the definitions of HOMO and LUMO energy levels follow a different convention than work functions.

More details on OLEDs, and the definitions described above, can be found in U.S. Pat. No. 7,279,704, which is incorporated herein by reference in its entirety.

SUMMARY OF THE INVENTION

In one aspect, a compound having the formula:



is provided. R_1 and R_2 are optionally linked and the sum of the number of carbon atoms in R_1 and R_2 is at least 2. R_3 , R_4 , R_5 , R_6 are optionally linked, and R_a and R_b represent mono-, di-, tri- or tetra-substitution. X is selected from the group consisting of BR, NR, PR, O, S, Se, C=O, S=O, SO₂, CRR', SiRR', and GeRR', and R_a , R_b , R, R', R_1 , R_2 , R_3 , R_4 , R_5 , and R_6 are independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; and wherein n is 1 or 2.

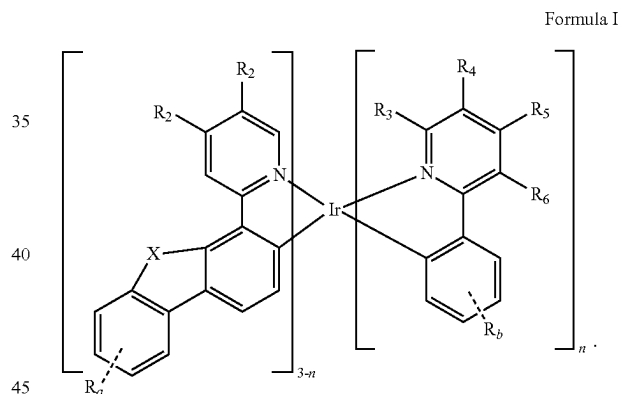
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In one aspect, n is Z. In one aspect, X is O. In one aspect, R_1 is hydrogen and R_2 is alkyl. In another aspect, R_1 is alkyl and R_2 is hydrogen. In one aspect, R_1 and R_2 are alkyl. In one aspect, R_1 and R_2 contain one or more deuterium atoms. In another aspect, R_1 or R_2 are independently selected from the group consisting of branched alkyl, cyclic alkyl, bicyclic alkyl, and multicyclic alkyl. In one aspect, R_1 or R_2 is iso-propyl.

In one aspect, R_1 or R_2 contain one or more deuterium atoms. In one aspect, R_3 , R_4 , R_5 and R_6 are independently selected from the group consisting of hydrogen, deuterium, alkyl, aryl, and combinations thereof. In another aspect, at least one of R_3 , R_4 , R_5 and R_6 comprises a branched alkyl, cyclic alkyl, bicyclic alkyl, or a multicyclic alkyl. In one aspect, R_3 , R_4 , R_5 or R_6 contain one or more deuterium atoms.

In one aspect, the compound is selected from the group consisting of Compound 53, Compounds 157-159, Compound 165, Compound 174, Compound 175, Compounds 184-185, Compound 314, Compound 321, Compounds 625-628, Compound 633, Compound 643, Compounds 652-653, and Compounds 1145-1146.

In one aspect, a first device is provided. The first device comprises a first organic light-emitting device, further comprising an anode, a cathode, and an organic layer, disposed between the anode and the cathode, comprising a compound having the formula:



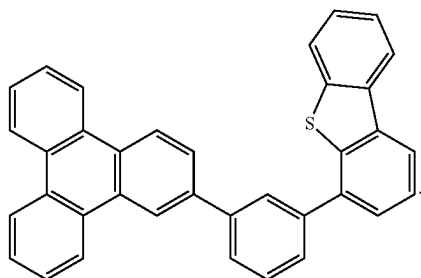
R_1 and R_2 are optionally linked and the sum of the number of carbon atoms in R_1 and R_2 is at least 2. R_3 , R_4 , R_5 , R_6 are optionally linked, and R_a and R_b represent mono-, di-, tri- or tetra-substitution. X is selected from the group consisting of BR, NR, PR, O, S, Se, C=O, S=O, SO₂, CRR', SiRR', and GeRR', and R_a , R_b , R, R', R_1 , R_2 , R_3 , R_4 , R_5 , and R_6 are independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof, and wherein n is 1 or 2.

In one aspect, the first device is a consumer product. In another aspect, the first device is an organic light-emitting device. In another aspect, the first device comprises a lighting panel. In one aspect, the organic layer is an emissive layer and the compound is an emissive dopant. In another aspect, the organic layer is an emissive layer and the compound is a non-emissive dopant.

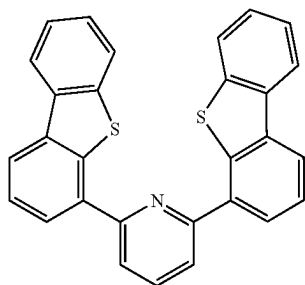
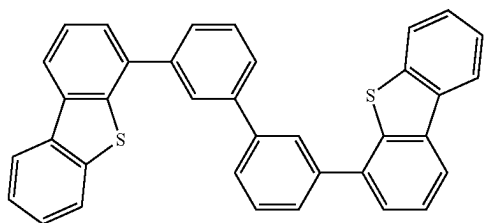
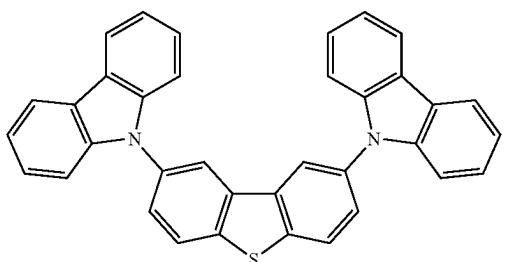
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In one aspect, the organic layer further comprises a host. In another aspect, the host comprises a triphenylene containing benzo-fused thiophene or benzo-fused furan, wherein any substituent in the host is an unfused substituent independently selected from the group consisting of C_nH_{2n+1} , OC_nH_{2n+1} , OAr_1 , $N(C_nH_{2n+1})_2$, $N(Ar_1)(Ar_2)$, $CH=CH-C_nH_{2n+1}$, $C\equiv CHC_nH_{2n+1}$, Ar_1 , Ar_1-Ar_2 , $C_nH_{2n}-Ar_1$, or no substitution, wherein Ar_1 and Ar_2 are independently selected from the group consisting of benzene, biphenyl, naphthalene, triphenylene, carbazole, and heteroaromatic analogs thereof, and wherein n is from 1 to 10.

In one aspect, the host has the formula

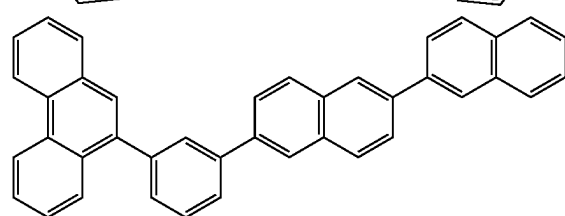
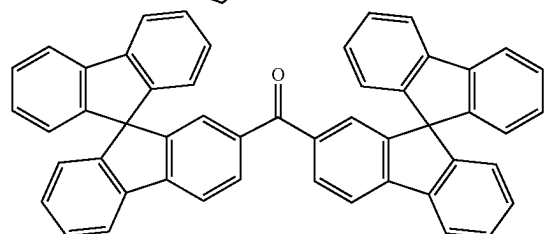
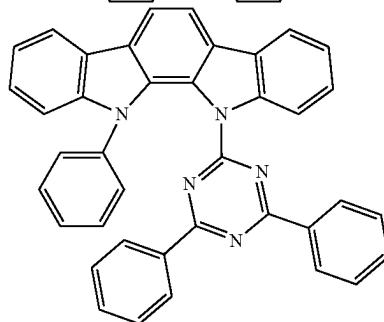
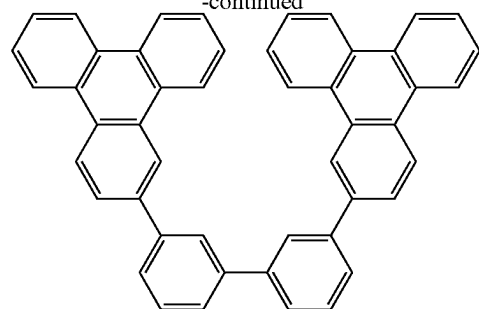


In another aspect, the host is selected from the group consisting of:



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-continued



and combinations thereof.

In one aspect, the host is a metal complex.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows an organic light emitting device.

FIG. 2 shows an inverted organic light-emitting device that does not have a separate electron transport layer.

FIG. 3 shows a compound of Formula I.

DETAILED DESCRIPTION

Generally, an OLED comprises at least one organic layer disposed between and electrically connected to an anode and a cathode. When a current is applied, the anode injects holes and the cathode injects electrons into the organic layer(s). The injected holes and electrons each migrate toward the oppositely charged electrode. When an electron and hole localize on the same molecule, an "exciton," which is a localized electron-hole pair having an excited energy state, is formed. Light is emitted when the exciton relaxes via a photoemissive mechanism. In some cases, the exciton may be localized on an excimer or an exciplex. Non-radiative mechanisms, such as thermal relaxation, may also occur, but are generally considered undesirable.

The initial OLEDs used emissive molecules that emitted light from their singlet states ("fluorescence") as disclosed, for example, in U.S. Pat. No. 4,769,292, which is incorporated by reference in its entirety. Fluorescent emission generally occurs in a time frame of less than 10 nanoseconds.

More recently, OLEDs having emissive materials that emit light from triplet states ("phosphorescence") have been demonstrated. Baldo et al., "Highly Efficient Phosphorescent Emission from Organic Electroluminescent Devices," *Nature*, vol. 395, 151-154, 1998; ("Baldo-I") and Baldo et al., "Very high-efficiency green organic light-emitting devices based on electrophosphorescence," *Appl. Phys. Lett.*, vol. 75, No. 3, 4-6 (1999) ("Baldo-II"), which are incorporated by reference in their entireties. Phosphorescence is described in more detail in U.S. Pat. No. 7,279,704 at cols. 5-6, which are incorporated by reference.

FIG. 1 shows an organic light emitting device **100**. The figures are not necessarily drawn to scale. Device **100** may include a substrate **110**, an anode **115**, a hole injection layer **120**, a hole transport layer **125**, an electron blocking layer **130**, an emissive layer **135**, a hole blocking layer **140**, an electron transport layer **145**, an electron injection layer **150**, a protective layer **155**, and a cathode **160**. Cathode **160** is a compound cathode having a first conductive layer **162** and a second conductive layer **164**. Device **100** may be fabricated by depositing the layers described, in order. The properties and functions of these various layers, as well as example materials, are described in more detail in U.S. Pat. No. 7,279,704 at cols. 6-10, which are incorporated by reference.

More examples for each of these layers are available. For example, a flexible and transparent substrate-anode combination is disclosed in U.S. Pat. No. 5,844,363, which is incorporated by reference in its entirety. An example of a p-doped hole transport layer is m-MTDATA doped with F.sub.4-TCNQ at a molar ratio of 50:1, as disclosed in U.S. Patent Application Publication No. 2003/0230980, which is incorporated by reference in its entirety. Examples of emissive and host materials are disclosed in U.S. Pat. No. 6,303,238 to Thompson et al., which is incorporated by reference in its entirety. An example of an n-doped electron transport layer is Bphen doped with Li at a molar ratio of 1:1, as disclosed in U.S. Patent Application Publication No. 2003/0230980, which is incorporated by reference in its entirety. U.S. Pat. Nos. 5,703,436 and 5,707,745, which are incorporated by reference in their entireties, disclose examples of cathodes including compound cathodes having a thin layer of metal such as Mg:Ag with an overlying transparent, electrically-conductive, sputter-deposited ITO layer. The theory and use of blocking layers is described in more detail in U.S. Pat. No. 6,097,147 and U.S. Patent Application Publication No. 2003/0230980, which are incorporated by reference in their entireties. Examples of injection layers are provided in U.S. Patent Application Publication No. 2004/0174116, which is incorporated by reference in its entirety. A description of protective layers may be found in U.S. Patent Application Publication No. 2004/0174116, which is incorporated by reference in its entirety.

FIG. 2 shows an inverted OLED **200**. The device includes a substrate **210**, a cathode **215**, an emissive layer **220**, a hole transport layer **225**, and an anode **230**. Device **200** may be fabricated by depositing the layers described, in order. Because the most common OLED configuration has a cathode disposed over the anode, and device **200** has cathode **215** disposed under anode **230**, device **200** may be referred

to as an "inverted" OLED. Materials similar to those described with respect to device **100** may be used in the corresponding layers of device **200**. FIG. 2 provides one example of how some layers may be omitted from the structure of device **100**.

The simple layered structure illustrated in FIGS. 1 and 2 is provided by way of non-limiting example, and it is understood that embodiments of the invention may be used in connection with a wide variety of other structures. The specific materials and structures described are exemplary in nature, and other materials and structures may be used. Functional OLEDs may be achieved by combining the various layers described in different ways, or layers may be omitted entirely, based on design, performance, and cost factors. Other layers not specifically described may also be included. Materials other than those specifically described may be used. Although many of the examples provided herein describe various layers as comprising a single material, it is understood that combinations of materials, such as a mixture of host and dopant, or more generally a mixture, may be used. Also, the layers may have various sublayers. The names given to the various layers herein are not intended to be strictly limiting. For example, in device **200**, hole transport layer **225** transports holes and injects holes into emissive layer **220**, and may be described as a hole transport layer or a hole injection layer. In one embodiment, an OLED may be described as having an "organic layer" disposed between a cathode and an anode. This organic layer may comprise a single layer, or may further comprise multiple layers of different organic materials as described, for example, with respect to FIGS. 1 and 2.

Structures and materials not specifically described may also be used, such as OLEDs comprised of polymeric materials (PLEDs) such as disclosed in U.S. Pat. No. 5,247,190 to Friend et al., which is incorporated by reference in its entirety. By way of further example, OLEDs having a single organic layer may be used. OLEDs may be stacked, for example as described in U.S. Pat. No. 5,707,745 to Forrest et al., which is incorporated by reference in its entirety. The OLED structure may deviate from the simple layered structure illustrated in FIGS. 1 and 2. For example, the substrate may include an angled reflective surface to improve out-coupling, such as a mesa structure as described in U.S. Pat. No. 6,091,195 to Forrest et al., and/or a pit structure as described in U.S. Pat. No. 5,834,893 to Bulovic et al., which are incorporated by reference in their entireties.

Unless otherwise specified, any of the layers of the various embodiments may be deposited by any suitable method. For the organic layers, preferred methods include thermal evaporation, ink-jet, such as described in U.S. Pat. Nos. 6,013,982 and 6,087,196, which are incorporated by reference in their entireties, organic vapor phase deposition (OVPD), such as described in U.S. Pat. No. 6,337,102 to Forrest et al., which is incorporated by reference in its entirety, and deposition by organic vapor jet printing (OVJP), such as described in U.S. patent application Ser. No. 10/233,470, which is incorporated by reference in its entirety. Other suitable deposition methods include spin coating and other solution based processes. Solution based processes are preferably carried out in nitrogen or an inert atmosphere. For the other layers, preferred methods include thermal evaporation. Preferred patterning methods include deposition through a mask, cold welding such as described in U.S. Pat. Nos. 6,294,398 and 6,468,819, which are incorporated by reference in their entireties, and patterning associated with some of the deposition methods such as ink-jet and OVJD. Other methods may also be used. The

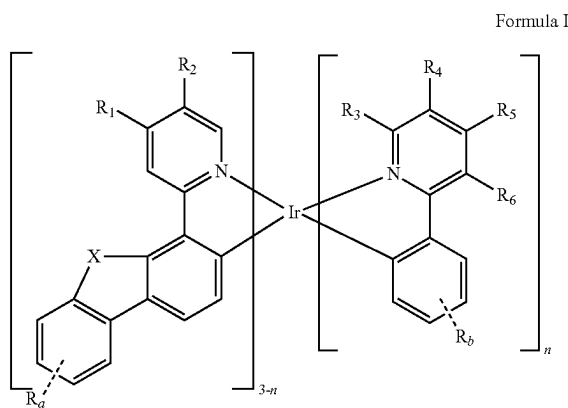
materials to be deposited may be modified to make them compatible with a particular deposition method. For example, substituents such as alkyl and aryl groups, branched or unbranched, and preferably containing at least 3 carbons, may be used in small molecules to enhance their ability to undergo solution processing. Substituents having 20 carbons or more may be used, and 3-20 carbons is a preferred range. Materials with asymmetric structures may have better solution processability than those having symmetric structures, because asymmetric materials may have a lower tendency to recrystallize. Dendrimer substituents may be used to enhance the ability of small molecules to undergo solution processing.

Devices fabricated in accordance with embodiments of the invention may be incorporated into a wide variety of consumer products, including flat panel displays, computer monitors, televisions, billboards, lights for interior or exterior illumination and/or signaling, heads up displays, fully transparent displays, flexible displays, laser printers, telephones, cell phones, personal digital assistants (PDAs), laptop computers, digital cameras, camcorders, viewfinders, micro-displays, vehicles, a large area wall, theater or stadium screen, or a sign. Various control mechanisms may be used to control devices fabricated in accordance with the present invention, including passive matrix and active matrix. Many of the devices are intended for use in a temperature range comfortable to humans, such as 18 degrees C. to 30 degrees C., and more preferably at room temperature (20-25 degrees C.).

The materials and structures described herein may have applications in devices other than OLEDs. For example, other optoelectronic devices such as organic solar cells and organic photodetectors may employ the materials and structures. More generally, organic devices, such as organic transistors, may employ the materials and structures.

The terms halo, halogen, alkyl, cycloalkyl, alkenyl, alkynyl, aryl, heterocyclic group, aryl, aromatic group, and heteroaryl are known to the art, and are defined in U.S. Pat. No. 7,279,704 at cols. 31-32, which are incorporated herein by reference.

In one embodiment, a compound having the formula:

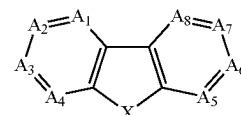


is provided. R_1 and R_2 are optionally linked and the sum of the number of carbon atoms in R_1 and R_2 is at least 2. Thus, both R_1 and R_2 both represent a substituent with at least one carbon. If R_1 does not represent a substituent containing carbon, then R_2 must represent a substituent containing at least two carbons and vice versa. R_3 , R_4 , R_5 , R_6 are optionally linked, and R_a and R_b represent mono-, di-, tri- or

tetra-substitution. X is selected from the group consisting of BR, NR, PR, O, S, Se, C=O, S=O, SO₂, CRR', SiRR', and GeRR', and R_a , R_b , R, R', R_1 , R_2 , R_3 , R_4 , R_5 , and R_6 are independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, aryl-alkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; and wherein n is 1 or 2.

In one embodiment, n is 2. In one embodiment, X is O. In one embodiment, R_1 is hydrogen and R_2 is alkyl. In another embodiment, R_1 is alkyl and R_2 is hydrogen. In one embodiment, R_1 and R_2 are alkyl. In another embodiment, R_1 or R_2 are independently selected from the group consisting of branched alkyl, cyclic alkyl, bicyclic alkyl, and multicyclic alkyl. In one embodiment, R_1 or R_2 is iso-propyl. Substitution at the 4- and 5-positions of either pyridine ring in the compounds of Formula I can give rise to compounds with desirable properties such as saturated green emission, high efficiencies, and long device lifetimes when incorporated in OLED devices. The photophysical and device properties of devices incorporating these compounds may be tuned by varying the nature of the substituent at the 4- or 5-position on the pyridine. The 4-position on a pyridine ring in the compound of Formula I is the position occupied by the R_5 or R_1 substituent, whereas the 5-position is the position occupied by the R_4 or R_2 substituent.

As used herein, fragments containing the following structure:

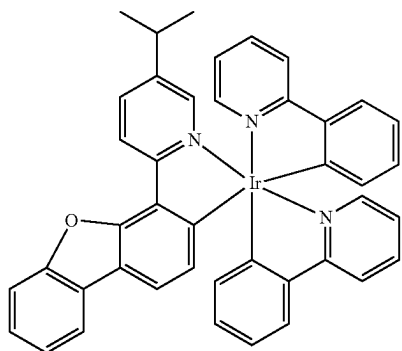


are called DBX groups, i.e. dibenzo X, where X is any of the atoms or groups described herein. Atoms A1-A8 can comprise nitrogen or carbon.

In one embodiment, R_1 or R_2 contain one or more deuterium atoms. In one embodiment, R_1 and R_2 contain one or more deuterium atoms. In one embodiment, R_3 , R_4 , R_5 and R_6 are independently selected from the group consisting of hydrogen, deuterium, alkyl, aryl, and combinations thereof. In another embodiment, at least one of R_3 , R_4 , R_5 and R_6 comprises a branched alkyl, cyclic alkyl, bicyclic alkyl, or a multicyclic alkyl. In one embodiment, R_3 , R_4 , R_5 or R_6 contain one or more deuterium atoms. Without being bound by theory, incorporation of deuterium is thought to improve stability of compounds due the greater bond strength of the carbon-deuterium (C-D) bond versus the carbon-hydrogen (C-H) bond. Therefore, compounds wherein labile C-H bonds are replaced by C-D bonds, higher stability can be expected. Without being bound by theory, it is believed that incorporation of deuterium atoms on the alkyl groups of ligands for iridium complexes, the resulting complexes can have longer device lifetimes.

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In one embodiment, the compound is selected from the group consisting of:

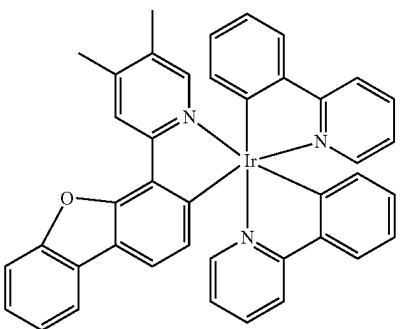


Compound 53 5

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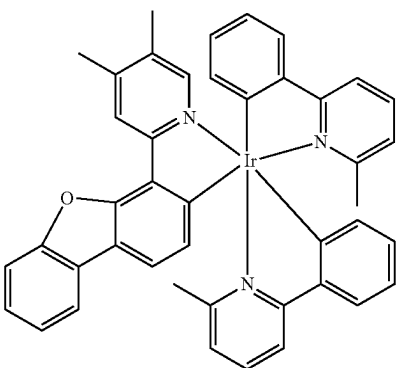


Compound 157

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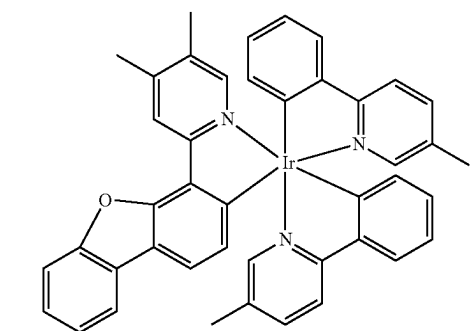


Compound 158

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Compound 159

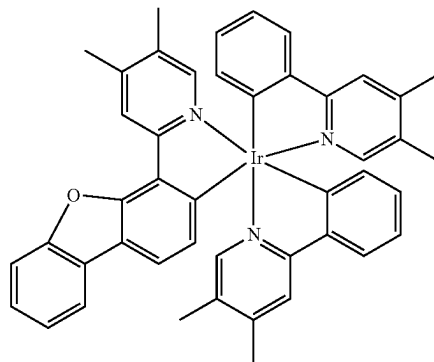
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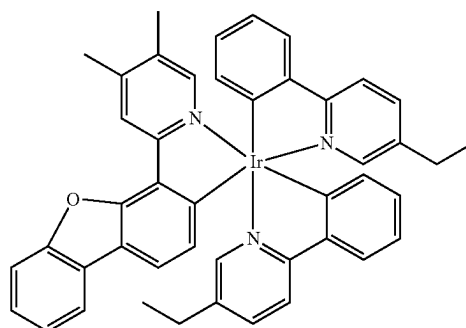
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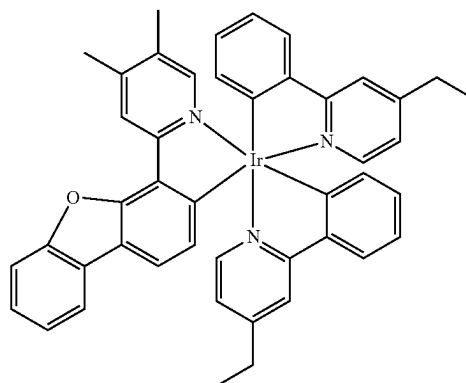
Compound 165



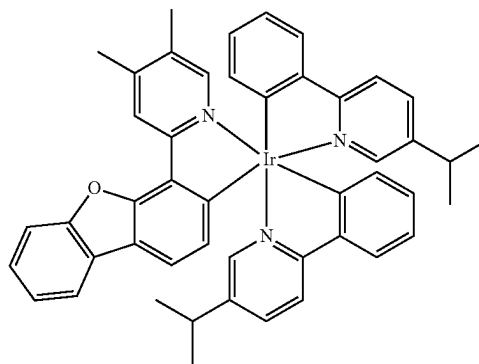
Compound 174



Compound 175



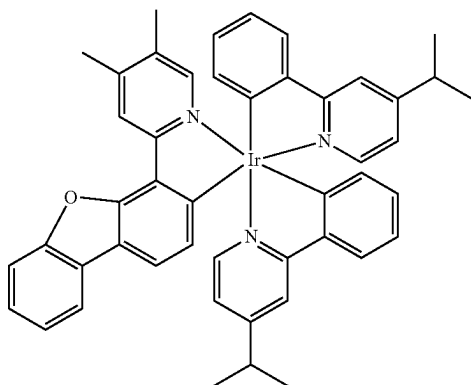
Compound 184



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-continued

Compound 185



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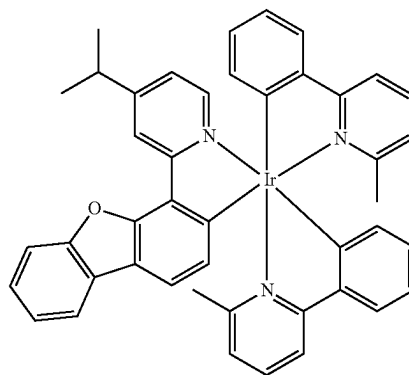
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Compound 626



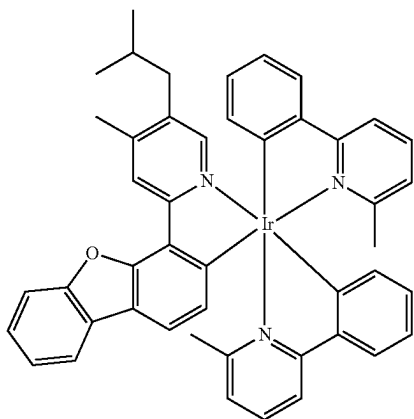
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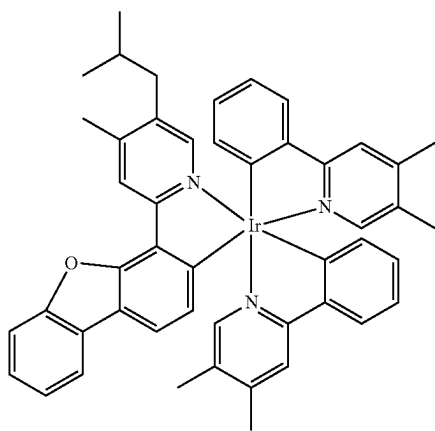
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Compound 314



Compound 321

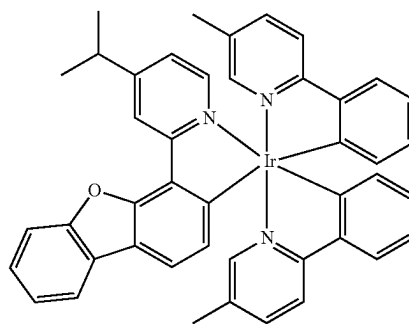


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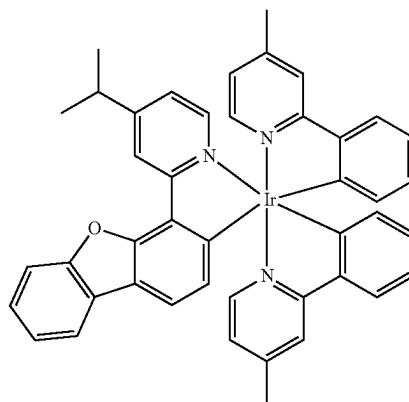
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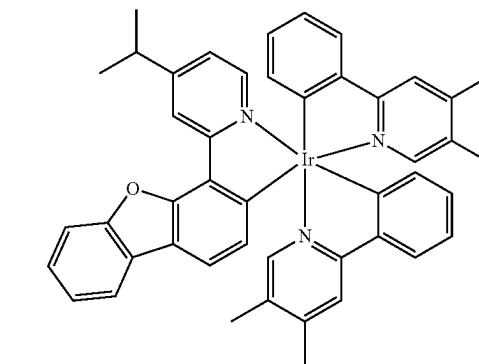
Compound 627



Compound 628



Compound 633

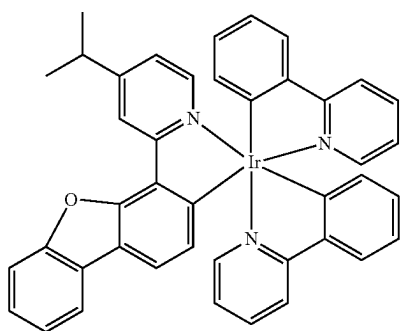


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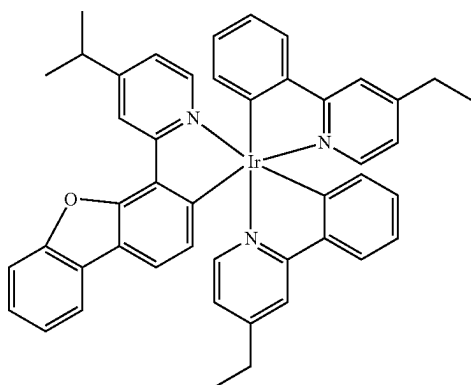
Compound 625



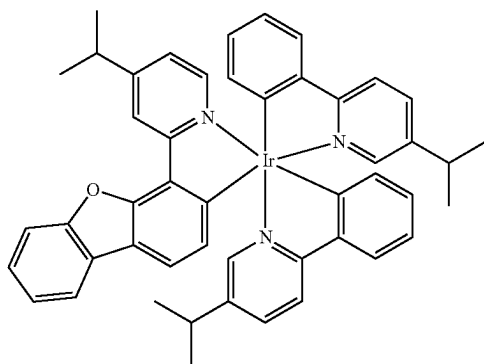
15

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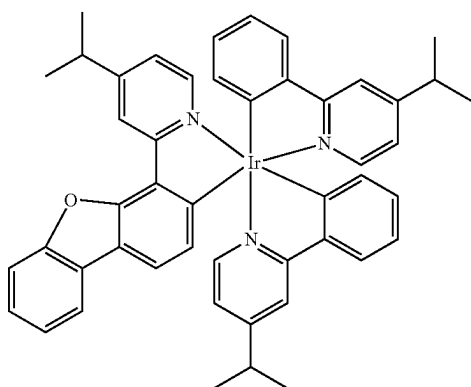
Compound 643



Compound 652

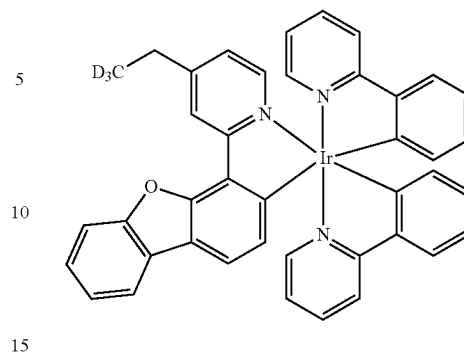


Compound 653

**16**

-continued

Compound 1145



20

25

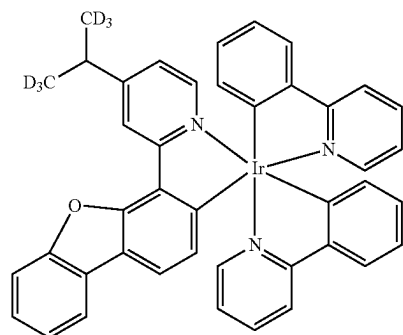
30

35

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45

Compound 1146



In one embodiment, suitable R₁-R₆ groups in the compounds of Formula I include the configurations of substituents in Table 1.

TABLE 1

Compound #	R ₁	R ₂	R ₃	R ₄	R ₅	R ₆
1.	H	Ethyl	H	H	H	H
2.	H	Ethyl	Methyl	H	H	H
3.	H	Ethyl	H	Methyl	H	H
4.	H	Ethyl	H	H	Methyl	H
5.	H	Ethyl	H	H	H	Methyl
6.	H	Ethyl	Methyl	Methyl	H	H
7.	H	Ethyl	Methyl	H	Methyl	H
8.	H	Ethyl	Methyl	H	H	Methyl
9.	H	Ethyl	H	Methyl	Methyl	H
10.	H	Ethyl	H	Methyl	H	Methyl
11.	H	Ethyl	H	H	Methyl	Methyl
12.	H	Ethyl	Methyl	Methyl	Methyl	H
13.	H	Ethyl	Methyl	Methyl	H	Methyl
14.	H	Ethyl	Methyl	H	Methyl	Methyl
15.	H	Ethyl	H	Methyl	Methyl	Methyl

TABLE 1-continued

Compound #	R ₁	R ₂	R ₃	R ₄	R ₅	R ₆
16.	H	Ethyl	Methyl	Methyl	Methyl	Methyl
17.	H	Ethyl	Ethyl	H	H	H
18.	H	Ethyl	H	Ethyl	H	H
19.	H	Ethyl	H	H	Ethyl	H
20.	H	Ethyl	H	H	H	Ethyl
21.	H	Ethyl	Methyl	Ethyl	H	H
22.	H	Ethyl	Methyl	H	Ethyl	H
23.	H	Ethyl	Methyl	H	H	Ethyl
24.	H	Ethyl	H	Ethyl	Methyl	H
25.	H	Ethyl	H	Ethyl	Ethyl	H
26.	H	Ethyl	H	Methyl	Ethyl	H
27.	H	Ethyl	Isopropyl	H	H	H
28.	H	Ethyl	H	Isopropyl	H	H
29.	H	Ethyl	H	H	Isopropyl	H
30.	H	Ethyl	H	H	H	Isopropyl
31.	H	Ethyl	Methyl	Isopropyl	H	H
32.	H	Ethyl	Methyl	H	Isopropyl	H
33.	H	Ethyl	Methyl	H	H	Isopropyl
34.	H	Ethyl	H	Isopropyl	Methyl	H
35.	H	Ethyl	H	Isopropyl	Isopropyl	H
36.	H	Ethyl	H	Methyl	Isopropyl	H
37.	H	Ethyl	H	Isopropyl	Ethyl	H
38.	H	Ethyl	H	Ethyl	Isopropyl	H
39.	H	Ethyl	Isobutyl	H	H	H
40.	H	Ethyl	H	Isobutyl	H	H
41.	H	Ethyl	H	H	Isobutyl	H
42.	H	Ethyl	H	H	H	Isobutyl
43.	H	Ethyl	Methyl	Isobutyl	H	H
44.	H	Ethyl	Methyl	H	Isobutyl	H
45.	H	Ethyl	Methyl	H	H	Isobutyl
46.	H	Ethyl	H	Isobutyl	Methyl	H
47.	H	Ethyl	H	Isobutyl	Isobutyl	H
48.	H	Ethyl	H	Methyl	Isobutyl	H
49.	H	Ethyl	H	Isobutyl	Ethyl	H
50.	H	Ethyl	H	Ethyl	Isobutyl	H
51.	H	Ethyl	H	Isobutyl	Isopropyl	H
52.	H	Ethyl	H	Isopropyl	Isobutyl	H
53.	H	Isopropyl	H	H	H	H
54.	H	Isopropyl	Methyl	H	H	H
55.	H	Isopropyl	H	Methyl	H	H
56.	H	Isopropyl	H	H	Methyl	H
57.	H	Isopropyl	H	H	H	Methyl
58.	H	Isopropyl	Methyl	Methyl	H	H
59.	H	Isopropyl	Methyl	H	Methyl	H
60.	H	Isopropyl	Methyl	H	H	Methyl
61.	H	Isopropyl	H	Methyl	Methyl	H
62.	H	Isopropyl	H	Methyl	H	Methyl
63.	H	Isopropyl	H	H	Methyl	Methyl
64.	H	Isopropyl	Methyl	Methyl	Methyl	H
65.	H	Isopropyl	Methyl	Methyl	H	Methyl
66.	H	Isopropyl	Methyl	H	Methyl	Methyl
67.	H	Isopropyl	H	Methyl	Methyl	Methyl
68.	H	Isopropyl	Methyl	Methyl	Methyl	Methyl
69.	H	Isopropyl	Ethyl	H	H	H
70.	H	Isopropyl	H	Ethyl	H	H
71.	H	Isopropyl	H	H	Ethyl	H
72.	H	Isopropyl	H	H	H	Ethyl
73.	H	Isopropyl	Methyl	Ethyl	H	H
74.	H	Isopropyl	Methyl	H	Ethyl	H
75.	H	Isopropyl	Methyl	H	H	Ethyl
76.	H	Isopropyl	H	Ethyl	Methyl	H
77.	H	Isopropyl	H	Ethyl	Ethyl	H
78.	H	Isopropyl	H	Methyl	Ethyl	H
79.	H	Isopropyl	Isopropyl	H	H	H
80.	H	Isopropyl	H	Isopropyl	H	H
81.	H	Isopropyl	H	H	Isopropyl	H
82.	H	Isopropyl	H	H	H	Isopropyl
83.	H	Isopropyl	Methyl	Isopropyl	H	H
84.	H	Isopropyl	Methyl	H	Isopropyl	H
85.	H	Isopropyl	Methyl	H	H	Isopropyl
86.	H	Isopropyl	H	Isopropyl	Methyl	H
87.	H	Isopropyl	H	Isopropyl	Isopropyl	H
88.	H	Isopropyl	H	Methyl	Isopropyl	H
89.	H	Isopropyl	H	Isopropyl	Ethyl	H
90.	H	Isopropyl	H	Ethyl	Isopropyl	H
91.	H	Isopropyl	Isobutyl	H	H	H
92.	H	Isopropyl	H	Isobutyl	H	H
93.	H	Isopropyl	H	H	Isobutyl	H

TABLE 1-continued

Compound #	R ₁	R ₂	R ₃	R ₄	R ₅	R ₆
94.	H	Isopropyl	H	H	H	Isobutyl
95.	H	Isopropyl	Methyl	Isobutyl	H	H
96.	H	Isopropyl	Methyl	H	Isobutyl	H
97.	H	Isopropyl	Methyl	H	H	Isobutyl
98.	H	Isopropyl	H	Isobutyl	Methyl	H
99.	H	Isopropyl	H	Isobutyl	Isobutyl	H
100.	H	Isopropyl	H	Methyl	Isobutyl	H
101.	H	Isopropyl	H	Isobutyl	Ethyl	H
102.	H	Isopropyl	H	Ethyl	Isobutyl	H
103.	H	Isopropyl	H	Isobutyl	Isopropyl	H
104.	H	Isopropyl	H	Isopropyl	Isobutyl	H
105.	H	Isobutyl	H	H	H	H
106.	H	Isobutyl	Methyl	H	H	H
107.	H	Isobutyl	H	Methyl	H	H
108.	H	Isobutyl	H	H	Methyl	H
109.	H	Isobutyl	H	H	H	Methyl
110.	H	Isobutyl	Methyl	Methyl	H	H
111.	H	Isobutyl	Methyl	H	Methyl	H
112.	H	Isobutyl	Methyl	H	H	Methyl
113.	H	Isobutyl	H	Methyl	Methyl	H
114.	H	Isobutyl	H	Methyl	H	Methyl
115.	H	Isobutyl	H	H	Methyl	Methyl
116.	H	Isobutyl	Methyl	Methyl	Methyl	H
117.	H	Isobutyl	Methyl	Methyl	H	Methyl
118.	H	Isobutyl	Methyl	H	Methyl	Methyl
119.	H	Isobutyl	H	Methyl	Methyl	Methyl
120.	H	Isobutyl	Methyl	Methyl	Methyl	Methyl
121.	H	Isobutyl	Ethyl	H	H	H
122.	H	Isobutyl	H	Ethyl	H	H
123.	H	Isobutyl	H	H	Ethyl	H
124.	H	Isobutyl	H	H	H	Ethyl
125.	H	Isobutyl	Methyl	Ethyl	H	H
126.	H	Isobutyl	Methyl	H	Ethyl	H
127.	H	Isobutyl	Methyl	H	H	Ethyl
128.	H	Isobutyl	H	Ethyl	Methyl	H
129.	H	Isobutyl	H	Ethyl	Ethyl	H
130.	H	Isobutyl	H	Methyl	Ethyl	H
131.	H	Isobutyl	Isopropyl	H	H	H
132.	H	Isobutyl	H	Isopropyl	H	H
133.	H	Isobutyl	H	H	Isopropyl	H
134.	H	Isobutyl	H	H	H	Isopropyl
135.	H	Isobutyl	Methyl	Isopropyl	H	H
136.	H	Isobutyl	Methyl	H	Isopropyl	H
137.	H	Isobutyl	Methyl	H	H	Isopropyl
138.	H	Isobutyl	H	Isopropyl	Methyl	H
139.	H	Isobutyl	H	Isopropyl	Isopropyl	H
140.	H	Isobutyl	H	Methyl	Isopropyl	H
141.	H	Isobutyl	H	Isopropyl	Ethyl	H
142.	H	Isobutyl	H	Ethyl	Isopropyl	H
143.	H	Isobutyl	Isobutyl	H	H	H
144.	H	Isobutyl	H	Isobutyl	H	H
145.	H	Isobutyl	H	H	Isobutyl	H
146.	H	Isobutyl	H	H	H	Isobutyl
147.	H	Isobutyl	Methyl	Isobutyl	H	H
148.	H	Isobutyl	Methyl	H	Isobutyl	H
149.	H	Isobutyl	Methyl	H	H	Isobutyl
150.	H	Isobutyl	H	Isobutyl	Methyl	H
151.	H	Isobutyl	H	Isobutyl	Isobutyl	H
152.	H	Isobutyl	H	Methyl	Isobutyl	H
153.	H	Isobutyl	H	Isobutyl	Ethyl	H
154.	H	Isobutyl	H	Ethyl	Isobutyl	H
155.	H	Isobutyl	H	Isobutyl	Isopropyl	H
156.	H	Isobutyl	H	Isopropyl	Isobutyl	H
157.	Methyl	Methyl	H	H	H	H
158.	Methyl	Methyl	Methyl	H	H	H
159.	Methyl	Methyl	H	Methyl	H	H
160.	Methyl	Methyl	H	H	Methyl	H
161.	Methyl	Methyl	H	H	H	Methyl
162.	Methyl	Methyl	Methyl	Methyl	H	H
163.	Methyl	Methyl	Methyl	H	Methyl	H
164.	Methyl	Methyl	Methyl	H	H	Methyl
165.	Methyl	Methyl	H	Methyl	Methyl	H
166.	Methyl	Methyl	H	Methyl	H	Methyl
167.	Methyl	Methyl	H	H	Methyl	Methyl
168.	Methyl	Methyl	Methyl	Methyl	Methyl	H
169.	Methyl	Methyl	Methyl	Methyl	H	Methyl
170.	Methyl	Methyl	Methyl	H	Methyl	Methyl
171.	Methyl	Methyl	H	Methyl	Methyl	Methyl

TABLE 1-continued

Compound #	R ₁	R ₂	R ₃	R ₄	R ₅	R ₆
172.	Methyl	Methyl	Methyl	Methyl	Methyl	Methyl
173.	Methyl	Methyl	Ethyl	H	H	H
174.	Methyl	Methyl	H	Ethyl	H	H
175.	Methyl	Methyl	H	H	Ethyl	H
176.	Methyl	Methyl	H	H	H	Ethyl
177.	Methyl	Methyl	Methyl	Ethyl	H	H
178.	Methyl	Methyl	Methyl	H	Ethyl	H
179.	Methyl	Methyl	Methyl	H	H	Ethyl
180.	Methyl	Methyl	H	Ethyl	Methyl	H
181.	Methyl	Methyl	H	Ethyl	Ethyl	H
182.	Methyl	Methyl	H	Methyl	Ethyl	H
183.	Methyl	Methyl	Isopropyl	H	H	H
184.	Methyl	Methyl	H	Isopropyl	H	H
185.	Methyl	Methyl	H	H	Isopropyl	H
186.	Methyl	Methyl	H	H	H	Isopropyl
187.	Methyl	Methyl	Methyl	Isopropyl	H	H
188.	Methyl	Methyl	Methyl	H	Isopropyl	H
189.	Methyl	Methyl	Methyl	H	H	Isopropyl
190.	Methyl	Methyl	H	Isopropyl	Methyl	H
191.	Methyl	Methyl	H	Isopropyl	Isopropyl	H
192.	Methyl	Methyl	H	Methyl	Isopropyl	H
193.	Methyl	Methyl	H	Isopropyl	Ethyl	H
194.	Methyl	Methyl	H	Ethyl	Isopropyl	H
195.	Methyl	Methyl	Isobutyl	H	H	H
196.	Methyl	Methyl	H	Isobutyl	H	H
197.	Methyl	Methyl	H	H	Isobutyl	H
198.	Methyl	Methyl	H	H	H	Isobutyl
199.	Methyl	Methyl	Methyl	Isobutyl	H	H
200.	Methyl	Methyl	Methyl	H	Isobutyl	H
201.	Methyl	Methyl	Methyl	H	H	Isobutyl
202.	Methyl	Methyl	H	Isobutyl	Methyl	H
203.	Methyl	Methyl	H	Isobutyl	Isobutyl	H
204.	Methyl	Methyl	H	Methyl	Isobutyl	H
205.	Methyl	Methyl	H	Isobutyl	Ethyl	H
206.	Methyl	Methyl	H	Ethyl	Isobutyl	H
207.	Methyl	Methyl	H	Isobutyl	Isopropyl	H
208.	Methyl	Methyl	H	Isopropyl	Isobutyl	H
209.	Methyl	Ethyl	H	H	H	H
210.	Methyl	Ethyl	Methyl	H	H	H
211.	Methyl	Ethyl	H	Methyl	H	H
212.	Methyl	Ethyl	H	H	Methyl	H
213.	Methyl	Ethyl	H	H	H	Methyl
214.	Methyl	Ethyl	Methyl	Methyl	H	H
215.	Methyl	Ethyl	Methyl	H	Methyl	H
216.	Methyl	Ethyl	Methyl	H	H	Methyl
217.	Methyl	Ethyl	H	Methyl	Methyl	H
218.	Methyl	Ethyl	H	Methyl	H	Methyl
219.	Methyl	Ethyl	H	H	Methyl	Methyl
220.	Methyl	Ethyl	Methyl	Methyl	Methyl	H
221.	Methyl	Ethyl	Methyl	Methyl	H	Methyl
222.	Methyl	Ethyl	Methyl	H	Methyl	Methyl
223.	Methyl	Ethyl	H	Methyl	Methyl	Methyl
224.	Methyl	Ethyl	Methyl	Methyl	Methyl	Methyl
225.	Methyl	Ethyl	Ethyl	H	H	H
226.	Methyl	Ethyl	H	Ethyl	H	H
227.	Methyl	Ethyl	H	H	Ethyl	H
228.	Methyl	Ethyl	H	H	H	Ethyl
229.	Methyl	Ethyl	Methyl	Ethyl	H	H
230.	Methyl	Ethyl	Methyl	H	Ethyl	H
231.	Methyl	Ethyl	Methyl	H	H	Ethyl
232.	Methyl	Ethyl	H	Ethyl	Methyl	H
233.	Methyl	Ethyl	H	Ethyl	Ethyl	H
234.	Methyl	Ethyl	H	Methyl	Ethyl	H
235.	Methyl	Ethyl	Isopropyl	H	H	H
236.	Methyl	Ethyl	H	Isopropyl	H	H
237.	Methyl	Ethyl	H	H	Isopropyl	H
238.	Methyl	Ethyl	H	H	H	Isopropyl
239.	Methyl	Ethyl	Methyl	Isopropyl	H	H
240.	Methyl	Ethyl	Methyl	H	Isopropyl	H
241.	Methyl	Ethyl	Methyl	H	H	Isopropyl
242.	Methyl	Ethyl	H	Isopropyl	Methyl	H
243.	Methyl	Ethyl	H	Isopropyl	Isopropyl	H
244.	Methyl	Ethyl	H	Methyl	Isopropyl	H
245.	Methyl	Ethyl	H	Isopropyl	Ethyl	H
246.	Methyl	Ethyl	H	Ethyl	Isopropyl	H
247.	Methyl	Ethyl	Isobutyl	H	H	H
248.	Methyl	Ethyl	H	Isobutyl	H	H
249.	Methyl	Ethyl	H	H	Isobutyl	H

TABLE 1-continued

Compound #	R ₁	R ₂	R ₃	R ₄	R ₅	R ₆
250.	Methyl	Ethyl	H	H	H	Isobutyl
251.	Methyl	Ethyl	Methyl	Isobutyl	H	H
252.	Methyl	Ethyl	Methyl	H	Isobutyl	H
253.	Methyl	Ethyl	Methyl	H	H	Isobutyl
254.	Methyl	Ethyl	H	Isobutyl	Methyl	H
255.	Methyl	Ethyl	H	Isobutyl	Isobutyl	H
256.	Methyl	Ethyl	H	Methyl	Isobutyl	H
257.	Methyl	Ethyl	H	Isobutyl	Ethyl	H
258.	Methyl	Ethyl	H	Ethyl	Isobutyl	H
259.	Methyl	Ethyl	H	Isobutyl	Isopropyl	H
260.	Methyl	Ethyl	H	Isopropyl	Isobutyl	H
261.	Methyl	Isopropyl	H	H	H	H
262.	Methyl	Isopropyl	Methyl	H	H	H
263.	Methyl	Isopropyl	H	Methyl	H	H
264.	Methyl	Isopropyl	H	H	Methyl	H
265.	Methyl	Isopropyl	H	H	H	Methyl
266.	Methyl	Isopropyl	Methyl	Methyl	H	H
267.	Methyl	Isopropyl	Methyl	H	Methyl	H
268.	Methyl	Isopropyl	Methyl	H	H	Methyl
269.	Methyl	Isopropyl	H	Methyl	Methyl	H
270.	Methyl	Isopropyl	H	Methyl	H	Methyl
271.	Methyl	Isopropyl	H	H	Methyl	Methyl
272.	Methyl	Isopropyl	Methyl	Methyl	Methyl	H
273.	Methyl	Isopropyl	Methyl	Methyl	H	Methyl
274.	Methyl	Isopropyl	Methyl	H	Methyl	Methyl
275.	Methyl	Isopropyl	H	Methyl	Methyl	Methyl
276.	Methyl	Isopropyl	Methyl	Methyl	Methyl	Methyl
277.	Methyl	Isopropyl	Ethyl	H	H	H
278.	Methyl	Isopropyl	H	Ethyl	H	H
279.	Methyl	Isopropyl	H	H	Ethyl	H
280.	Methyl	Isopropyl	H	H	H	Ethyl
281.	Methyl	Isopropyl	Methyl	Ethyl	H	H
282.	Methyl	Isopropyl	Methyl	H	Ethyl	H
283.	Methyl	Isopropyl	Methyl	H	H	Ethyl
284.	Methyl	Isopropyl	H	Ethyl	Methyl	H
285.	Methyl	Isopropyl	H	Ethyl	Ethyl	H
286.	Methyl	Isopropyl	H	Methyl	Ethyl	H
287.	Methyl	Isopropyl	Isopropyl	H	H	H
288.	Methyl	Isopropyl	H	Isopropyl	H	H
289.	Methyl	Isopropyl	H	H	Isopropyl	H
290.	Methyl	Isopropyl	H	H	H	Isopropyl
291.	Methyl	Isopropyl	Methyl	Isopropyl	H	H
292.	Methyl	Isopropyl	Methyl	H	Isopropyl	H
293.	Methyl	Isopropyl	Methyl	H	H	Isopropyl
294.	Methyl	Isopropyl	H	Isopropyl	Methyl	H
295.	Methyl	Isopropyl	H	Isopropyl	Isopropyl	H
296.	Methyl	Isopropyl	H	Methyl	Isopropyl	H
297.	Methyl	Isopropyl	H	Isopropyl	Ethyl	H
298.	Methyl	Isopropyl	H	Ethyl	Isopropyl	H
299.	Methyl	Isopropyl	Isobutyl	H	H	H
300.	Methyl	Isopropyl	H	Isobutyl	H	H
301.	Methyl	Isopropyl	H	H	Isobutyl	H
302.	Methyl	Isopropyl	H	H	H	Isobutyl
303.	Methyl	Isopropyl	Methyl	Isobutyl	H	H
304.	Methyl	Isopropyl	Methyl	H	Isobutyl	H
305.	Methyl	Isopropyl	Methyl	H	H	Isobutyl
306.	Methyl	Isopropyl	H	Isobutyl	Methyl	H
307.	Methyl	Isopropyl	H	Isobutyl	Isobutyl	H
308.	Methyl	Isopropyl	H	Methyl	Isobutyl	H
309.	Methyl	Isopropyl	H	Isobutyl	Ethyl	H
310.	Methyl	Isopropyl	H	Ethyl	Isobutyl	H
311.	Methyl	Isopropyl	H	Isobutyl	Isopropyl	H
312.	Methyl	Isopropyl	H	Isopropyl	Isobutyl	H
313.	Methyl	Isobutyl	H	H	H	H
314.	Methyl	Isobutyl	Methyl	H	H	H
315.	Methyl	Isobutyl	H	Methyl	H	H
316.	Methyl	Isobutyl	H	H	Methyl	H
317.	Methyl	Isobutyl	H	H	H	Methyl
318.	Methyl	Isobutyl	Methyl	Methyl	H	H
319.	Methyl	Isobutyl	Methyl	H	Methyl	H
320.	Methyl	Isobutyl	Methyl	H	H	Methyl
321.	Methyl	Isobutyl	H	Methyl	Methyl	H
322.	Methyl	Isobutyl	H	Methyl	H	Methyl
323.	Methyl	Isobutyl	H	H	Methyl	Methyl
324.	Methyl	Isobutyl	Methyl	Methyl	Methyl	H
325.	Methyl	Isobutyl	Methyl	Methyl	H	Methyl
326.	Methyl	Isobutyl	Methyl	H	Methyl	Methyl
327.	Methyl	Isobutyl	H	Methyl	Methyl	Methyl

TABLE 1-continued

Compound #	R ₁	R ₂	R ₃	R ₄	R ₅	R ₆
328.	Methyl	Isobutyl	Methyl	Methyl	Methyl	Methyl
329.	Methyl	Isobutyl	Ethyl	H	H	H
330.	Methyl	Isobutyl	H	Ethyl	H	H
331.	Methyl	Isobutyl	H	H	Ethyl	H
332.	Methyl	Isobutyl	H	H	H	Ethyl
333.	Methyl	Isobutyl	Methyl	Ethyl	H	H
334.	Methyl	Isobutyl	Methyl	H	Ethyl	H
335.	Methyl	Isobutyl	Methyl	H	H	Ethyl
336.	Methyl	Isobutyl	H	Ethyl	Methyl	H
337.	Methyl	Isobutyl	H	Ethyl	Ethyl	H
338.	Methyl	Isobutyl	H	Methyl	Ethyl	H
339.	Methyl	Isobutyl	Isopropyl	H	H	H
340.	Methyl	Isobutyl	H	Isopropyl	H	H
341.	Methyl	Isobutyl	H	H	Isopropyl	H
342.	Methyl	Isobutyl	H	H	H	Isopropyl
343.	Methyl	Isobutyl	Methyl	Isopropyl	H	H
344.	Methyl	Isobutyl	Methyl	H	Isopropyl	H
345.	Methyl	Isobutyl	Methyl	H	H	Isopropyl
346.	Methyl	Isobutyl	H	Isopropyl	Methyl	H
347.	Methyl	Isobutyl	H	Isopropyl	Isopropyl	H
348.	Methyl	Isobutyl	H	Methyl	Isopropyl	H
349.	Methyl	Isobutyl	H	Isopropyl	Ethyl	H
350.	Methyl	Isobutyl	H	Ethyl	Isopropyl	H
351.	Methyl	Isobutyl	Isobutyl	H	H	H
352.	Methyl	Isobutyl	H	Isobutyl	H	H
353.	Methyl	Isobutyl	H	H	Isobutyl	H
354.	Methyl	Isobutyl	H	H	H	Isobutyl
355.	Methyl	Isobutyl	Methyl	Isobutyl	H	H
356.	Methyl	Isobutyl	Methyl	H	Isobutyl	H
357.	Methyl	Isobutyl	Methyl	H	H	Isobutyl
358.	Methyl	Isobutyl	H	Isobutyl	Methyl	H
359.	Methyl	Isobutyl	H	Isobutyl	Isobutyl	H
360.	Methyl	Isobutyl	H	Methyl	Isobutyl	H
361.	Methyl	Isobutyl	H	Isobutyl	Ethyl	H
362.	Methyl	Isobutyl	H	Ethyl	Isobutyl	H
363.	Methyl	Isobutyl	H	Isobutyl	Isopropyl	H
364.	Methyl	Isobutyl	H	Isopropyl	Isobutyl	H
365.	Ethyl	H	H	H	H	H
366.	Ethyl	H	Methyl	H	H	H
367.	Ethyl	H	H	Methyl	H	H
368.	Ethyl	H	H	H	Methyl	H
369.	Ethyl	H	H	H	H	Methyl
370.	Ethyl	H	Methyl	Methyl	H	H
371.	Ethyl	H	Methyl	H	Methyl	H
372.	Ethyl	H	Methyl	H	H	Methyl
373.	Ethyl	H	H	Methyl	Methyl	H
374.	Ethyl	H	H	Methyl	H	Methyl
375.	Ethyl	H	H	H	Methyl	Methyl
376.	Ethyl	H	Methyl	Methyl	Methyl	H
377.	Ethyl	H	Methyl	Methyl	H	Methyl
378.	Ethyl	H	Methyl	H	Methyl	Methyl
379.	Ethyl	H	H	Methyl	Methyl	Methyl
380.	Ethyl	H	Methyl	Methyl	Methyl	Methyl
381.	Ethyl	H	Ethyl	H	H	H
382.	Ethyl	H	H	Ethyl	H	H
383.	Ethyl	H	H	H	Ethyl	H
384.	Ethyl	H	H	H	H	Ethyl
385.	Ethyl	H	Methyl	Ethyl	H	H
386.	Ethyl	H	Methyl	H	Ethyl	H
387.	Ethyl	H	Methyl	H	H	Ethyl
388.	Ethyl	H	H	Ethyl	Methyl	H
389.	Ethyl	H	H	Ethyl	Ethyl	H
390.	Ethyl	H	H	Methyl	Ethyl	H
391.	Ethyl	H	Isopropyl	H	H	H
392.	Ethyl	H	H	Isopropyl	H	H
393.	Ethyl	H	H	H	Isopropyl	H
394.	Ethyl	H	H	H	H	Isopropyl
395.	Ethyl	H	Methyl	Isopropyl	H	H
396.	Ethyl	H	Methyl	H	Isopropyl	H
397.	Ethyl	H	Methyl	H	H	Isopropyl
398.	Ethyl	H	H	Isopropyl	Methyl	H
399.	Ethyl	H	H	Isopropyl	Isopropyl	H
400.	Ethyl	H	H	Methyl	Isopropyl	H
401.	Ethyl	H	H	Isopropyl	Ethyl	H
402.	Ethyl	H	H	Ethyl	Isopropyl	H
403.	Ethyl	H	Isobutyl	H	H	H
404.	Ethyl	H	H	Isobutyl	H	H
405.	Ethyl	H	H	H	Isobutyl	H

TABLE 1-continued

Compound #	R ₁	R ₂	R ₃	R ₄	R ₅	R ₆
406.	Ethyl	H	H	H	H	Isobutyl
407.	Ethyl	H	Methyl	Isobutyl	H	H
408.	Ethyl	H	Methyl	H	Isobutyl	H
409.	Ethyl	H	Methyl	H	H	Isobutyl
410.	Ethyl	H	H	Isobutyl	Methyl	H
411.	Ethyl	H	H	Isobutyl	Isobutyl	H
412.	Ethyl	H	H	Methyl	Isobutyl	H
413.	Ethyl	H	H	Isobutyl	Ethyl	H
414.	Ethyl	H	H	Ethyl	Isobutyl	H
415.	Ethyl	H	H	Isobutyl	Isopropyl	H
416.	Ethyl	H	H	Isopropyl	Isobutyl	H
417.	Ethyl	Methyl	H	H	H	H
418.	Ethyl	Methyl	Methyl	H	H	H
419.	Ethyl	Methyl	H	Methyl	H	H
420.	Ethyl	Methyl	H	H	Methyl	H
421.	Ethyl	Methyl	H	H	H	Methyl
422.	Ethyl	Methyl	Methyl	Methyl	H	H
423.	Ethyl	Methyl	Methyl	H	Methyl	H
424.	Ethyl	Methyl	Methyl	H	H	Methyl
425.	Ethyl	Methyl	H	Methyl	Methyl	H
426.	Ethyl	Methyl	H	Methyl	H	Methyl
427.	Ethyl	Methyl	H	H	Methyl	Methyl
428.	Ethyl	Methyl	Methyl	Methyl	Methyl	H
429.	Ethyl	Methyl	Methyl	Methyl	H	Methyl
430.	Ethyl	Methyl	Methyl	H	Methyl	Methyl
431.	Ethyl	Methyl	H	Methyl	Methyl	Methyl
432.	Ethyl	Methyl	Methyl	Methyl	Methyl	Methyl
433.	Ethyl	Methyl	Ethyl	H	H	H
434.	Ethyl	Methyl	H	Ethyl	H	H
435.	Ethyl	Methyl	H	H	Ethyl	H
436.	Ethyl	Methyl	H	H	H	Ethyl
437.	Ethyl	Methyl	Methyl	Ethyl	H	H
438.	Ethyl	Methyl	Methyl	H	Ethyl	H
439.	Ethyl	Methyl	Methyl	H	H	Ethyl
440.	Ethyl	Methyl	H	Ethyl	Methyl	H
441.	Ethyl	Methyl	H	Ethyl	Ethyl	H
442.	Ethyl	Methyl	H	Methyl	Ethyl	H
443.	Ethyl	Methyl	Isopropyl	H	H	H
444.	Ethyl	Methyl	H	Isopropyl	H	H
445.	Ethyl	Methyl	H	H	Isopropyl	H
446.	Ethyl	Methyl	H	H	H	Isopropyl
447.	Ethyl	Methyl	Methyl	Isopropyl	H	H
448.	Ethyl	Methyl	Methyl	H	Isopropyl	H
449.	Ethyl	Methyl	Methyl	H	H	Isopropyl
450.	Ethyl	Methyl	H	Isopropyl	Methyl	H
451.	Ethyl	Methyl	H	Isopropyl	Isopropyl	H
452.	Ethyl	Methyl	H	Methyl	Isopropyl	H
453.	Ethyl	Methyl	H	Isopropyl	Ethyl	H
454.	Ethyl	Methyl	H	Ethyl	Isopropyl	H
455.	Ethyl	Methyl	Isobutyl	H	H	H
456.	Ethyl	Methyl	H	Isobutyl	H	H
457.	Ethyl	Methyl	H	H	Isobutyl	H
458.	Ethyl	Methyl	H	H	H	Isobutyl
459.	Ethyl	Methyl	Methyl	Isobutyl	H	H
460.	Ethyl	Methyl	Methyl	H	Isobutyl	H
461.	Ethyl	Methyl	Methyl	H	H	Isobutyl
462.	Ethyl	Methyl	H	Isobutyl	Methyl	H
463.	Ethyl	Methyl	H	Isobutyl	Isobutyl	H
464.	Ethyl	Methyl	H	Methyl	Isobutyl	H
465.	Ethyl	Methyl	H	Isobutyl	Ethyl	H
466.	Ethyl	Methyl	H	Ethyl	Isobutyl	H
467.	Ethyl	Methyl	H	Isobutyl	Isopropyl	H
468.	Ethyl	Methyl	H	Isopropyl	Isobutyl	H
469.	Ethyl	Ethyl	H	H	H	H
470.	Ethyl	Ethyl	Methyl	H	H	H
471.	Ethyl	Ethyl	H	Methyl	H	H
472.	Ethyl	Ethyl	H	H	Methyl	H
473.	Ethyl	Ethyl	H	H	H	Methyl
474.	Ethyl	Ethyl	Methyl	Methyl	H	H
475.	Ethyl	Ethyl	Methyl	H	Methyl	H
476.	Ethyl	Ethyl	Methyl	H	H	Methyl
477.	Ethyl	Ethyl	H	Methyl	Methyl	H
478.	Ethyl	Ethyl	H	Methyl	H	Methyl
479.	Ethyl	Ethyl	H	H	Methyl	Methyl
480.	Ethyl	Ethyl	Methyl	Methyl	Methyl	H
481.	Ethyl	Ethyl	Methyl	Methyl	H	Methyl
482.	Ethyl	Ethyl	Methyl	H	Methyl	Methyl
483.	Ethyl	Ethyl	H	Methyl	Methyl	Methyl

TABLE 1-continued

Compound #	R ₁	R ₂	R ₃	R ₄	R ₅	R ₆
484.	Ethyl	Ethyl	Methyl	Methyl	Methyl	Methyl
485.	Ethyl	Ethyl	Ethyl	H	H	H
486.	Ethyl	Ethyl	H	Ethyl	H	H
487.	Ethyl	Ethyl	H	H	Ethyl	H
488.	Ethyl	Ethyl	H	H	H	Ethyl
489.	Ethyl	Ethyl	Methyl	Ethyl	H	H
490.	Ethyl	Ethyl	Methyl	H	Ethyl	H
491.	Ethyl	Ethyl	Methyl	H	H	Ethyl
492.	Ethyl	Ethyl	H	Ethyl	Methyl	H
493.	Ethyl	Ethyl	H	Ethyl	Ethyl	H
494.	Ethyl	Ethyl	H	Methyl	Ethyl	H
495.	Ethyl	Ethyl	Isopropyl	H	H	H
496.	Ethyl	Ethyl	H	Isopropyl	H	H
497.	Ethyl	Ethyl	H	H	Isopropyl	H
498.	Ethyl	Ethyl	H	H	H	Isopropyl
499.	Ethyl	Ethyl	Methyl	Isopropyl	H	H
500.	Ethyl	Ethyl	Methyl	H	Isopropyl	H
501.	Ethyl	Ethyl	Methyl	H	H	Isopropyl
502.	Ethyl	Ethyl	H	Isopropyl	Methyl	H
503.	Ethyl	Ethyl	H	Isopropyl	Isopropyl	H
504.	Ethyl	Ethyl	H	Methyl	Isopropyl	H
505.	Ethyl	Ethyl	H	Isopropyl	Ethyl	H
506.	Ethyl	Ethyl	H	Ethyl	Isopropyl	H
507.	Ethyl	Ethyl	Isobutyl	H	H	H
508.	Ethyl	Ethyl	H	Isobutyl	H	H
509.	Ethyl	Ethyl	H	H	Isobutyl	H
510.	Ethyl	Ethyl	H	H	H	Isobutyl
511.	Ethyl	Ethyl	Methyl	Isobutyl	H	H
512.	Ethyl	Ethyl	Methyl	H	Isobutyl	H
513.	Ethyl	Ethyl	Methyl	H	H	Isobutyl
514.	Ethyl	Ethyl	H	Isobutyl	Methyl	H
515.	Ethyl	Ethyl	H	Isobutyl	Isobutyl	H
516.	Ethyl	Ethyl	H	Methyl	Isobutyl	H
517.	Ethyl	Ethyl	H	Isobutyl	Ethyl	H
518.	Ethyl	Ethyl	H	Ethyl	Isobutyl	H
519.	Ethyl	Ethyl	H	Isobutyl	Isopropyl	H
520.	Ethyl	Ethyl	H	Isopropyl	Isobutyl	H
521.	Ethyl	Isopropyl	H	H	H	H
522.	Ethyl	Isopropyl	Methyl	H	H	H
523.	Ethyl	Isopropyl	H	Methyl	H	H
524.	Ethyl	Isopropyl	H	H	Methyl	H
525.	Ethyl	Isopropyl	H	H	H	Methyl
526.	Ethyl	Isopropyl	Methyl	Methyl	H	H
527.	Ethyl	Isopropyl	Methyl	H	Methyl	H
528.	Ethyl	Isopropyl	Methyl	H	H	Methyl
529.	Ethyl	Isopropyl	H	Methyl	Methyl	H
530.	Ethyl	Isopropyl	H	Methyl	H	Methyl
531.	Ethyl	Isopropyl	H	H	Methyl	Methyl
532.	Ethyl	Isopropyl	Methyl	Methyl	Methyl	H
533.	Ethyl	Isopropyl	Methyl	Methyl	H	Methyl
534.	Ethyl	Isopropyl	Methyl	H	Methyl	Methyl
535.	Ethyl	Isopropyl	H	Methyl	Methyl	Methyl
536.	Ethyl	Isopropyl	Methyl	Methyl	Methyl	Methyl
537.	Ethyl	Isopropyl	Ethyl	H	H	H
538.	Ethyl	Isopropyl	H	Ethyl	H	H
539.	Ethyl	Isopropyl	H	H	Ethyl	H
540.	Ethyl	Isopropyl	H	H	H	Ethyl
541.	Ethyl	Isopropyl	Methyl	Ethyl	H	H
542.	Ethyl	Isopropyl	Methyl	H	Ethyl	H
543.	Ethyl	Isopropyl	Methyl	H	H	Ethyl
544.	Ethyl	Isopropyl	H	Ethyl	Methyl	H
545.	Ethyl	Isopropyl	H	Ethyl	Ethyl	H
546.	Ethyl	Isopropyl	H	Methyl	Ethyl	H
547.	Ethyl	Isopropyl	Isopropyl	H	H	H
548.	Ethyl	Isopropyl	H	Isopropyl	H	H
549.	Ethyl	Isopropyl	H	H	Isopropyl	H
550.	Ethyl	Isopropyl	H	H	H	Isopropyl
551.	Ethyl	Isopropyl	Methyl	Isopropyl	H	H
552.	Ethyl	Isopropyl	Methyl	H	Isopropyl	H
553.	Ethyl	Isopropyl	Methyl	H	H	Isopropyl
554.	Ethyl	Isopropyl	H	Isopropyl	Methyl	H
555.	Ethyl	Isopropyl	H	Isopropyl	Isopropyl	H
556.	Ethyl	Isopropyl	H	Methyl	Isopropyl	H
557.	Ethyl	Isopropyl	H	Isopropyl	Ethyl	H
558.	Ethyl	Isopropyl	H	Ethyl	Isopropyl	H
559.	Ethyl	Isopropyl	Isobutyl	H	H	H
560.	Ethyl	Isopropyl	H	Isobutyl	H	H
561.	Ethyl	Isopropyl	H	H	Isobutyl	H

TABLE 1-continued

Compound #	R ₁	R ₂	R ₃	R ₄	R ₅	R ₆
562.	Ethyl	Isopropyl	H	H	H	Isobutyl
563.	Ethyl	Isopropyl	Methyl	Isobutyl	H	H
564.	Ethyl	Isopropyl	Methyl	H	Isobutyl	H
565.	Ethyl	Isopropyl	Methyl	H	H	Isobutyl
566.	Ethyl	Isopropyl	H	Isobutyl	Methyl	H
567.	Ethyl	Isopropyl	H	Isobutyl	Isobutyl	H
568.	Ethyl	Isopropyl	H	Methyl	Isobutyl	H
569.	Ethyl	Isopropyl	H	Isobutyl	Ethyl	H
570.	Ethyl	Isopropyl	H	Ethyl	Isobutyl	H
571.	Ethyl	Isopropyl	H	Isobutyl	Isopropyl	H
572.	Ethyl	Isopropyl	H	Isopropyl	Isobutyl	H
573.	Ethyl	Isobutyl	H	H	H	H
574.	Ethyl	Isobutyl	Methyl	H	H	H
575.	Ethyl	Isobutyl	H	Methyl	H	H
576.	Ethyl	Isobutyl	H	H	Methyl	H
577.	Ethyl	Isobutyl	H	H	H	Methyl
578.	Ethyl	Isobutyl	Methyl	Methyl	H	H
579.	Ethyl	Isobutyl	Methyl	H	Methyl	H
580.	Ethyl	Isobutyl	Methyl	H	H	Methyl
581.	Ethyl	Isobutyl	H	Methyl	Methyl	H
582.	Ethyl	Isobutyl	H	Methyl	H	Methyl
583.	Ethyl	Isobutyl	H	H	Methyl	Methyl
584.	Ethyl	Isobutyl	Methyl	Methyl	Methyl	H
585.	Ethyl	Isobutyl	Methyl	Methyl	H	Methyl
586.	Ethyl	Isobutyl	Methyl	H	Methyl	Methyl
587.	Ethyl	Isobutyl	H	Methyl	Methyl	Methyl
588.	Ethyl	Isobutyl	Methyl	Methyl	Methyl	Methyl
589.	Ethyl	Isobutyl	Ethyl	H	H	H
590.	Ethyl	Isobutyl	H	Ethyl	H	H
591.	Ethyl	Isobutyl	H	H	Ethyl	H
592.	Ethyl	Isobutyl	H	H	H	Ethyl
593.	Ethyl	Isobutyl	Methyl	Ethyl	H	H
594.	Ethyl	Isobutyl	Methyl	H	Ethyl	H
595.	Ethyl	Isobutyl	Methyl	H	H	Ethyl
596.	Ethyl	Isobutyl	H	Ethyl	Methyl	H
597.	Ethyl	Isobutyl	H	Ethyl	Ethyl	H
598.	Ethyl	Isobutyl	H	Methyl	Ethyl	H
599.	Ethyl	Isobutyl	Isopropyl	H	H	H
600.	Ethyl	Isobutyl	H	Isopropyl	H	H
601.	Ethyl	Isobutyl	H	H	Isopropyl	H
602.	Ethyl	Isobutyl	H	H	H	Isopropyl
603.	Ethyl	Isobutyl	Methyl	Isopropyl	H	H
604.	Ethyl	Isobutyl	Methyl	H	Isopropyl	H
605.	Ethyl	Isobutyl	Methyl	H	H	Isopropyl
606.	Ethyl	Isobutyl	H	Isopropyl	Methyl	H
607.	Ethyl	Isobutyl	H	Isopropyl	Isopropyl	H
608.	Ethyl	Isobutyl	H	Methyl	Isopropyl	H
609.	Ethyl	Isobutyl	H	Isopropyl	Ethyl	H
610.	Ethyl	Isobutyl	H	Ethyl	Isopropyl	H
611.	Ethyl	Isobutyl	Isobutyl	H	H	H
612.	Ethyl	Isobutyl	H	Isobutyl	H	H
613.	Ethyl	Isobutyl	H	H	Isobutyl	H
614.	Ethyl	Isobutyl	H	H	H	Isobutyl
615.	Ethyl	Isobutyl	Methyl	Isobutyl	H	H
616.	Ethyl	Isobutyl	Methyl	H	Isobutyl	H
617.	Ethyl	Isobutyl	Methyl	H	H	Isobutyl
618.	Ethyl	Isobutyl	H	Isobutyl	Methyl	H
619.	Ethyl	Isobutyl	H	Isobutyl	Isobutyl	H
620.	Ethyl	Isobutyl	H	Methyl	Isobutyl	H
621.	Ethyl	Isobutyl	H	Isobutyl	Ethyl	H
622.	Ethyl	Isobutyl	H	Ethyl	Isobutyl	H
623.	Ethyl	Isobutyl	H	Isobutyl	Isopropyl	H
624.	Ethyl	Isobutyl	H	Isopropyl	Isobutyl	H
625.	Isopropyl	H	H	H	H	H
626.	Isopropyl	H	Methyl	H	H	H
627.	Isopropyl	H	H	Methyl	H	H
628.	Isopropyl	H	H	H	Methyl	H
629.	Isopropyl	H	H	H	H	Methyl
630.	Isopropyl	H	Methyl	Methyl	H	H
631.	Isopropyl	H	Methyl	H	Methyl	H
632.	Isopropyl	H	Methyl	H	H	Methyl
633.	Isopropyl	H	H	Methyl	Methyl	H
634.	Isopropyl	H	H	Methyl	H	Methyl
635.	Isopropyl	H	H	H	Methyl	Methyl
636.	Isopropyl	H	Methyl	Methyl	Methyl	H
637.	Isopropyl	H	Methyl	Methyl	H	Methyl
638.	Isopropyl	H	Methyl	H	Methyl	Methyl
639.	Isopropyl	H	H	Methyl	Methyl	Methyl

TABLE 1-continued

Compound #	R ₁	R ₂	R ₃	R ₄	R ₅	R ₆
640.	Isopropyl	H	Methyl	Methyl	Methyl	Methyl
641.	Isopropyl	H	Ethyl	H	H	H
642.	Isopropyl	H	H	Ethyl	H	H
643.	Isopropyl	H	H	H	Ethyl	H
644.	Isopropyl	H	H	H	H	Ethyl
645.	Isopropyl	H	Methyl	Ethyl	H	H
646.	Isopropyl	H	Methyl	H	Ethyl	H
647.	Isopropyl	H	Methyl	H	H	Ethyl
648.	Isopropyl	H	H	Ethyl	Methyl	H
649.	Isopropyl	H	H	Ethyl	Ethyl	H
650.	Isopropyl	H	H	Methyl	Ethyl	H
651.	Isopropyl	H	Isopropyl	H	H	H
652.	Isopropyl	H	H	Isopropyl	H	H
653.	Isopropyl	H	H	H	Isopropyl	H
654.	Isopropyl	H	H	H	H	Isopropyl
655.	Isopropyl	H	Methyl	Isopropyl	H	H
656.	Isopropyl	H	Methyl	H	Isopropyl	H
657.	Isopropyl	H	Methyl	H	H	Isopropyl
658.	Isopropyl	H	H	Isopropyl	Methyl	H
659.	Isopropyl	H	H	Isopropyl	Isopropyl	H
660.	Isopropyl	H	H	Methyl	Isopropyl	H
661.	Isopropyl	H	H	Isopropyl	Ethyl	H
662.	Isopropyl	H	H	Ethyl	Isopropyl	H
663.	Isopropyl	H	Isobutyl	H	H	H
664.	Isopropyl	H	H	Isobutyl	H	H
665.	Isopropyl	H	H	H	Isobutyl	H
666.	Isopropyl	H	H	H	H	Isobutyl
667.	Isopropyl	H	Methyl	Isobutyl	H	H
668.	Isopropyl	H	Methyl	H	Isobutyl	H
669.	Isopropyl	H	Methyl	H	H	Isobutyl
670.	Isopropyl	H	H	Isobutyl	Methyl	H
671.	Isopropyl	H	H	Isobutyl	Isobutyl	H
672.	Isopropyl	H	H	Methyl	Isobutyl	H
673.	Isopropyl	H	H	Isobutyl	Ethyl	H
674.	Isopropyl	H	H	Ethyl	Isobutyl	H
675.	Isopropyl	H	H	Isobutyl	Isopropyl	H
676.	Isopropyl	H	H	Isopropyl	Isobutyl	H
677.	Isopropyl	Methyl	H	H	H	H
678.	Isopropyl	Methyl	Methyl	H	H	H
679.	Isopropyl	Methyl	H	Methyl	H	H
680.	Isopropyl	Methyl	H	H	Methyl	H
681.	Isopropyl	Methyl	H	H	H	Methyl
682.	Isopropyl	Methyl	Methyl	Methyl	H	H
683.	Isopropyl	Methyl	Methyl	H	Methyl	H
684.	Isopropyl	Methyl	Methyl	H	H	Methyl
685.	Isopropyl	Methyl	H	Methyl	Methyl	H
686.	Isopropyl	Methyl	H	Methyl	H	Methyl
687.	Isopropyl	Methyl	H	H	Methyl	Methyl
688.	Isopropyl	Methyl	Methyl	Methyl	Methyl	H
689.	Isopropyl	Methyl	Methyl	Methyl	H	Methyl
690.	Isopropyl	Methyl	Methyl	H	Methyl	Methyl
691.	Isopropyl	Methyl	H	Methyl	Methyl	Methyl
692.	Isopropyl	Methyl	Methyl	Methyl	Methyl	Methyl
693.	Isopropyl	Methyl	Ethyl	H	H	H
694.	Isopropyl	Methyl	H	Ethyl	H	H
695.	Isopropyl	Methyl	H	H	Ethyl	H
696.	Isopropyl	Methyl	H	H	H	Ethyl
697.	Isopropyl	Methyl	Methyl	Ethyl	H	H
698.	Isopropyl	Methyl	Methyl	H	Ethyl	H
699.	Isopropyl	Methyl	Methyl	H	H	Ethyl
700.	Isopropyl	Methyl	H	Ethyl	Methyl	H
701.	Isopropyl	Methyl	H	Ethyl	Ethyl	H
702.	Isopropyl	Methyl	H	Methyl	Ethyl	H
703.	Isopropyl	Methyl	Isopropyl	H	H	H
704.	Isopropyl	Methyl	H	Isopropyl	H	H
705.	Isopropyl	Methyl	H	H	Isopropyl	H
706.	Isopropyl	Methyl	H	H	H	Isopropyl
707.	Isopropyl	Methyl	Methyl	Isopropyl	H	H
708.	Isopropyl	Methyl	Methyl	H	Isopropyl	H
709.	Isopropyl	Methyl	Methyl	H	H	Isopropyl
710.	Isopropyl	Methyl	H	Isopropyl	Methyl	H
711.	Isopropyl	Methyl	H	Isopropyl	Isopropyl	H
712.	Isopropyl	Methyl	H	Methyl	Isopropyl	H
713.	Isopropyl	Methyl	H	Isopropyl	Ethyl	H
714.	Isopropyl	Methyl	H	Ethyl	Isopropyl	H
715.	Isopropyl	Methyl	Isobutyl	H	H	H
716.	Isopropyl	Methyl	H	Isobutyl	H	H
717.	Isopropyl	Methyl	H	H	Isobutyl	H

TABLE 1-continued

Compound #	R ₁	R ₂	R ₃	R ₄	R ₅	R ₆
718.	Isopropyl	Methyl	H	H	H	Isobutyl
719.	Isopropyl	Methyl	Methyl	Isobutyl	H	H
720.	Isopropyl	Methyl	Methyl	H	Isobutyl	H
721.	Isopropyl	Methyl	Methyl	H	H	Isobutyl
722.	Isopropyl	Methyl	H	Isobutyl	Methyl	H
723.	Isopropyl	Methyl	H	Isobutyl	Isobutyl	H
724.	Isopropyl	Methyl	H	Methyl	Isobutyl	H
725.	Isopropyl	Methyl	H	Isobutyl	Ethyl	H
726.	Isopropyl	Methyl	H	Ethyl	Isobutyl	H
727.	Isopropyl	Methyl	H	Isobutyl	Isopropyl	H
728.	Isopropyl	Methyl	H	Isopropyl	Isobutyl	H
729.	Isopropyl	Ethyl	H	H	H	H
730.	Isopropyl	Ethyl	Methyl	H	H	H
731.	Isopropyl	Ethyl	H	Methyl	H	H
732.	Isopropyl	Ethyl	H	H	Methyl	H
733.	Isopropyl	Ethyl	H	H	H	Methyl
734.	Isopropyl	Ethyl	Methyl	Methyl	H	H
735.	Isopropyl	Ethyl	Methyl	H	Methyl	H
736.	Isopropyl	Ethyl	Methyl	H	H	Methyl
737.	Isopropyl	Ethyl	H	Methyl	Methyl	H
738.	Isopropyl	Ethyl	H	Methyl	H	Methyl
739.	Isopropyl	Ethyl	H	H	Methyl	Methyl
740.	Isopropyl	Ethyl	Methyl	Methyl	Methyl	H
741.	Isopropyl	Ethyl	Methyl	Methyl	H	Methyl
742.	Isopropyl	Ethyl	Methyl	H	Methyl	Methyl
743.	Isopropyl	Ethyl	H	Methyl	Methyl	Methyl
744.	Isopropyl	Ethyl	Methyl	Methyl	Methyl	Methyl
745.	Isopropyl	Ethyl	Ethyl	H	H	H
746.	Isopropyl	Ethyl	H	Ethyl	H	H
747.	Isopropyl	Ethyl	H	H	Ethyl	H
748.	Isopropyl	Ethyl	H	H	H	Ethyl
749.	Isopropyl	Ethyl	Methyl	Ethyl	H	H
750.	Isopropyl	Ethyl	Methyl	H	Ethyl	H
751.	Isopropyl	Ethyl	Methyl	H	H	Ethyl
752.	Isopropyl	Ethyl	H	Ethyl	Methyl	H
753.	Isopropyl	Ethyl	H	Ethyl	Ethyl	H
754.	Isopropyl	Ethyl	H	Methyl	Ethyl	H
755.	Isopropyl	Ethyl	Isopropyl	H	H	H
756.	Isopropyl	Ethyl	H	Isopropyl	H	H
757.	Isopropyl	Ethyl	H	H	Isopropyl	H
758.	Isopropyl	Ethyl	H	H	H	Isopropyl
759.	Isopropyl	Ethyl	Methyl	Isopropyl	H	H
760.	Isopropyl	Ethyl	Methyl	H	Isopropyl	H
761.	Isopropyl	Ethyl	Methyl	H	H	Isopropyl
762.	Isopropyl	Ethyl	H	Isopropyl	Methyl	H
763.	Isopropyl	Ethyl	H	Isopropyl	Isopropyl	H
764.	Isopropyl	Ethyl	H	Methyl	Isopropyl	H
765.	Isopropyl	Ethyl	H	Isopropyl	Ethyl	H
766.	Isopropyl	Ethyl	H	Ethyl	Isopropyl	H
767.	Isopropyl	Ethyl	Isobutyl	H	H	H
768.	Isopropyl	Ethyl	H	Isobutyl	H	H
769.	Isopropyl	Ethyl	H	H	Isobutyl	H
770.	Isopropyl	Ethyl	H	H	H	Isobutyl
771.	Isopropyl	Ethyl	Methyl	Isobutyl	H	H
772.	Isopropyl	Ethyl	Methyl	H	Isobutyl	H
773.	Isopropyl	Ethyl	Methyl	H	H	Isobutyl
774.	Isopropyl	Ethyl	H	Isobutyl	Methyl	H
775.	Isopropyl	Ethyl	H	Isobutyl	Isobutyl	H
776.	Isopropyl	Ethyl	H	Methyl	Isobutyl	H
777.	Isopropyl	Ethyl	H	Isobutyl	Ethyl	H
778.	Isopropyl	Ethyl	H	Ethyl	Isobutyl	H
779.	Isopropyl	Ethyl	H	Isobutyl	Isopropyl	H
780.	Isopropyl	Ethyl	H	Isopropyl	Isobutyl	H
781.	Isopropyl	Isopropyl	H	H	H	H
782.	Isopropyl	Isopropyl	Methyl	H	H	H
783.	Isopropyl	Isopropyl	H	Methyl	H	H
784.	Isopropyl	Isopropyl	H	H	Methyl	H
785.	Isopropyl	Isopropyl	H	H	H	Methyl
786.	Isopropyl	Isopropyl	Methyl	Methyl	H	H
787.	Isopropyl	Isopropyl	Methyl	H	Methyl	H
788.	Isopropyl	Isopropyl	Methyl	H	H	Methyl
789.	Isopropyl	Isopropyl	H	Methyl	Methyl	H
790.	Isopropyl	Isopropyl	H	Methyl	H	Methyl
791.	Isopropyl	Isopropyl	H	H	Methyl	Methyl
792.	Isopropyl	Isopropyl	Methyl	Methyl	Methyl	H
793.	Isopropyl	Isopropyl	Methyl	Methyl	H	Methyl
794.	Isopropyl	Isopropyl	Methyl	H	Methyl	Methyl
795.	Isopropyl	Isopropyl	H	Methyl	Methyl	Methyl

TABLE 1-continued

Compound #	R ₁	R ₂	R ₃	R ₄	R ₅	R ₆
796.	Isopropyl	Isopropyl	Methyl	Methyl	Methyl	Methyl
797.	Isopropyl	Isopropyl	Ethyl	H	H	H
798.	Isopropyl	Isopropyl	H	Ethyl	H	H
799.	Isopropyl	Isopropyl	H	H	Ethyl	H
800.	Isopropyl	Isopropyl	H	H	H	Ethyl
801.	Isopropyl	Isopropyl	Methyl	Ethyl	H	H
802.	Isopropyl	Isopropyl	Methyl	H	Ethyl	H
803.	Isopropyl	Isopropyl	Methyl	H	H	Ethyl
804.	Isopropyl	Isopropyl	H	Ethyl	Methyl	H
805.	Isopropyl	Isopropyl	H	Ethyl	Ethyl	H
806.	Isopropyl	Isopropyl	H	Methyl	Ethyl	H
807.	Isopropyl	Isopropyl	Isopropyl	H	H	H
808.	Isopropyl	Isopropyl	H	Isopropyl	H	H
809.	Isopropyl	Isopropyl	H	H	Isopropyl	H
810.	Isopropyl	Isopropyl	H	H	H	Isopropyl
811.	Isopropyl	Isopropyl	Methyl	Isopropyl	H	H
812.	Isopropyl	Isopropyl	Methyl	H	Isopropyl	H
813.	Isopropyl	Isopropyl	Methyl	H	H	Isopropyl
814.	Isopropyl	Isopropyl	H	Isopropyl	Methyl	H
815.	Isopropyl	Isopropyl	H	Isopropyl	Isopropyl	H
816.	Isopropyl	Isopropyl	H	Methyl	Isopropyl	H
817.	Isopropyl	Isopropyl	H	Isopropyl	Ethyl	H
818.	Isopropyl	Isopropyl	H	Ethyl	Isopropyl	H
819.	Isopropyl	Isopropyl	Isobutyl	H	H	H
820.	Isopropyl	Isopropyl	H	Isobutyl	H	H
821.	Isopropyl	Isopropyl	H	H	Isobutyl	H
822.	Isopropyl	Isopropyl	H	H	H	Isobutyl
823.	Isopropyl	Isopropyl	Methyl	Isobutyl	H	H
824.	Isopropyl	Isopropyl	Methyl	H	Isobutyl	H
825.	Isopropyl	Isopropyl	Methyl	H	H	Isobutyl
826.	Isopropyl	Isopropyl	H	Isobutyl	Methyl	H
827.	Isopropyl	Isopropyl	H	Isobutyl	Isobutyl	H
828.	Isopropyl	Isopropyl	H	Methyl	Isobutyl	H
829.	Isopropyl	Isopropyl	H	Isobutyl	Ethyl	H
830.	Isopropyl	Isopropyl	H	Ethyl	Isobutyl	H
831.	Isopropyl	Isopropyl	H	Isobutyl	Isopropyl	H
832.	Isopropyl	Isopropyl	H	Isopropyl	Isobutyl	H
833.	Isopropyl	Isobutyl	H	H	H	H
834.	Isopropyl	Isobutyl	Methyl	H	H	H
835.	Isopropyl	Isobutyl	H	Methyl	H	H
836.	Isopropyl	Isobutyl	H	H	Methyl	H
837.	Isopropyl	Isobutyl	H	H	H	Methyl
838.	Isopropyl	Isobutyl	Methyl	Methyl	H	H
839.	Isopropyl	Isobutyl	Methyl	H	Methyl	H
840.	Isopropyl	Isobutyl	Methyl	H	H	Methyl
841.	Isopropyl	Isobutyl	H	Methyl	Methyl	H
842.	Isopropyl	Isobutyl	H	Methyl	H	Methyl
843.	Isopropyl	Isobutyl	H	H	Methyl	Methyl
844.	Isopropyl	Isobutyl	Methyl	Methyl	Methyl	H
845.	Isopropyl	Isobutyl	Methyl	Methyl	H	Methyl
846.	Isopropyl	Isobutyl	Methyl	H	Methyl	Methyl
847.	Isopropyl	Isobutyl	H	Methyl	Methyl	Methyl
848.	Isopropyl	Isobutyl	Methyl	Methyl	Methyl	Methyl
849.	Isopropyl	Isobutyl	Ethyl	H	H	H
850.	Isopropyl	Isobutyl	H	Ethyl	H	H
851.	Isopropyl	Isobutyl	H	H	Ethyl	H
852.	Isopropyl	Isobutyl	H	H	H	Ethyl
853.	Isopropyl	Isobutyl	Methyl	Ethyl	H	H
854.	Isopropyl	Isobutyl	Methyl	H	Ethyl	H
855.	Isopropyl	Isobutyl	Methyl	H	H	Ethyl
856.	Isopropyl	Isobutyl	H	Ethyl	Methyl	H
857.	Isopropyl	Isobutyl	H	Ethyl	Ethyl	H
858.	Isopropyl	Isobutyl	H	Methyl	Ethyl	H
859.	Isopropyl	Isobutyl	Isopropyl	H	H	H
860.	Isopropyl	Isobutyl	H	Isopropyl	H	H
861.	Isopropyl	Isobutyl	H	H	Isopropyl	H
862.	Isopropyl	Isobutyl	H	H	H	Isopropyl
863.	Isopropyl	Isobutyl	Methyl	Isopropyl	H	H
864.	Isopropyl	Isobutyl	Methyl	H	Isopropyl	H
865.	Isopropyl	Isobutyl	Methyl	H	H	Isopropyl
866.	Isopropyl	Isobutyl	H	Isopropyl	Methyl	H
867.	Isopropyl	Isobutyl	H	Isopropyl	Isopropyl	H
868.	Isopropyl	Isobutyl	H	Methyl	Isopropyl	H
869.	Isopropyl	Isobutyl	H	Isopropyl	Ethyl	H
870.	Isopropyl	Isobutyl	H	Ethyl	Isopropyl	H
871.	Isopropyl	Isobutyl	Isobutyl	H	H	H
872.	Isopropyl	Isobutyl	H	Isobutyl	H	H
873.	Isopropyl	Isobutyl	H	H	Isobutyl	H

TABLE 1-continued

Compound #	R ₁	R ₂	R ₃	R ₄	R ₅	R ₆
874.	Isopropyl	Isobutyl	H	H	H	Isobutyl
875.	Isopropyl	Isobutyl	Methyl	Isobutyl	H	H
876.	Isopropyl	Isobutyl	Methyl	H	Isobutyl	H
877.	Isopropyl	Isobutyl	Methyl	H	H	Isobutyl
878.	Isopropyl	Isobutyl	H	Isobutyl	Methyl	H
879.	Isopropyl	Isobutyl	H	Isobutyl	Isobutyl	H
880.	Isopropyl	Isobutyl	H	Methyl	Isobutyl	H
881.	Isopropyl	Isobutyl	H	Isobutyl	Ethyl	H
882.	Isopropyl	Isobutyl	H	Ethyl	Isobutyl	H
883.	Isopropyl	Isobutyl	H	Isobutyl	Isopropyl	H
884.	Isopropyl	Isobutyl	H	Isopropyl	Isobutyl	H
885.	Isobutyl	H	H	H	H	H
886.	Isobutyl	H	Methyl	H	H	H
887.	Isobutyl	H	H	Methyl	H	H
888.	Isobutyl	H	H	H	Methyl	H
889.	Isobutyl	H	H	H	H	Methyl
890.	Isobutyl	H	Methyl	Methyl	H	H
891.	Isobutyl	H	Methyl	H	Methyl	H
892.	Isobutyl	H	Methyl	H	H	Methyl
893.	Isobutyl	H	H	Methyl	Methyl	H
894.	Isobutyl	H	H	Methyl	H	Methyl
895.	Isobutyl	H	H	H	Methyl	Methyl
896.	Isobutyl	H	Methyl	Methyl	Methyl	H
897.	Isobutyl	H	Methyl	Methyl	H	Methyl
898.	Isobutyl	H	Methyl	H	Methyl	Methyl
899.	Isobutyl	H	H	Methyl	Methyl	Methyl
900.	Isobutyl	H	Methyl	Methyl	Methyl	Methyl
901.	Isobutyl	H	Ethyl	H	H	H
902.	Isobutyl	H	H	Ethyl	H	H
903.	Isobutyl	H	H	H	Ethyl	H
904.	Isobutyl	H	H	H	H	Ethyl
905.	Isobutyl	H	Methyl	Ethyl	H	H
906.	Isobutyl	H	Methyl	H	Ethyl	H
907.	Isobutyl	H	Methyl	H	H	Ethyl
908.	Isobutyl	H	H	Ethyl	Methyl	H
909.	Isobutyl	H	H	Ethyl	Ethyl	H
910.	Isobutyl	H	H	Methyl	Ethyl	H
911.	Isobutyl	H	Isopropyl	H	H	H
912.	Isobutyl	H	H	Isopropyl	H	H
913.	Isobutyl	H	H	H	Isopropyl	H
914.	Isobutyl	H	H	H	H	Isopropyl
915.	Isobutyl	H	Methyl	Isopropyl	H	H
916.	Isobutyl	H	Methyl	H	Isopropyl	H
917.	Isobutyl	H	Methyl	H	H	Isopropyl
918.	Isobutyl	H	H	Isopropyl	Methyl	H
919.	Isobutyl	H	H	Isopropyl	Isopropyl	H
920.	Isobutyl	H	H	Methyl	Isopropyl	H
921.	Isobutyl	H	H	Isopropyl	Ethyl	H
922.	Isobutyl	H	H	Ethyl	Isopropyl	H
923.	Isobutyl	H	Isobutyl	H	H	H
924.	Isobutyl	H	H	Isobutyl	H	H
925.	Isobutyl	H	H	H	Isobutyl	H
926.	Isobutyl	H	H	H	H	Isobutyl
927.	Isobutyl	H	Methyl	Isobutyl	H	H
928.	Isobutyl	H	Methyl	H	Isobutyl	H
929.	Isobutyl	H	Methyl	H	H	Isobutyl
930.	Isobutyl	H	H	Isobutyl	Methyl	H
931.	Isobutyl	H	H	Isobutyl	Isobutyl	H
932.	Isobutyl	H	H	Methyl	Isobutyl	H
933.	Isobutyl	H	H	Isobutyl	Ethyl	H
934.	Isobutyl	H	H	Ethyl	Isobutyl	H
935.	Isobutyl	H	H	Isobutyl	Isopropyl	H
936.	Isobutyl	H	H	Isopropyl	Isobutyl	H
937.	Isobutyl	Methyl	H	H	H	H
938.	Isobutyl	Methyl	Methyl	H	H	H
939.	Isobutyl	Methyl	H	Methyl	H	H
940.	Isobutyl	Methyl	H	H	Methyl	H
941.	Isobutyl	Methyl	H	H	H	Methyl
942.	Isobutyl	Methyl	Methyl	Methyl	H	H
943.	Isobutyl	Methyl	Methyl	H	Methyl	H
944.	Isobutyl	Methyl	Methyl	H	H	Methyl
945.	Isobutyl	Methyl	H	Methyl	Methyl	H
946.	Isobutyl	Methyl	H	Methyl	H	Methyl
947.	Isobutyl	Methyl	H	H	Methyl	Methyl
948.	Isobutyl	Methyl	Methyl	Methyl	Methyl	H
949.	Isobutyl	Methyl	Methyl	Methyl	H	Methyl
950.	Isobutyl	Methyl	Methyl	H	Methyl	Methyl
951.	Isobutyl	Methyl	H	Methyl	Methyl	Methyl

TABLE 1-continued

Compound #	R ₁	R ₂	R ₃	R ₄	R ₅	R ₆
952.	Isobutyl	Methyl	Methyl	Methyl	Methyl	Methyl
953.	Isobutyl	Methyl	Ethyl	H	H	H
954.	Isobutyl	Methyl	H	Ethyl	H	H
955.	Isobutyl	Methyl	H	H	Ethyl	H
956.	Isobutyl	Methyl	H	H	H	Ethyl
957.	Isobutyl	Methyl	Methyl	Ethyl	H	H
958.	Isobutyl	Methyl	Methyl	H	Ethyl	H
959.	Isobutyl	Methyl	Methyl	H	H	Ethyl
960.	Isobutyl	Methyl	H	Ethyl	Methyl	H
961.	Isobutyl	Methyl	H	Ethyl	Ethyl	H
962.	Isobutyl	Methyl	H	Methyl	Ethyl	H
963.	Isobutyl	Methyl	Isopropyl	H	H	H
964.	Isobutyl	Methyl	H	Isopropyl	H	H
965.	Isobutyl	Methyl	H	H	Isopropyl	H
966.	Isobutyl	Methyl	H	H	H	Isopropyl
967.	Isobutyl	Methyl	Methyl	Isopropyl	H	H
968.	Isobutyl	Methyl	Methyl	H	Isopropyl	H
969.	Isobutyl	Methyl	Methyl	H	H	Isopropyl
970.	Isobutyl	Methyl	H	Isopropyl	Methyl	H
971.	Isobutyl	Methyl	H	Isopropyl	Isopropyl	H
972.	Isobutyl	Methyl	H	Methyl	Isopropyl	H
973.	Isobutyl	Methyl	H	Isopropyl	Ethyl	H
974.	Isobutyl	Methyl	H	Ethyl	Isopropyl	H
975.	Isobutyl	Methyl	Isobutyl	H	H	H
976.	Isobutyl	Methyl	H	Isobutyl	H	H
977.	Isobutyl	Methyl	H	H	Isobutyl	H
978.	Isobutyl	Methyl	H	H	H	Isobutyl
979.	Isobutyl	Methyl	Methyl	Isobutyl	H	H
980.	Isobutyl	Methyl	Methyl	H	Isobutyl	H
981.	Isobutyl	Methyl	Methyl	H	H	Isobutyl
982.	Isobutyl	Methyl	H	Isobutyl	Methyl	H
983.	Isobutyl	Methyl	H	Isobutyl	Isobutyl	H
984.	Isobutyl	Methyl	H	Methyl	Isobutyl	H
985.	Isobutyl	Methyl	H	Isobutyl	Ethyl	H
986.	Isobutyl	Methyl	H	Ethyl	Isobutyl	H
987.	Isobutyl	Methyl	H	Isobutyl	Isopropyl	H
988.	Isobutyl	Methyl	H	Isopropyl	Isobutyl	H
989.	Isobutyl	Ethyl	H	H	H	H
990.	Isobutyl	Ethyl	Methyl	H	H	H
991.	Isobutyl	Ethyl	H	Methyl	H	H
992.	Isobutyl	Ethyl	H	H	Methyl	H
993.	Isobutyl	Ethyl	H	H	H	Methyl
994.	Isobutyl	Ethyl	Methyl	Methyl	H	H
995.	Isobutyl	Ethyl	Methyl	H	Methyl	H
996.	Isobutyl	Ethyl	Methyl	H	H	Methyl
997.	Isobutyl	Ethyl	H	Methyl	Methyl	H
998.	Isobutyl	Ethyl	H	Methyl	H	Methyl
999.	Isobutyl	Ethyl	H	H	Methyl	Methyl
1000.	Isobutyl	Ethyl	Methyl	Methyl	Methyl	H
1001.	Isobutyl	Ethyl	Methyl	Methyl	H	Methyl
1002.	Isobutyl	Ethyl	Methyl	H	Methyl	Methyl
1003.	Isobutyl	Ethyl	H	Methyl	Methyl	Methyl
1004.	Isobutyl	Ethyl	Methyl	Methyl	Methyl	Methyl
1005.	Isobutyl	Ethyl	Ethyl	H	H	H
1006.	Isobutyl	Ethyl	H	Ethyl	H	H
1007.	Isobutyl	Ethyl	H	H	Ethyl	H
1008.	Isobutyl	Ethyl	H	H	H	Ethyl
1009.	Isobutyl	Ethyl	Methyl	Ethyl	H	H
1010.	Isobutyl	Ethyl	Methyl	H	Ethyl	H
1011.	Isobutyl	Ethyl	Methyl	H	H	Ethyl
1012.	Isobutyl	Ethyl	H	Ethyl	Methyl	H
1013.	Isobutyl	Ethyl	H	Ethyl	Ethyl	H
1014.	Isobutyl	Ethyl	H	Methyl	Ethyl	H
1015.	Isobutyl	Ethyl	Isopropyl	H	H	H
1016.	Isobutyl	Ethyl	H	Isopropyl	H	H
1017.	Isobutyl	Ethyl	H	H	Isopropyl	H
1018.	Isobutyl	Ethyl	H	H	H	Isopropyl
1019.	Isobutyl	Ethyl	Methyl	Isopropyl	H	H
1020.	Isobutyl	Ethyl	Methyl	H	Isopropyl	H
1021.	Isobutyl	Ethyl	Methyl	H	H	Isopropyl
1022.	Isobutyl	Ethyl	H	Isopropyl	Methyl	H
1023.	Isobutyl	Ethyl	H	Isopropyl	Isopropyl	H
1024.	Isobutyl	Ethyl	H	Methyl	Isopropyl	H
1025.	Isobutyl	Ethyl	H	Isopropyl	Ethyl	H
1026.	Isobutyl	Ethyl	H	Ethyl	Isopropyl	H
1027.	Isobutyl	Ethyl	Isobutyl	H	H	H
1028.	Isobutyl	Ethyl	H	Isobutyl	H	H
1029.	Isobutyl	Ethyl	H	H	Isobutyl	H

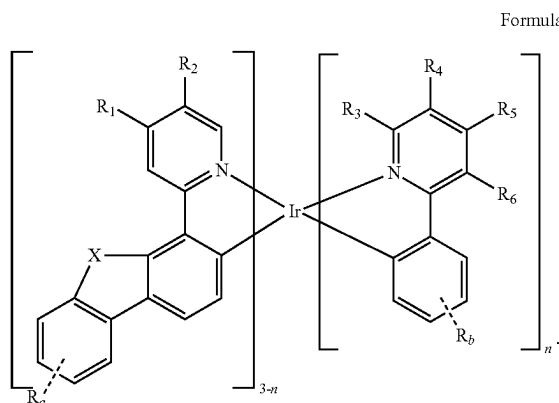
TABLE 1-continued

Compound #	R ₁	R ₂	R ₃	R ₄	R ₅	R ₆
1030.	Isobutyl	Ethyl	H	H	H	Isobutyl
1031.	Isobutyl	Ethyl	Methyl	Isobutyl	H	H
1032.	Isobutyl	Ethyl	Methyl	H	Isobutyl	H
1033.	Isobutyl	Ethyl	Methyl	H	H	Isobutyl
1034.	Isobutyl	Ethyl	H	Isobutyl	Methyl	H
1035.	Isobutyl	Ethyl	H	Isobutyl	Isobutyl	H
1036.	Isobutyl	Ethyl	H	Methyl	Isobutyl	H
1037.	Isobutyl	Ethyl	H	Isobutyl	Ethyl	H
1038.	Isobutyl	Ethyl	H	Ethyl	Isobutyl	H
1039.	Isobutyl	Ethyl	H	Isobutyl	Isopropyl	H
1040.	Isobutyl	Ethyl	H	Isopropyl	Isobutyl	H
1041.	Isobutyl	Isopropyl	H	H	H	H
1042.	Isobutyl	Isopropyl	Methyl	H	H	H
1043.	Isobutyl	Isopropyl	H	Methyl	H	H
1044.	Isobutyl	Isopropyl	H	H	Methyl	H
1045.	Isobutyl	Isopropyl	H	H	H	Methyl
1046.	Isobutyl	Isopropyl	Methyl	Methyl	H	H
1047.	Isobutyl	Isopropyl	Methyl	H	Methyl	H
1048.	Isobutyl	Isopropyl	Methyl	H	H	Methyl
1049.	Isobutyl	Isopropyl	H	Methyl	Methyl	H
1050.	Isobutyl	Isopropyl	H	Methyl	H	Methyl
1051.	Isobutyl	Isopropyl	H	H	Methyl	Methyl
1052.	Isobutyl	Isopropyl	Methyl	Methyl	Methyl	H
1053.	Isobutyl	Isopropyl	Methyl	Methyl	H	Methyl
1054.	Isobutyl	Isopropyl	Methyl	H	Methyl	Methyl
1055.	Isobutyl	Isopropyl	H	Methyl	Methyl	Methyl
1056.	Isobutyl	Isopropyl	Methyl	Methyl	Methyl	Methyl
1057.	Isobutyl	Isopropyl	Ethyl	H	H	H
1058.	Isobutyl	Isopropyl	H	Ethyl	H	H
1059.	Isobutyl	Isopropyl	H	H	Ethyl	H
1060.	Isobutyl	Isopropyl	H	H	H	Ethyl
1061.	Isobutyl	Isopropyl	Methyl	Ethyl	H	H
1062.	Isobutyl	Isopropyl	Methyl	H	Ethyl	H
1063.	Isobutyl	Isopropyl	Methyl	H	H	Ethyl
1064.	Isobutyl	Isopropyl	H	Ethyl	Methyl	H
1065.	Isobutyl	Isopropyl	H	Ethyl	Ethyl	H
1066.	Isobutyl	Isopropyl	H	Methyl	Ethyl	H
1067.	Isobutyl	Isopropyl	Isopropyl	H	H	H
1068.	Isobutyl	Isopropyl	H	Isopropyl	H	H
1069.	Isobutyl	Isopropyl	H	H	Isopropyl	H
1070.	Isobutyl	Isopropyl	H	H	H	Isopropyl
1071.	Isobutyl	Isopropyl	Methyl	Isopropyl	H	H
1072.	Isobutyl	Isopropyl	Methyl	H	Isopropyl	H
1073.	Isobutyl	Isopropyl	Methyl	H	H	Isopropyl
1074.	Isobutyl	Isopropyl	H	Isopropyl	Methyl	H
1075.	Isobutyl	Isopropyl	H	Isopropyl	Isopropyl	H
1076.	Isobutyl	Isopropyl	H	Methyl	Isopropyl	H
1077.	Isobutyl	Isopropyl	H	Isopropyl	Ethyl	H
1078.	Isobutyl	Isopropyl	H	Ethyl	Isopropyl	H
1079.	Isobutyl	Isopropyl	Isobutyl	H	H	H
1080.	Isobutyl	Isopropyl	H	Isobutyl	H	H
1081.	Isobutyl	Isopropyl	H	H	Isobutyl	H
1082.	Isobutyl	Isopropyl	H	H	H	Isobutyl
1083.	Isobutyl	Isopropyl	Methyl	Isobutyl	H	H
1084.	Isobutyl	Isopropyl	Methyl	H	Isobutyl	H
1085.	Isobutyl	Isopropyl	Methyl	H	H	Isobutyl
1086.	Isobutyl	Isopropyl	H	Isobutyl	Methyl	H
1087.	Isobutyl	Isopropyl	H	Isobutyl	Isobutyl	H
1088.	Isobutyl	Isopropyl	H	Methyl	Isobutyl	H
1089.	Isobutyl	Isopropyl	H	Isobutyl	Ethyl	H
1090.	Isobutyl	Isopropyl	H	Ethyl	Isobutyl	H
1091.	Isobutyl	Isopropyl	H	Isobutyl	Isopropyl	H
1092.	Isobutyl	Isopropyl	H	Isopropyl	Isobutyl	H
1093.	Isobutyl	Isobutyl	H	H	H	H
1094.	Isobutyl	Isobutyl	Methyl	H	H	H
1095.	Isobutyl	Isobutyl	H	Methyl	H	H
1096.	Isobutyl	Isobutyl	H	H	Methyl	H
1097.	Isobutyl	Isobutyl	H	H	H	Methyl
1098.	Isobutyl	Isobutyl	Methyl	Methyl	H	H
1099.	Isobutyl	Isobutyl	Methyl	H	Methyl	H
1100.	Isobutyl	Isobutyl	Methyl	H	H	Methyl
1101.	Isobutyl	Isobutyl	H	Methyl	Methyl	H
1102.	Isobutyl	Isobutyl	H	Methyl	H	Methyl
1103.	Isobutyl	Isobutyl	H	H	Methyl	Methyl
1104.	Isobutyl	Isobutyl	Methyl	Methyl	Methyl	H
1105.	Isobutyl	Isobutyl	Methyl	Methyl	H	Methyl
1106.	Isobutyl	Isobutyl	Methyl	H	Methyl	Methyl
1107.	Isobutyl	Isobutyl	H	Methyl	Methyl	Methyl

TABLE 1-continued

Compound #	R ₁	R ₂	R ₃	R ₄	R ₅	R ₆
1108.	Isobutyl	Isobutyl	Methyl	Methyl	Methyl	Methyl
1109.	Isobutyl	Isobutyl	Ethyl	H	H	H
1110.	Isobutyl	Isobutyl	H	Ethyl	H	H
1111.	Isobutyl	Isobutyl	H	H	Ethyl	H
1112.	Isobutyl	Isobutyl	H	H	H	Ethyl
1113.	Isobutyl	Isobutyl	Methyl	Ethyl	H	H
1114.	Isobutyl	Isobutyl	Methyl	H	Ethyl	H
1115.	Isobutyl	Isobutyl	Methyl	H	H	Ethyl
1116.	Isobutyl	Isobutyl	H	Ethyl	Methyl	H
1117.	Isobutyl	Isobutyl	H	Ethyl	Ethyl	H
1118.	Isobutyl	Isobutyl	H	Methyl	Ethyl	H
1119.	Isobutyl	Isobutyl	Isopropyl	H	H	H
1120.	Isobutyl	Isobutyl	H	Isopropyl	H	H
1121.	Isobutyl	Isobutyl	H	H	Isopropyl	H
1122.	Isobutyl	Isobutyl	H	H	H	Isopropyl
1123.	Isobutyl	Isobutyl	Methyl	Isopropyl	H	H
1124.	Isobutyl	Isobutyl	Methyl	H	Isopropyl	H
1125.	Isobutyl	Isobutyl	Methyl	H	H	Isopropyl
1126.	Isobutyl	Isobutyl	H	Isopropyl	Methyl	H
1127.	Isobutyl	Isobutyl	H	Isopropyl	Isopropyl	H
1128.	Isobutyl	Isobutyl	H	Methyl	Isopropyl	H
1129.	Isobutyl	Isobutyl	H	Isopropyl	Ethyl	H
1130.	Isobutyl	Isobutyl	H	Ethyl	Isopropyl	H
1131.	Isobutyl	Isobutyl	Isobutyl	H	H	H
1132.	Isobutyl	Isobutyl	H	Isobutyl	H	H
1133.	Isobutyl	Isobutyl	H	H	Isobutyl	H
1134.	Isobutyl	Isobutyl	H	H	H	Isobutyl
1135.	Isobutyl	Isobutyl	Methyl	Isobutyl	H	H
1136.	Isobutyl	Isobutyl	Methyl	H	Isobutyl	H
1137.	Isobutyl	Isobutyl	Methyl	H	H	Isobutyl
1138.	Isobutyl	Isobutyl	H	Isobutyl	Methyl	H
1139.	Isobutyl	Isobutyl	H	Isobutyl	Isobutyl	H
1140.	Isobutyl	Isobutyl	H	Methyl	Isobutyl	H
1141.	Isobutyl	Isobutyl	H	Isobutyl	Ethyl	H
1142.	Isobutyl	Isobutyl	H	Ethyl	Isobutyl	H
1143.	Isobutyl	Isobutyl	H	Isobutyl	Isopropyl	H
1144.	Isobutyl	Isobutyl	H	Isopropyl	Isobutyl	H
1145.	CD ₃ CH ₂	H	H	H	H	H
1146.	(CD ₃) ₂ CH	H	H	H	H	H

In one embodiment, a first device is provided. The first device comprises a first organic light emitting device, further comprising an anode, a cathode, and an organic layer, disposed between the anode and the cathode, comprising a compound having the formula:



R₁ and R₂ are optionally linked and the sum of the number of carbon atoms in R₁ and R₂ is at least 2. R₃, R₄, R₅, R₆ are optionally linked, and R_a and R_b represent mono-, di-, tri- or tetra-substitution. X is selected from the group consisting of

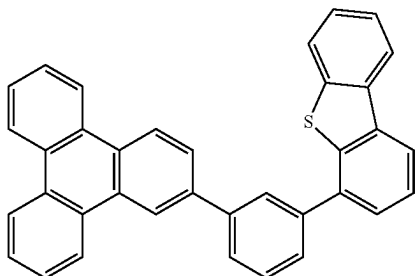
BR, NR, PR, O, S, Se, C=O, S=O, 502, CRR', SiRR', and GeRR', and R_a, R_b, R, R', R₁, R₂, R₃, R₄, R₅, and R₆ are independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, aryl-alkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof, and wherein n is 1 or 2.

In one embodiment, the first device is a consumer product. In another embodiment, the first device is an organic light-emitting device. In another embodiment, the first device comprises a lighting panel. In one embodiment, the organic layer is an emissive layer and the compound is an emissive dopant. In another embodiment, the organic layer is an emissive layer and the compound is a non-emissive dopant.

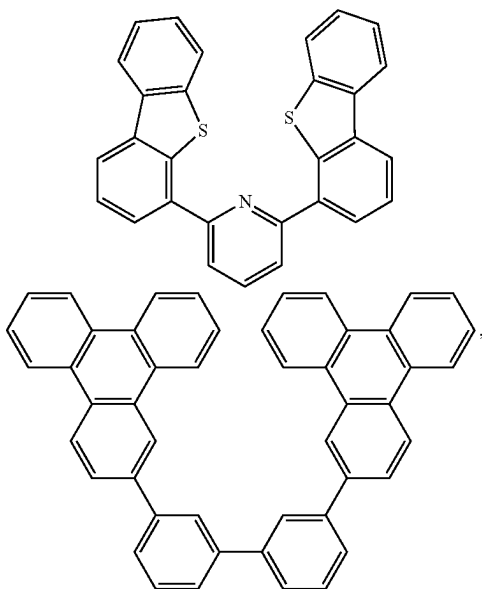
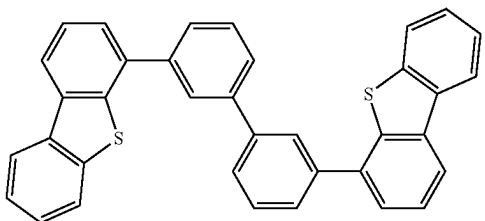
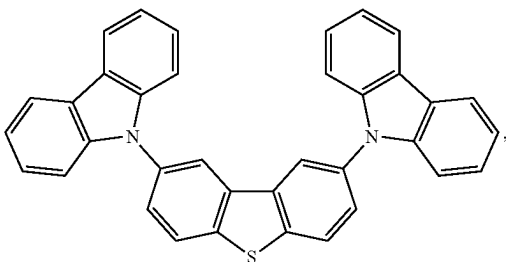
In one embodiment, the organic layer further comprises a host. In another embodiment, the host comprises a triphenylene containing benzo-fused thiophene or benzo-fused furan, wherein any substituent in the host is an unfused substituent independently selected from the group consisting of C_nH_{2n+1}, OC_nH_{2n+1}, Oar₁, N(C_nH_{2n+1})₂, N(Ar₁)(Ar₂), CH=CH—C_nH_{2n+1}, C≡CHC_nH_{2n+1}, Ar₁, Ar₁—Ar₂, C_nH_{2n}—Ar₁, or no substitution, wherein Ar₁ and Ar₂ are independently selected from the group consisting of benzene, biphenyl, naphthalene, triphenylene, carbazole, and heteroaromatic analogs thereof, and wherein n is from 1 to 10.

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In one embodiment, the host has the formula I

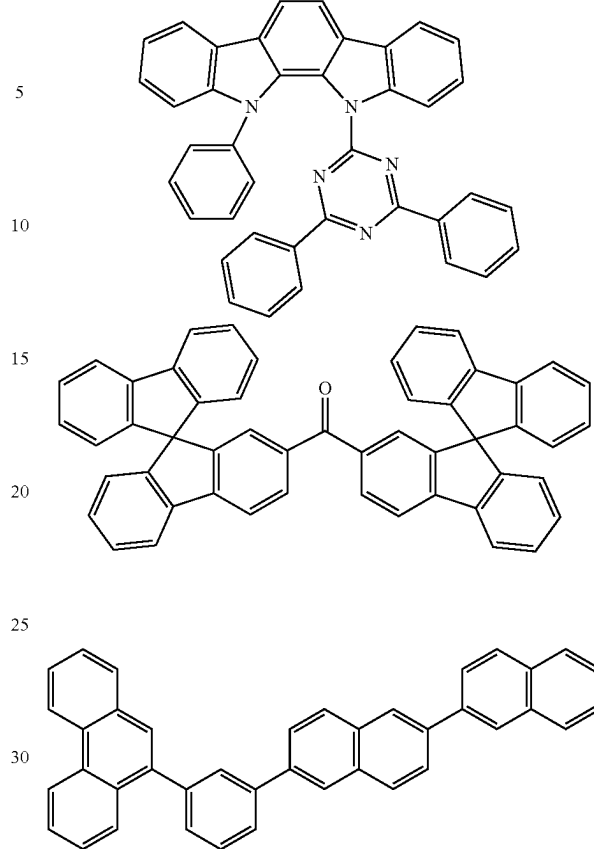


In another embodiment, the host is selected from the group consisting of:



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-continued



and combinations thereof.

In one embodiment, the host is a metal complex.

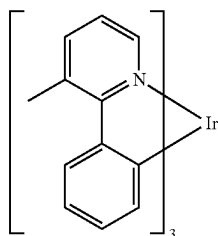
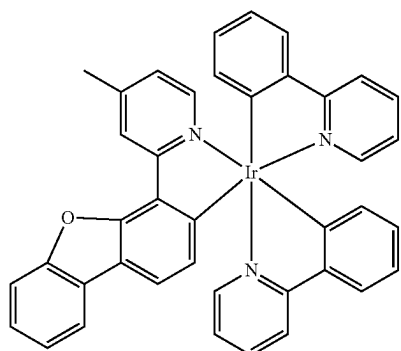
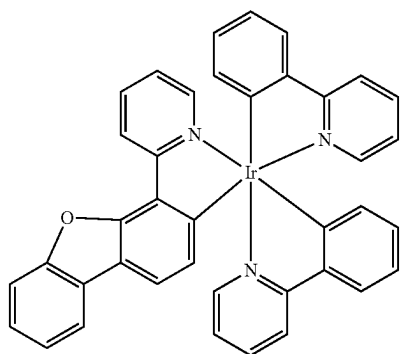
Device Examples

All example devices were fabricated by high vacuum (<10⁻⁷ Torr) thermal evaporation (VTE). The anode electrode is 1200 Å of indium tin oxide (ITO). The cathode consisted of 10 Å of LiF followed by 1,000 Å of Al. All devices are encapsulated with a glass lid sealed with an epoxy resin in a nitrogen glove box (<1 ppm of H₂O and O₂) immediately after fabrication, and a moisture getter was incorporated inside the package.

The organic stack of the device examples consisted of sequentially, from the ITO surface, 100 Å of Compound C as the hole injection layer (HIL), 300 Å of 4,4'-bis[N-(1-naphthyl)-N-phenylamino]biphenyl (alpha-NPD) as the hole transporting layer (HTL), 300 Å of the invention compound doped in Compound D as host with 5-15 weight percent of a compound of Formula I as the emissive layer (EML), 50 Å of Compound D as blocking layer (BL), and 400 Å of Alq (tris-8-hydroxyquinoline aluminum) as the ETL. Comparative Examples with Compound A and Compound B were fabricated similarly to the Device Examples except that Compound A and Compound B were used as the emitter in the EML.

The device results and data are summarized in Tables 1 from those devices. As used herein, NPD, Alq, Compound A, Compound B, Compound C, and Compound D have the following structures:

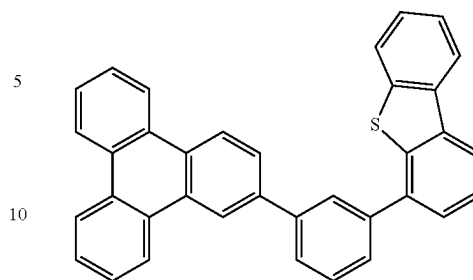
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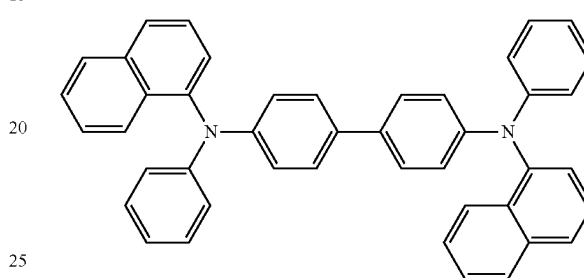
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Compound A



Compound D

Compound B



NPD

Compound C

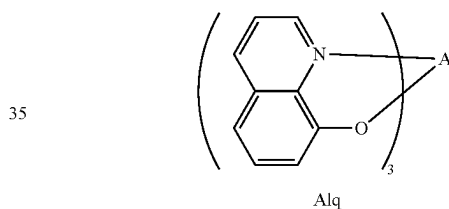


TABLE 2

VTE Phosphorescent OLEDs						
Example	HIL	HTL	EML (300 Å, doping %)		BL	ETL 1
Comparative Example 1	Compound C	NPD 300 Å	Compound D	Compound A 10%	Compound D	Alq 400 Å
Comparative Example 2	Compound C	NPD 300 Å	Compound D	Compound B 10%	Compound D	Alq 400 Å
Example 1	Compound C	NPD 300 Å	Compound D	Compound 53 12%	Compound D	Alq 400 Å
Example 2	Compound C	NPD 300 Å	Compound D	Compound 157 10%	Compound D	Alq 400 Å
Example 3	Compound C	NPD 300 Å	Compound D	Compound 158 12%	Compound D	Alq 400 Å
Example 4	Compound C	NPD 300 Å	Compound D	Compound 159 10%	Compound D	Alq 400 Å
Example 5	Compound C	NPD 300 Å	Compound D	Compound 165 12%	Compound D	Alq 400 Å
Example 6	Compound C	NPD 300 Å	Compound D	Compound 174 8%	Compound D	Alq 400 Å
Example 7	Compound C	NPD 300 Å	Compound D	Compound 175 8%	Compound D	Alq 400 Å
Example 8	Compound C	NPD 300 Å	Compound D	Compound 184 7%	Compound D	Alq 400 Å
Example 9	Compound C	NPD 300 Å	Compound D	Compound 185 10%	Compound D	Alq 400 Å
Example 10	Compound C	NPD 300 Å	Compound D	Compound 314 12%	Compound D	Alq 400 Å

TABLE 2-continued

VTE Phosphorescent OLEDs						
Example	HIL	HTL	EML (300 Å, doping %)		BL	ETL 1
Example 11	Compound C 100 Å	NPD 300 Å	Compound D	Compound 321 12%	Compound D 50 Å	Alq 400 Å
Example 12	Compound C 100 Å	NPD 300 Å	Compound D	Compound 625 12%	Compound D 50 Å	Alq 400 Å
Example 13	Compound C 100 Å	NPD 300 Å	Compound D	Compound 626 10%	Compound D 50 Å	Alq 400 Å
Example 14	Compound C 100 Å	NPD 300 Å	Compound D	Compound 627 8%	Compound D 50 Å	Alq 400 Å
Example 15	Compound C 100 Å	NPD 300 Å	Compound D	Compound 628 8%	Compound D 50 Å	Alq 400 Å
Example 16	Compound C 100 Å	NPD 300 Å	Compound D	Compound 633 10%	Compound D 50 Å	Alq 400 Å
Example 17	Compound C 100 Å	NPD 300 Å	Compound D	Compound 643 8%	Compound D 50 Å	Alq 400 Å
Example 18	Compound C 100 Å	NPD 300 Å	Compound D	Compound 652 8%	Compound D 50 Å	Alq 400 Å
Example 19	Compound C 100 Å	NPD 300 Å	Compound D	Compound 653 8%	Compound D 50 Å	Alq 400 Å
Example 20	Compound C 100 Å	NPD 300 Å	Compound D	Compound 1145 10%	Compound D 50 Å	Alq 400 Å
Example 21	Compound C 100 Å	NPD 300 Å	Compound D	Compound 1146 10%	Compound D 50 Å	Alq 400 Å

TABLE 3

VTE Device Data									
	x	y	λ_{max} (nm)	FWHM (nm)	Voltage (V)	LE (Cd/A)	EQE (%)	PE (lm/W)	LT80% (h)
Comparative Example 1 Compound A	0.367	0.607	532	66	6.5	43.2	11.5	21	250
Comparative Example 2 Compound B	0.336	0.623	524	66	5.2	58.2	15.7	35	170
Example 1 Compound 53	0.345	0.619	526	64	5.1	69.2	18.7	42.7	20
Example 2 Compound 157	0.336	0.618	518	70	5.8	62.2	17.3	33.6	166
Example 3 Compound 158	0.318	0.632	518	60	4.7	73.4	20.2	49.1	55
Example 4 Compound 159	0.339	0.616	520	66	5.4	58.3	16.2	33.9	82
Example 5 Compound 165	0.333	0.626	522	62	4.6	56.1	15.2	38.3	54
Example 6 Compound 174	0.327	0.627	520	64	4.9	61.5	16.9	39.3	62
Example 7 Compound 175	0.314	0.635	518	60	5.3	71.2	19.5	42.2	77
Example 8 Compound 184	0.331	0.623	518	64	5.7	52.8	14.6	28.8	66
Example 9 Compound 185	0.331	0.623	518	64	5.4	56.7	15.7	32.9	53
Example 10 Compound 314	0.308	0.636	516	60	5.0	61.4	16.9	38.9	44
Example 11 Compound 321	0.326	0.630	520	60	5.1	61.4	16.6	37.7	73
Example 12 Compound 625	0.338	0.622	524	66	5.1	60.1	16.3	37.2	124
Example 13 Compound 626	0.336	0.624	522	64	5.5	67.4	18.2	38.2	71
Example 14 Compound 627	0.345	0.619	526	64	5.2	65.2	17.7	39.2	121
Example 15 Compound 628	0.351	0.615	526	64	5.6	68.6	18.6	38.7	124
Example 16 Compound 633	0.366	0.608	528	66	5.0	66.3	17.8	41.8	56
Example 17 Compound 643	0.339	0.625	526	60	5.2	71.5	19.1	43.3	108
Example 18 Compound 652	0.349	0.616	526	66	5.7	53.4	14.5	29.3	104

TABLE 3-continued

VTE Device Data									
	x	y	λ_{max} (nm)	FWHM (nm)	Voltage (V)	LE (Cd/A)	EQE (%)	PE (lm/W)	LT80% (h)
Example 19 Compound 653	0.337	0.626	524	62	5.1	65.6	17.6	40.8	84
Example 20 Compound 1145	0.343	0.618	524	68	5.6	61.3	16.7	34.4	59
Example 21 Compound 1146	0.343	0.618	524	68	5.6	63	17.2	35.5	157

Table 3 is a summary of the device data. The luminous efficiency (LE), external quantum efficiency (EQE) and power efficiency (PE) were measured at 1000 nits, while the lifetime (LT_{80%}) was defined as the time required for the device to decay to 80% of its initial luminance under a constant current density of 40 mA/cm².

The advantage of alkyl substitution on the 4- and 5-positions of the DBX-pyridine ring is clear from Table 3. Compared to Comparative Example 1, with no substitution at the 4- or 5-positions of the DBX-pyridine ring, the compounds of Formula I are more saturated (lower CIE x coordinate and shorter λ_{max}) and the broadness as measured by the FWHM is comparable. In all the inventive compounds, the voltage is lower and the LE, PE and EQE values are all higher. In the case of Compounds 53, 158, 175, 633 and 643, the PE is at least twice as high as Comparative Example 1.

Compared to Comparative Example 2 (Compound B), with only one-carbon substitution (methyl) at the 4-position of the DBX-pyridine ring, Compounds 53, 158, 174, 175, 184, 185, and 314 have more saturated color based CIE x coordinate and Compounds 157, 158, 159, 165, 174, 175, 184, 185, 314, 321 and 626 all have shorter λ_{max} values. Most of the compounds of Formula I have narrower emission profiles (as measured by FWHM) than Comparative Example 2. Compounds 53, 158, 165, 314, 321, 625, 633 and 653 all have lower driving voltages than Comparative Example 2. Most of the compounds of Formula I have greater LE, PE and EQE values than Comparative Example 2.

Combination with Other Materials

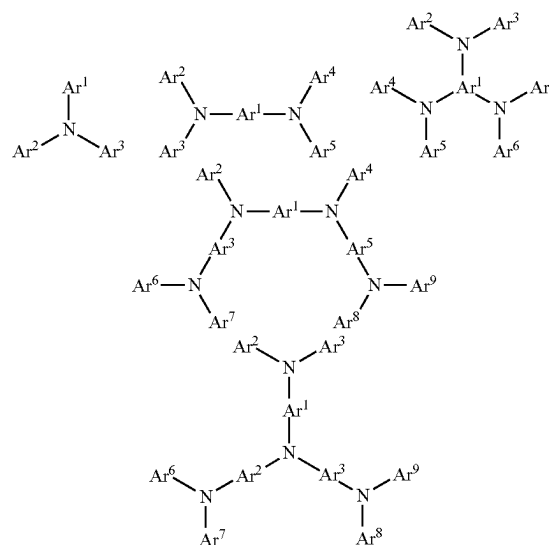
The materials described herein as useful for a particular layer in an organic light emitting device may be used in combination with a wide variety of other materials present in the device. For example, emissive dopants disclosed herein may be used in conjunction with a wide variety of hosts, transport layers, blocking layers, injection layers, electrodes and other layers that may be present. The materials described or referred to below are non-limiting examples of materials that may be useful in combination with the compounds disclosed herein, and one of skill in the art can readily consult the literature to identify other materials that may be useful in combination.

HIL/HTL:

A hole injecting/transporting material to be used in the present invention is not particularly limited, and any compound may be used as long as the compound is typically used as a hole injecting/transporting material. Examples of the material include, but not limit to: a phthalocyanine or porphyrin derivative; an aromatic amine derivative; an indolocarbazole derivative; a polymer containing fluoro-hydrocarbon; a polymer with conductivity dopants; a conducting polymer, such as PEDOT/PSS; a self-assembly monomer derived from compounds such as phosphonic acid and silane

derivatives; a metal oxide derivative, such as MoO_x; a p-type semiconducting organic compound, such as 1,4,5,8,9,12-Hexaazatriphenylenehexacarbonitrile; a metal complex, and a cross-linkable compounds.

Examples of aromatic amine derivatives used in HIL or HTL include, but not limit to the following general structures:

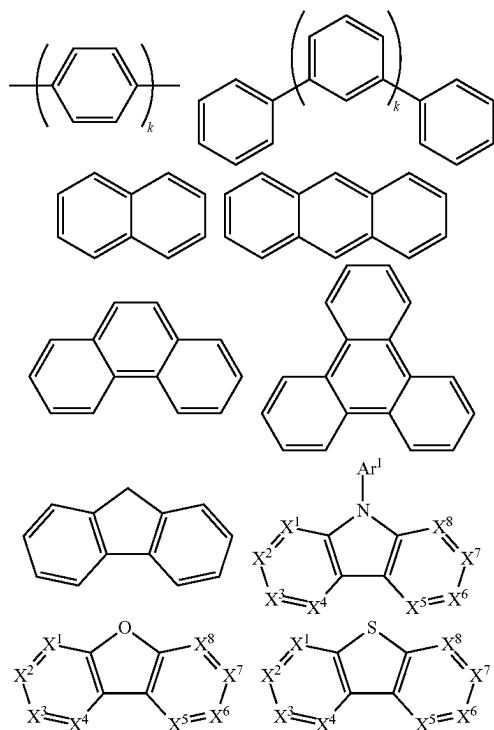


Each of Ar¹ to Ar⁹ is selected from the group consisting aromatic hydrocarbon cyclic compounds such as benzene, biphenyl, triphenyl, triphenylene, naphthalene, anthracene, phenalene, phenanthrene, fluorene, pyrene, chrysene, perylene, azulene; group consisting aromatic heterocyclic compounds such as dibenzothiophene, dibenzofuran, dibenzoselenophene, furan, thiophene, benzofuran, benzothiophene, benzoselenophene, carbazole, indolocarbazole, pyridylindole, pyrrolodipyridine, pyrazole, imidazole, triazole, oxazole, thiazole, oxadiazole, oxatriazole, dioxazole, thiadiazole, pyridine, pyridazine, pyrimidine, pyrazine, triazine, oxazine, oxathiazine, oxadiazine, indole, benzimidazole, indazole, indoxazine, benzoxazole, benzisoxazole, benzothiazole, quinoline, isoquinoline, cinnoline, quinoxaline, naphthyridine, phthalazine, pteridine, xanthene, acridine, phenazine, phenothiazine, phenoxazine, benzofuopyridine, furodipyridine, benzothienopyridine, thienodipyridine, benzoselenophenopyridine, and selenophenodipyridine; and group consisting 2 to 10 cyclic structural units which are groups of the same type or different types selected from the aromatic hydrocarbon cyclic group and the aromatic heterocyclic group and are bonded to each other directly or via at least one of oxygen atom, nitrogen

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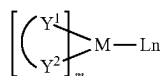
atom, sulfur atom, silicon atom, phosphorus atom, boron atom, chain structural unit and the aliphatic cyclic group. Wherein each Ar is further substituted by a substituent selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof.

In one aspect, Ar¹ to Ar⁹ is independently selected from the group consisting of:



k is an integer from 1 to 20; X¹ to X⁸ is C (including CH) or N; Ar¹ has the same group defined above.

Examples of metal complexes used in HIL or HTL include, but not limit to the following general formula:



M is a metal, having an atomic weight greater than 40; (Y¹—Y²) is a bidentate ligand, Y¹ and Y² are independently selected from C, N, O, P, and S; L is an ancillary ligand; m is an integer value from 1 to the maximum number of ligands that may be attached to the metal; and m+n is the maximum number of ligands that may be attached to the metal.

In one aspect, (Y¹—Y²) is a 2-phenylpyridine derivative.

In another aspect, (Y¹—Y²) is a carbene ligand.

In another aspect, M is selected from Ir, Pt, Os, and Zn.

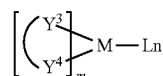
In a further aspect, the metal complex has a smallest oxidation potential in solution vs. Fc⁺/Fc couple less than about 0.6 V.

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Host:

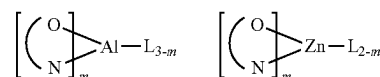
The light emitting layer of the organic EL device of the present invention preferably contains at least a metal complex as light emitting material, and may contain a host material using the metal complex as a dopant material. Examples of the host material are not particularly limited, and any metal complexes or organic compounds may be used as long as the triplet energy of the host is larger than that of the dopant.

Examples of metal complexes used as host are preferred to have the following general formula.



M is a metal; (Y³—Y⁴) is a bidentate ligand, Y³ and Y⁴ are independently selected from C, N, O, P, and S; L is an ancillary ligand; m is an integer value from 1 to the maximum number of ligands that may be attached to the metal; and m+n is the maximum number of ligands that may be attached to the metal.

In one aspect, the metal complexes are:



(O—N) is a bidentate ligand, having metal coordinated to atoms O and N.

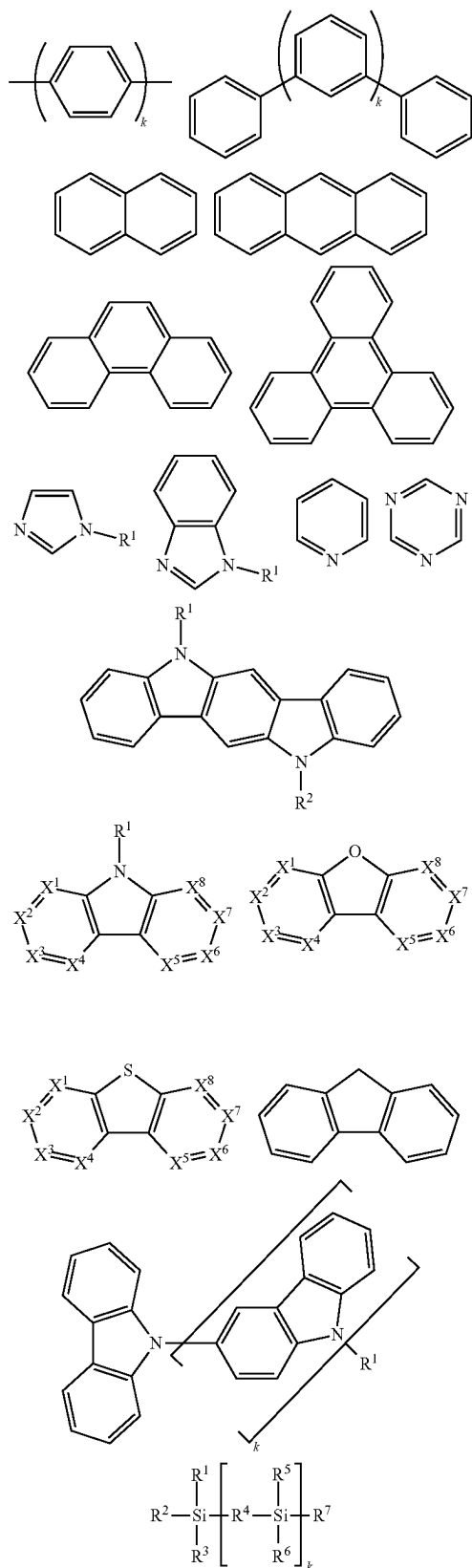
In another aspect, M is selected from Ir and Pt.

In a further aspect, (Y³—Y⁴) is a carbene ligand.

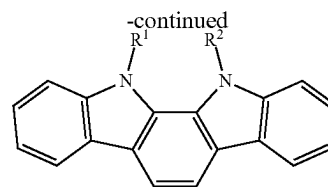
Examples of organic compounds used as host are selected from the group consisting aromatic hydrocarbon cyclic compounds such as benzene, biphenyl, triphenyl, triphenylene, naphthalene, anthracene, phenalene, phenanthrene, fluorene, pyrene, chrysene, perylene, azulene; group consisting aromatic heterocyclic compounds such as dibenzothiophene, dibenzofuran, dibenzoselenophene, furan, thiophene, benzofuran, benzothiophene, benzoselenophene, carbazole, indolocarbazole, pyridylindole, pyrrolodipyrindine, pyrazole, imidazole, triazole, oxazole, thiazole, oxadiazole, oxatriazole, dioxazole, thiadiazole, pyridine, pyridazine, pyrimidine, pyrazine, triazine, oxazine, oxathiazine, oxadiazine, indole, benzimidazole, indazole, indoxazine, benzoxazole, benzisoxazole, benzothiazole, quinoline, isoquinoline, cinnoline, quinazoline, quinoxaline, naphthyridine, phthalazine, pteridine, xanthene, acridine, phenazine, phenothiazine, phenoxazine, benzofuropyrindine, furodipyrindine, benzothienopyridine, thienodipyrindine, benzoselenophenopyridine, and selenophenodipyrindine; and group consisting 2 to 10 cyclic structural units which are groups of the same type or different types selected from the aromatic hydrocarbon cyclic group and the aromatic heterocyclic group and are bonded to each other directly or via at least one of oxygen atom, nitrogen atom, sulfur atom, silicon atom, phosphorus atom, boron atom, chain structural unit and the aliphatic cyclic group. Wherein each group is further substituted by a substituent selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof.

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In one aspect, host compound contains at least one of the following groups in the molecule:



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R¹ to R⁷ is independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof, when it is aryl or heteroaryl, it has the similar definition as Ar's mentioned above.

k is an integer from 0 to 20.

X¹ to X⁸ is selected from C (including CH) or N.

HBL:

A hole blocking layer (HBL) may be used to reduce the number of holes and/or excitons that leave the emissive layer. The presence of such a blocking layer in a device may result in substantially higher efficiencies as compared to a similar device lacking a blocking layer. Also, a blocking layer may be used to confine emission to a desired region of an OLED.

In one aspect, compound used in HBL contains the same molecule used as host described above.

In another aspect, compound used in HBL contains at least one of the following groups in the molecule:

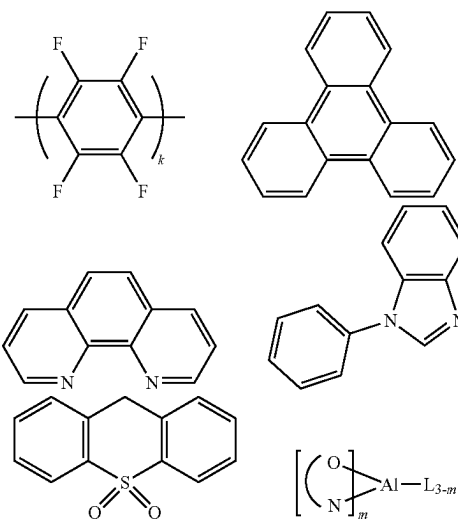
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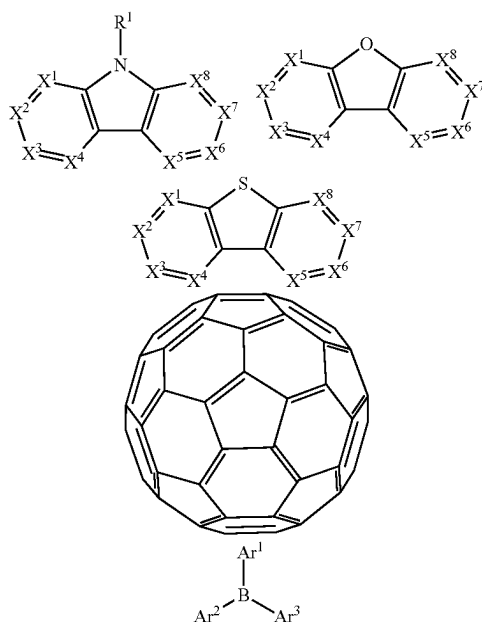
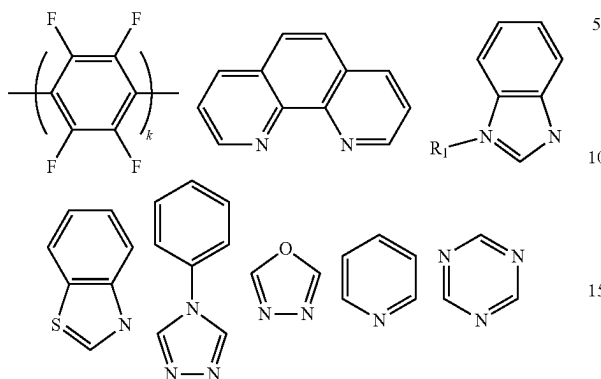
k is an integer from 0 to 20; L is an ancillary ligand, m is an integer from 1 to 3.

ETL:

Electron transport layer (ETL) may include a material capable of transporting electrons. Electron transport layer may be intrinsic (undoped), or doped. Doping may be used to enhance conductivity. Examples of the ETL material are not particularly limited, and any metal complexes or organic compounds may be used as long as they are typically used to transport electrons.

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In one aspect, compound used in ETL contains at least one of the following groups in the molecule:



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R^1 is selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, aryl-alkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof, it has the similar definition as Ar 's mentioned above.

Ar^1 to Ar^3 has the similar definition as Ar 's mentioned above.

k is an integer from 0 to 20.

X^1 to X^8 is selected from C (including CH) or N.

In another aspect, the metal complexes used in ETL contains, but not limit to the following general formula:



(O—N) or (N—N) is a bidentate ligand, having metal coordinated to atoms O, N or N, N; L is an ancillary ligand; m is an integer value from 1 to the maximum number of ligands that may be attached to the metal.

In any above-mentioned compounds used in each layer of the OLED device, the hydrogen atoms can be partially or fully deuterated.

In addition to and/or in combination with the materials disclosed herein, many hole injection materials, hole transporting materials, host materials, dopant materials, exciton/hole blocking layer materials, electron transporting and electron injecting materials may be used in an OLED. Non-limiting examples of the materials that may be used in an OLED in combination with materials disclosed herein are listed in Table 3 below. Table 3 lists non-limiting classes of materials, non-limiting examples of compounds for each class, and references that disclose the materials.

TABLE 2

MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Hole injection materials		
Phthalocyanine and porphyrin compounds		Appl. Phys. Lett. 69, 2160 (1996)

TABLE 2-continued

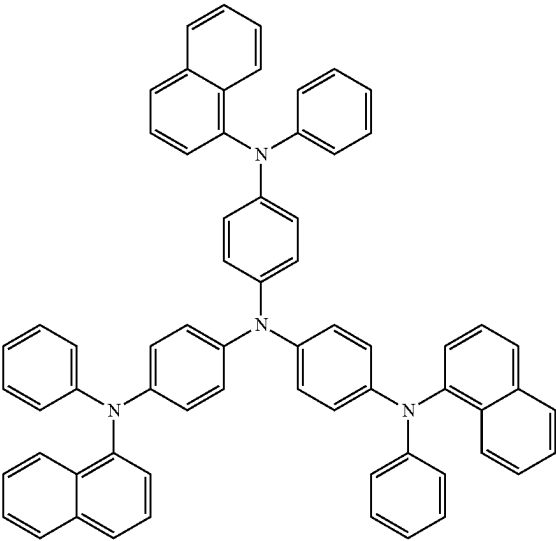
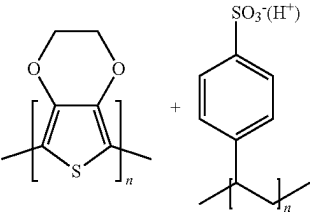
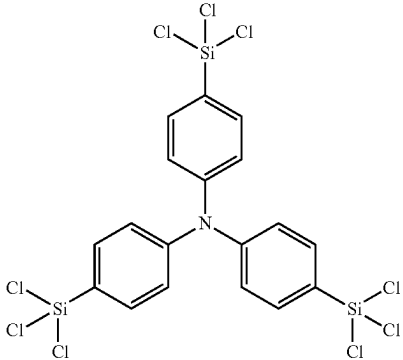
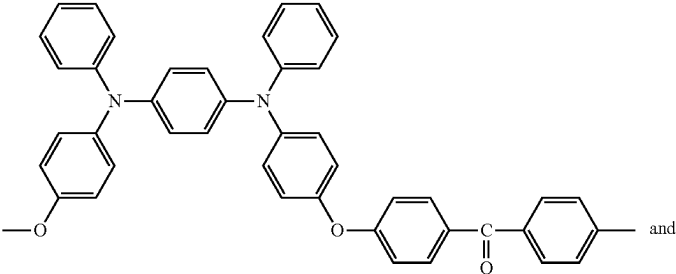
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Starburst triarylamines		J. Lumin. 72-74, 985 (1997)
CF _x Fluorohydrocarbon polymer	$\text{---}[\text{CH}_x\text{F}_y]_n\text{---}$	Appl. Phys. Lett. 78, 673 (2001)
Conducting polymers (e.g., PEDOT:PSS, polyaniline, polythiophene)		Synth. Met. 87, 171 (1997) WO2007002683
Phosphonic acid and silane SAMs		US20030162053
Triaryamine or polythiophene polymers with conductivity dopants		EA01725079A1

TABLE 2-continued

MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Arylamines complexed with metal oxides such as molybdenum and tungsten oxides		SID Symposium Digest, 37, 923 (2006) WO2009018009
p-type semiconducting organic complexes		US20020158242

TABLE 2-continued

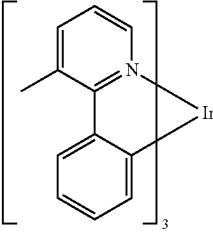
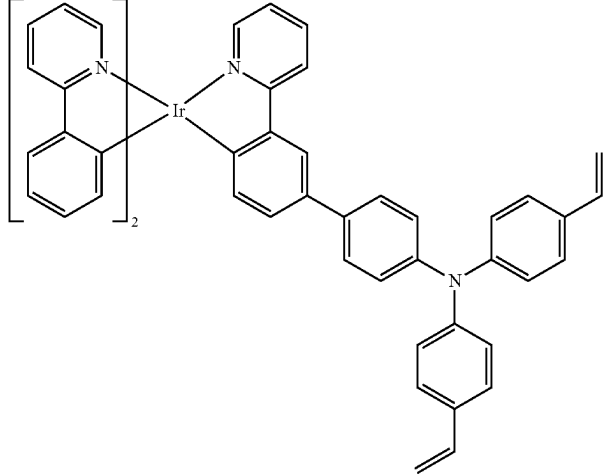
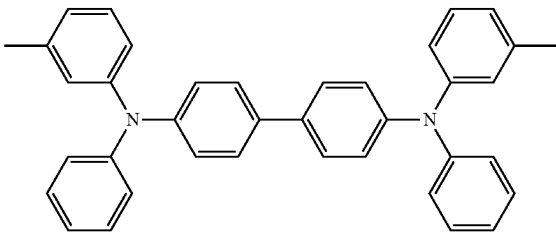
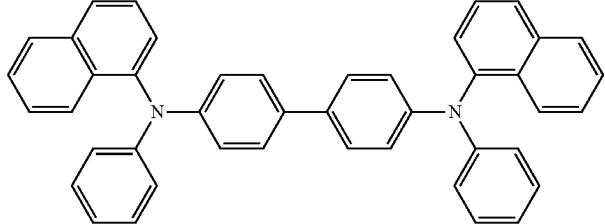
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Metal organometallic complexes		US20060240279
Cross-linkable compounds		US20080220265
Hole transporting materials		
Triarylamines (e.g., TPD, α -NPD)		Appl. Phys. Lett. 51, 913 (1987)
		U.S. Pat. No. 5,061,569

TABLE 2-continued

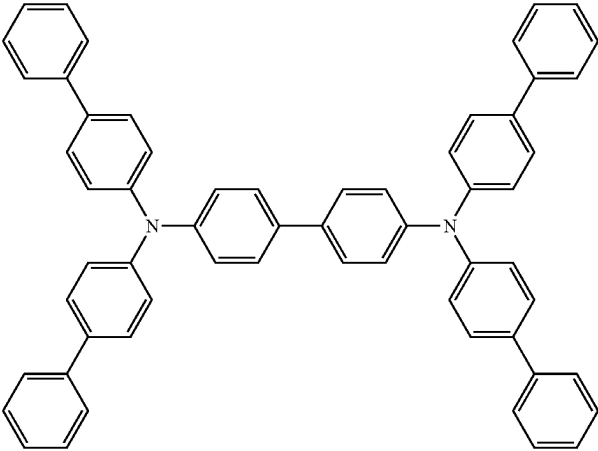
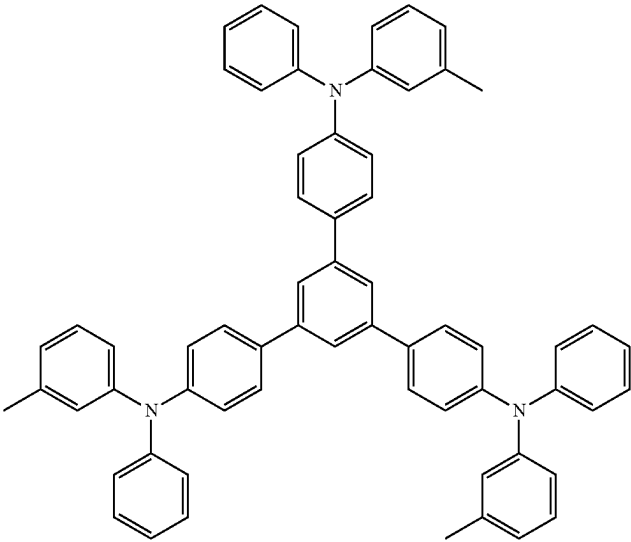
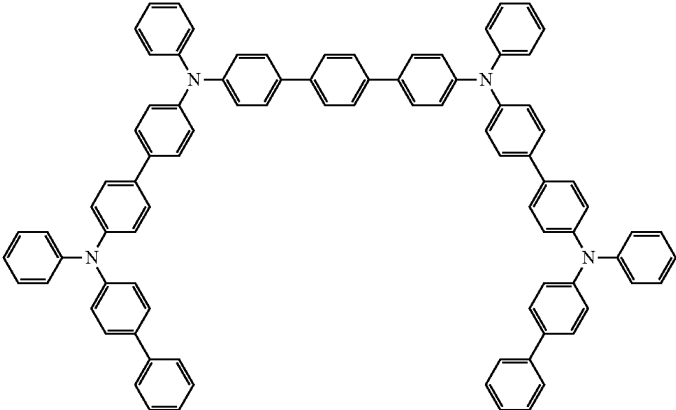
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		EP650955
		J. Mater. Chem. 3, 319 (1993)
		Appl. Phys. Lett. 90, 183503 (2007)

TABLE 2-continued

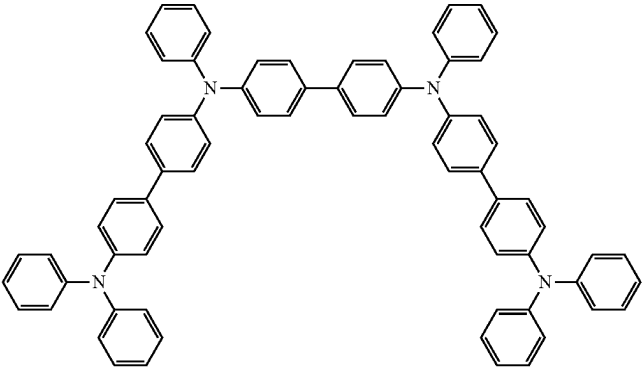
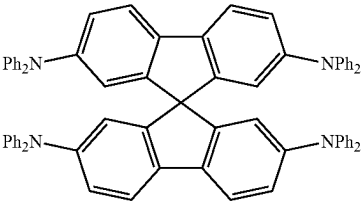
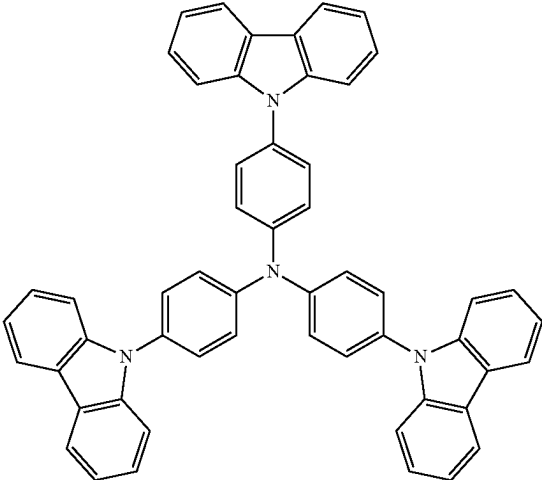
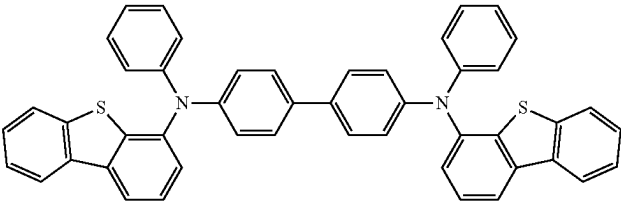
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		Appl. Phys. Lett. 90, 183503 (2007)
Triarylamine on spirofluorene core		Synth. Met. 91, 209 (1997)
Arylamine carbazole compounds		Adv. Mater. 6, 677 (1994), US20080124572
Triarylamine with (di)benzothiophene/ (di)benzofuran		US20070278938, US20080106190

TABLE 2-continued

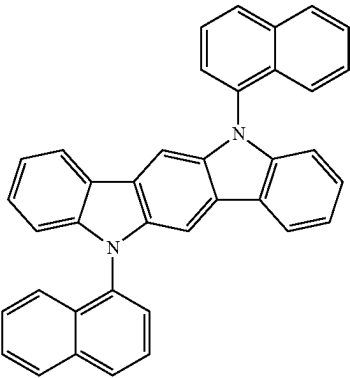
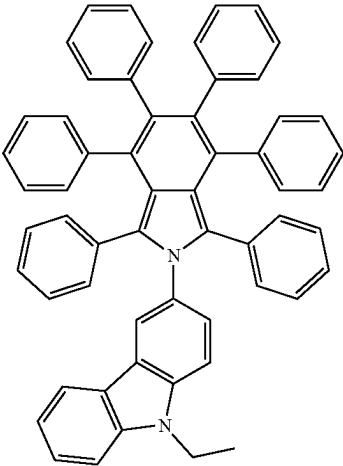
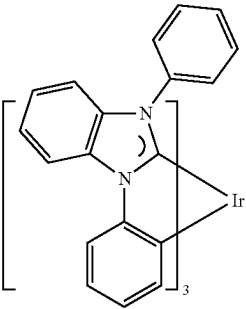
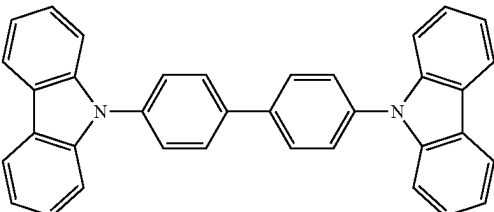
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Indolocarbazoles		Synth. Met. 111, 421 (2000)
Isoindole compounds		Chem. Mater. 15, 3148 (2003)
Metal carbene complexes		US20080018221
Phosphorescent OLED host materials		
Red hosts		
Arylcarbazoles		Appl. Phys. Lett. 78, 1622 (2001)

TABLE 2-continued

MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Metal 8-hydroxyquinolates (e.g., Alq ₃ , BAlq)		Nature 395, 151 (1998)
		US20060202194
		WO2005014551
		WO2006072002
Metal phenoxybenzothiazole compounds		Appl. Phys. Lett. 90, 123509 (2007)
Conjugated oligomers and polymers (e.g., polyfluorene)		Org. Electron. 1, 15 (2000)
Aromatic fused rings		WO2009066779, WO2009066778, WO2009063833, US20090045731, US20090045730, WO2009008311, US2009008605, US2009009065

TABLE 2-continued

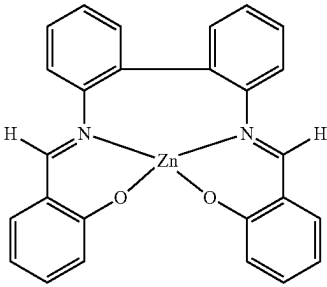
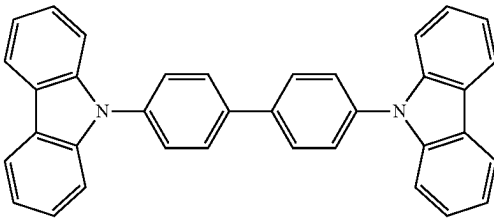
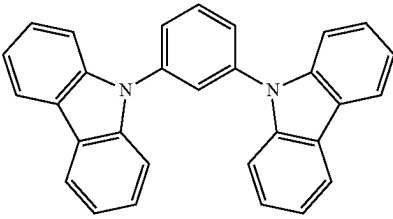
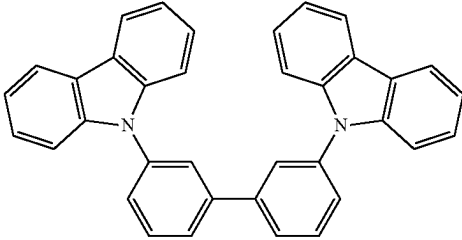
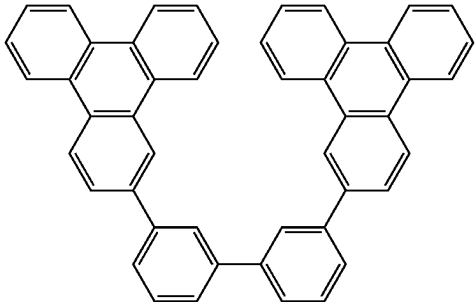
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Zinc complexes		WO2009062578
Green hosts		
Arylcarbazoles		Appl. Phys. Lett. 78, 1622 (2001)
		US20030175553
		WO2001039234
Aryltriphenylene compounds		US20060280965

TABLE 2-continued

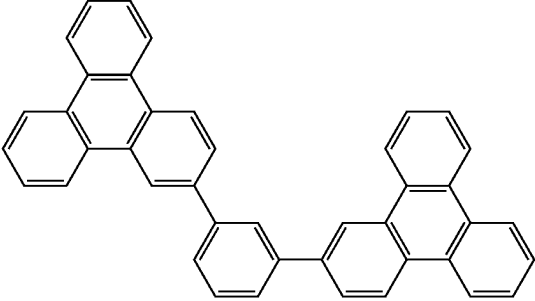
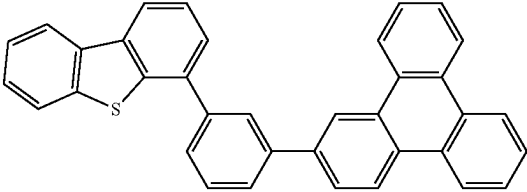
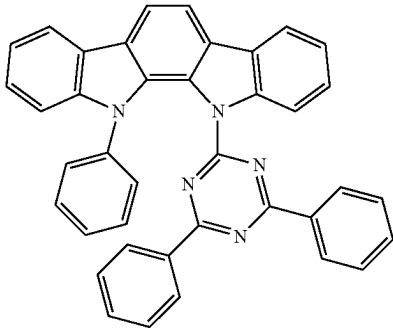
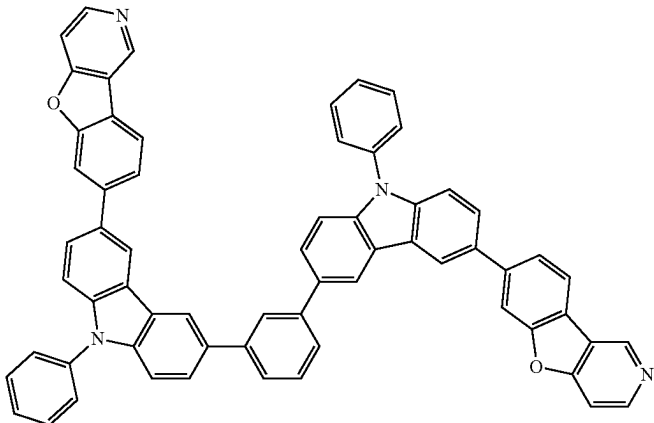
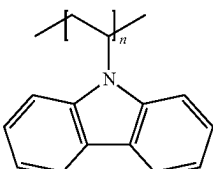
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		US20060280965
		WO2009021126
Donor acceptor type molecules		WO2008056746
Aza-carbazole/DBT/DBF		JP2008074939
Polymers (e.g., PVK)		Appl. Phys. Lett. 77, 2280 (2000)

TABLE 2-continued

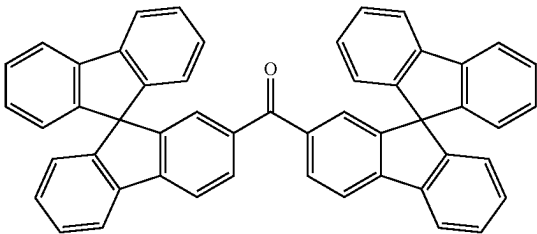
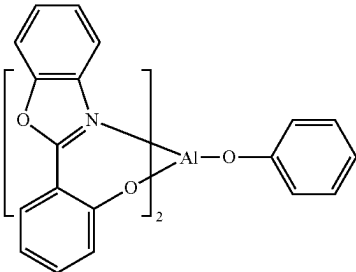
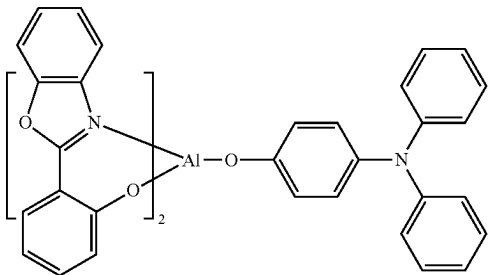
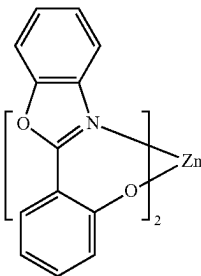
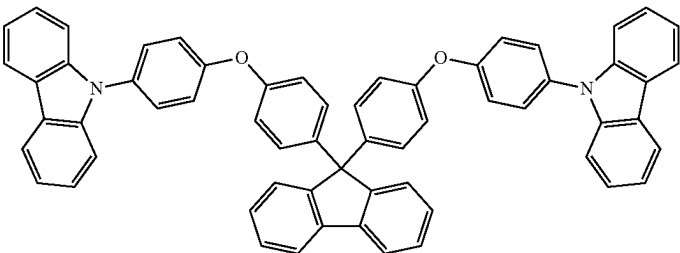
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Spirofluorene compounds		WO2004093207
Metal phenoxybenzoxazole compounds		WO2005089025
		WO2006132173
		JP200511610
Spirofluorene-carbazole compounds		JP2007254297

TABLE 2-continued

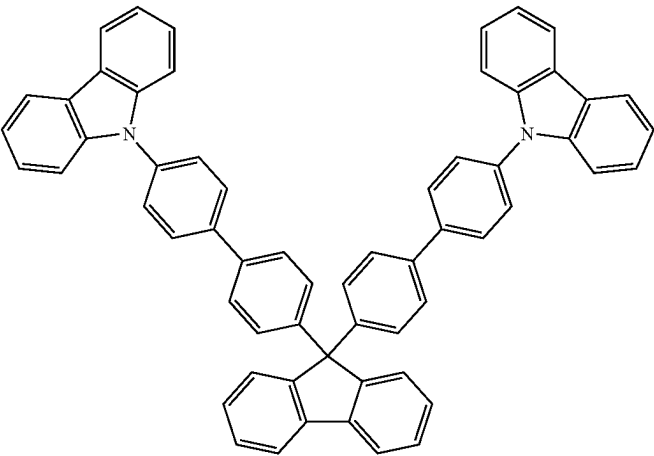
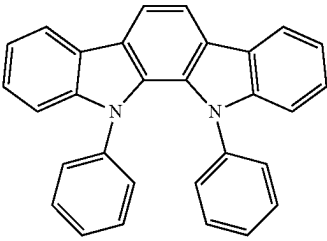
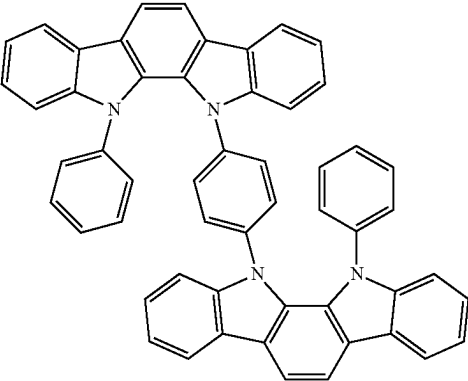
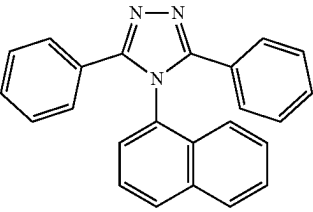
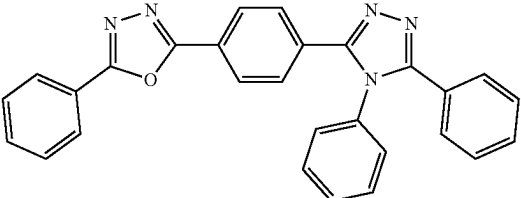
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
		JP2007254297
Indolocarbazoles		WO2007063796
		WO2007063754
5-member ring electron deficient heterocycles (e.g., triazole, oxadiazole)		J. Appl. Phys. 90, 5048 (2001)
		WO2004107822

TABLE 2-continued

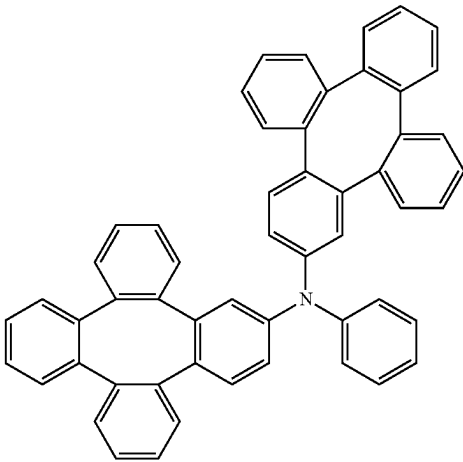
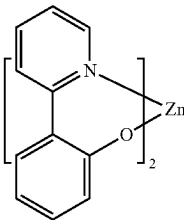
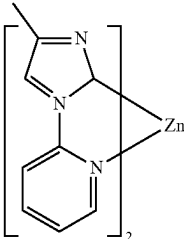
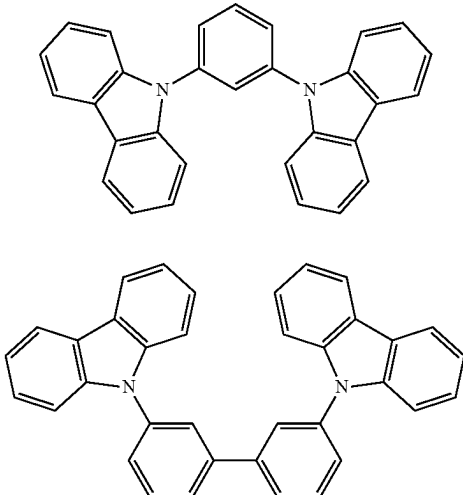
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Tetraphenylene complexes		US20050112407
Metal phenoxypyridine compounds		WO2005030900
Metal coordination complexes (e.g., Zn, Al with N'N ligands)		US20040137268, US20040137267
Blue hosts		Appl. Phys. Lett, 82, 2422 (2003)
Arylcarbazoles		US20070190359

TABLE 2-continued

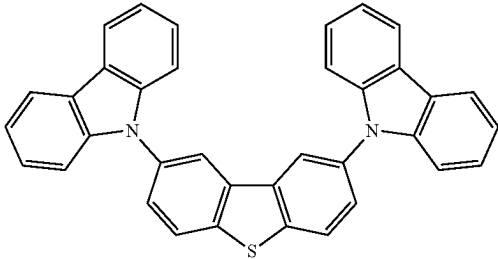
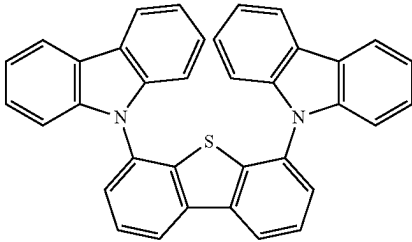
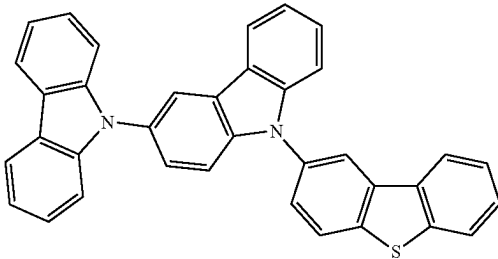
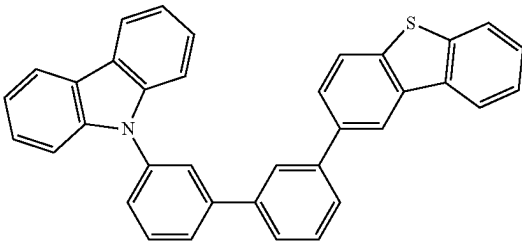
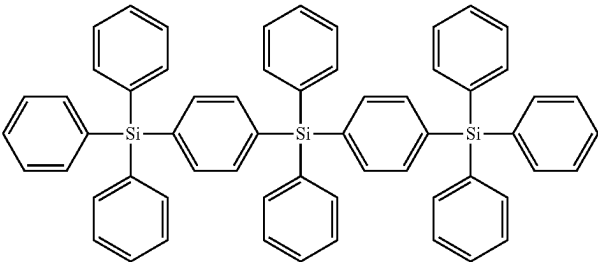
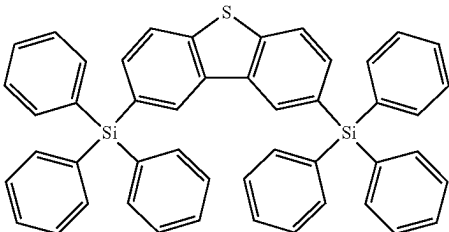
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Dibenzothiophene/ Dibenzofuran-carbazole compounds		WO2006114966, US20090167162
		US20090167162
		WO2009086028
		US20090030202, US20090017330
Silicon aryl compounds		US20050238919
		WO2009003898

TABLE 2-continued

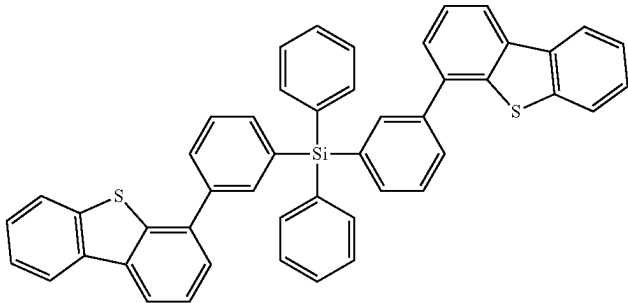
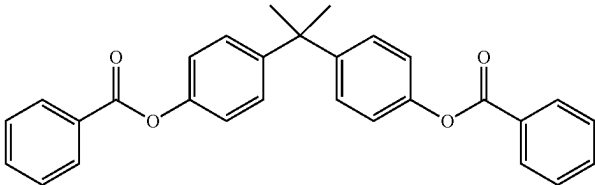
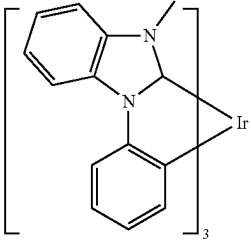
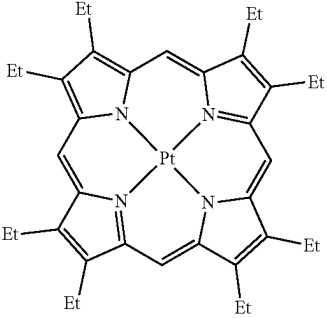
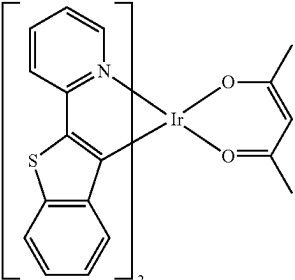
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Silicon/Germanium aryl compounds		EP2034538A
Aryl benzoyl ester		WO2006100298
High triplet metal organometallic complex		U.S. Pat. No. 7,154,114
	Phosphorescent dopants Red dopants	
Heavy metal porphyrins (e.g., PtOEP)		Nature 395, 151 (1998)
Iridium(III) organometallic complexes		Appl. Phys. Lett. 78, 1622 (2001)

TABLE 2-continued

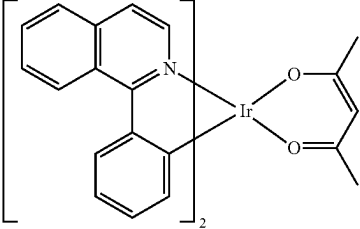
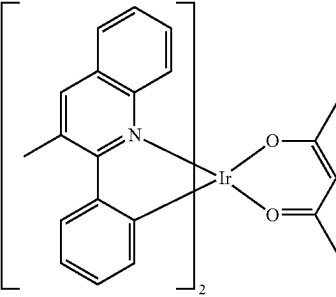
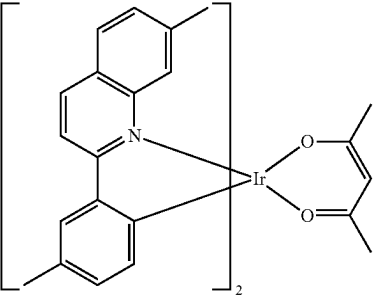
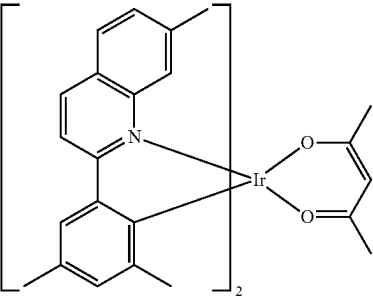
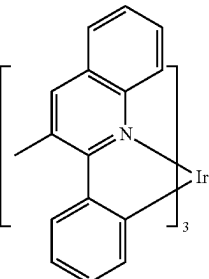
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		US20070087321

TABLE 2-continued

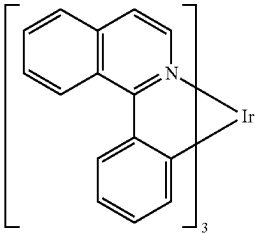
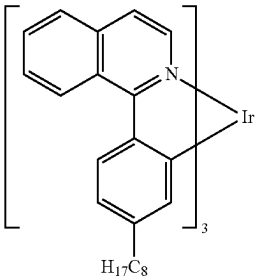
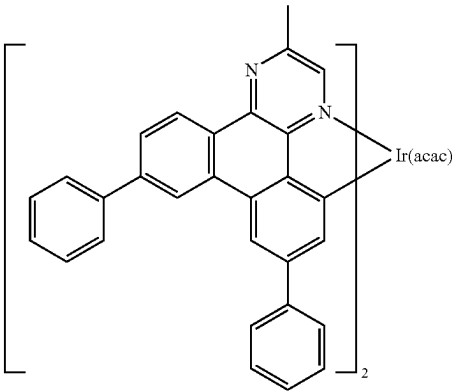
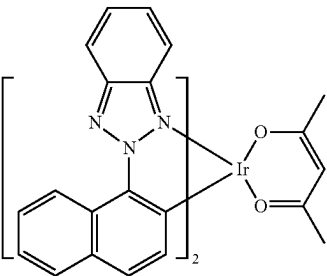
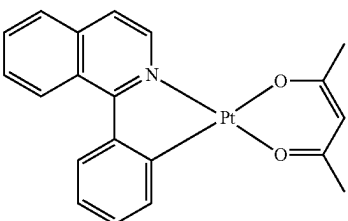
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		Adv. Mater. 19, 739 (2007)
		WO2009100991
		WO2008101842
Platinum(II) organometallic complexes		WO2003040257

TABLE 2-continued

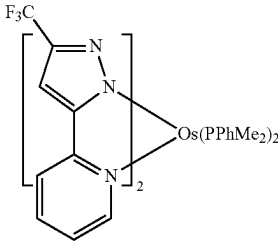
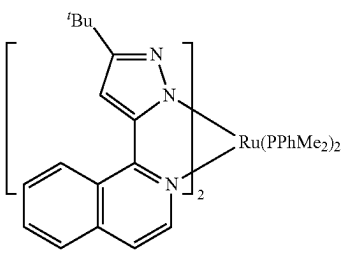
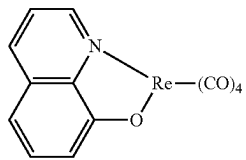
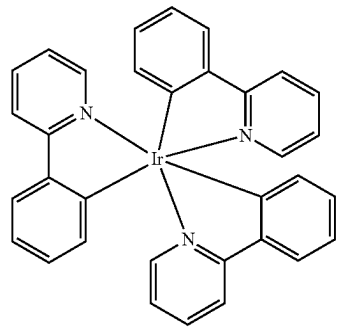
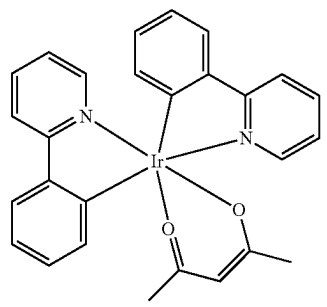
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Osmium(III) complexes		Chem. Mater. 17, 3532 (2005)
Ruthenium(II) complexes		Adv. Mater. 17, 1059 (2005)
Rhenium (I), (II), and (III) complexes		US20050244673
Green dopants		
Iridium(III) organometallic complexes		Inorg. Chem. 40, 1704 (2001)
	and its derivatives	
		US20020034656

TABLE 2-continued

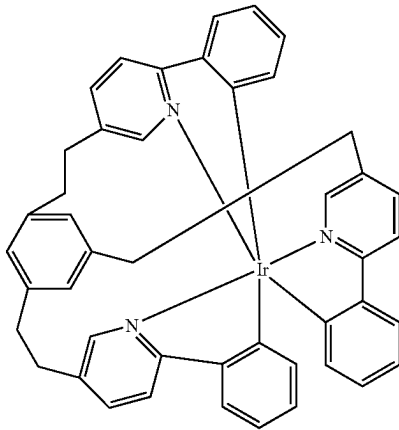
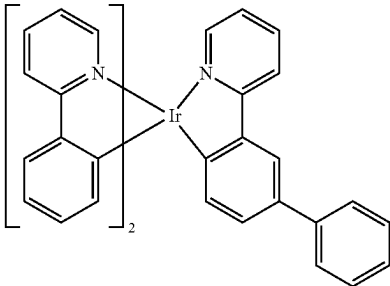
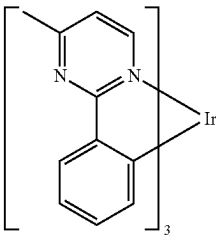
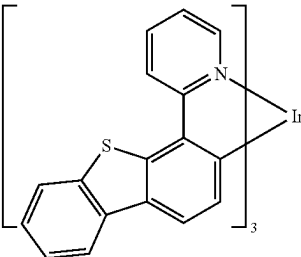
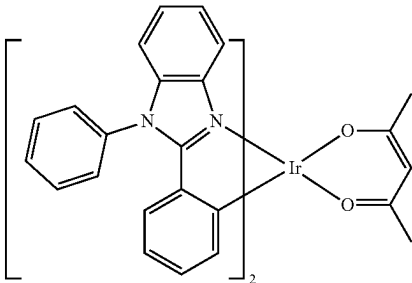
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		US20090108737
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		U.S. Pat. No. 6,921,915
		U.S. Pat. No. 6,687,266

TABLE 2-continued

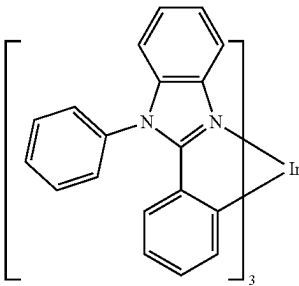
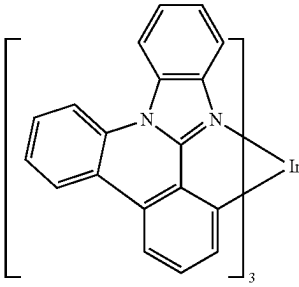
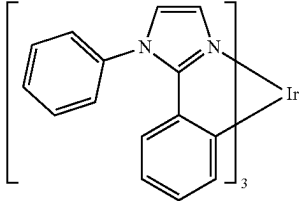
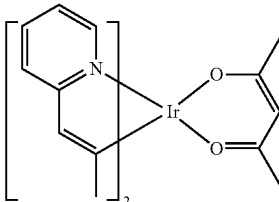
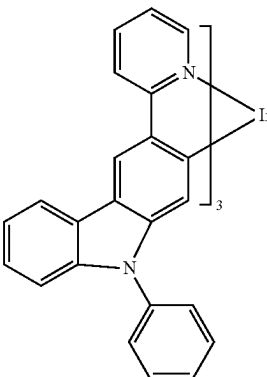
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
		Chem. Mater. 16, 2480 (2004)
		US20070190359
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		Adv. Mater. 16, 2003 (2004)
		Angew. Chem. Int. Ed. 2006, 45, 7800

TABLE 2-continued

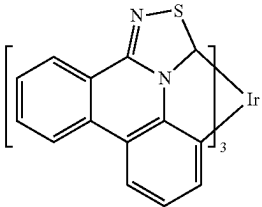
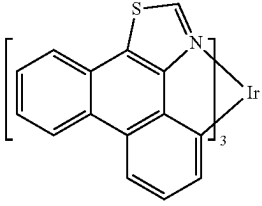
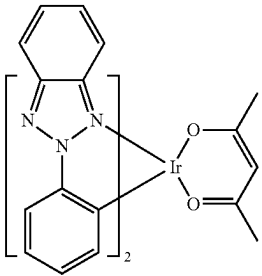
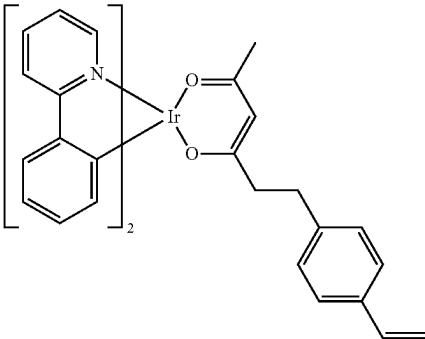
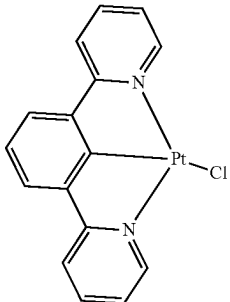
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
		WO2009050290
		US20090165846
		US20080015355
Monomer for polymeric metal organometallic compounds		U.S. Pat. No. 7,250,226, U.S. Pat. No. 7,396,598
Pt(II) organometallic complexes, including polydentate ligands		Appl. Phys. Lett. 86, 153505 (2005)

TABLE 2-continued

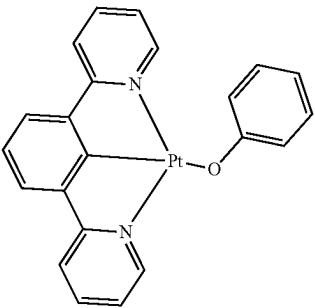
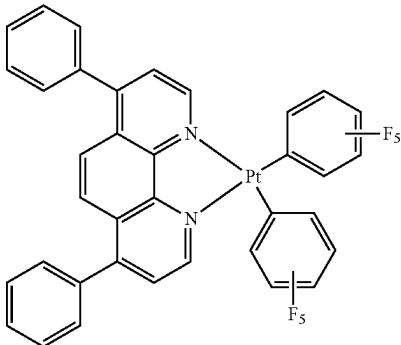
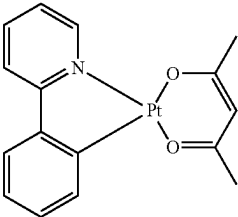
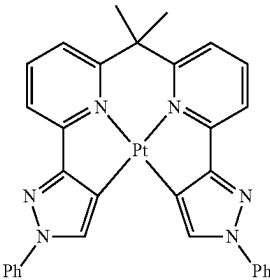
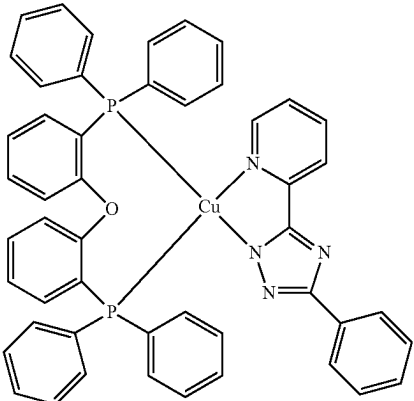
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		Appl. Phys. Lett. 86, 153505 (2005)
		Chem. Lett. 34, 592 (2005)
		WO2002015645
		US20060263635
Cu complexes		WO2009000673

TABLE 2-continued

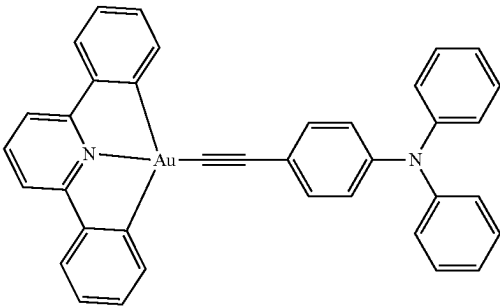
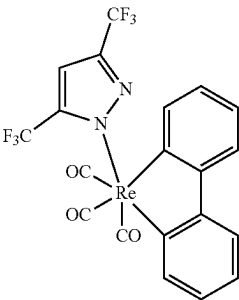
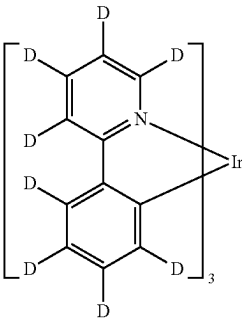
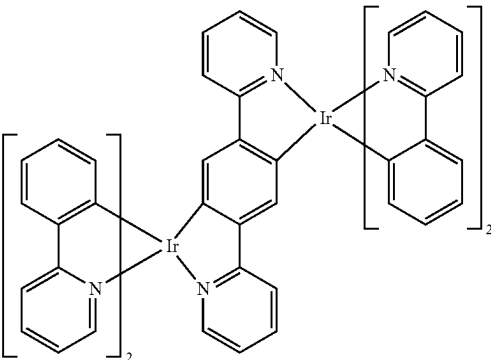
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Gold complexes		Chem. Commun. 2906 (2005)
Rhenium(III) complexes		Inorg. Chem. 42, 1248 (2003)
Deuterated organometallic complexes		US20030138657
Organometallic complexes with two or more metal centers		US20030152802

TABLE 2-continued

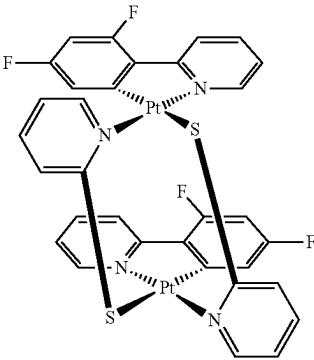
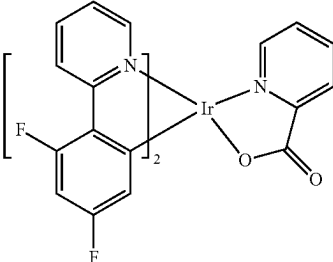
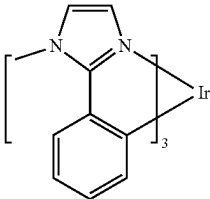
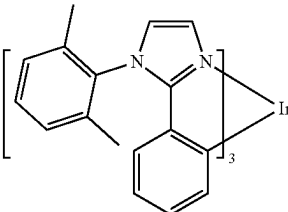
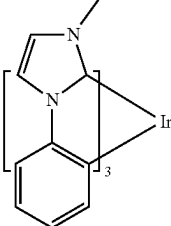
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
		U.S. Pat. No. 7,090,928
	Blue dopants	
Iridium(III) organometallic complexes		WO2002002714
		WO2006009024
		US20060251923
		U.S. Pat. No. 7,393,599, WO2006056418, US20050260441, WO2005019373

TABLE 2-continued

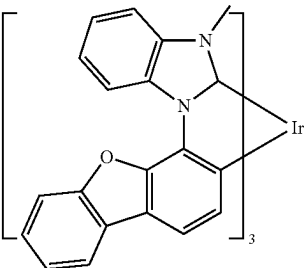
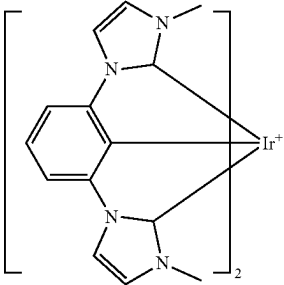
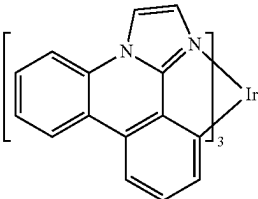
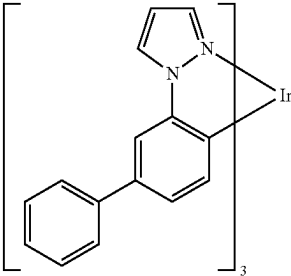
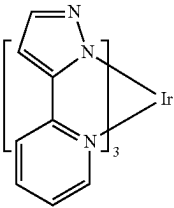
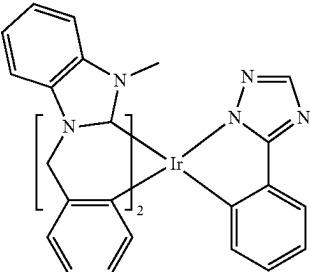
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		U.S. Pat. No. 7,534,505
		U.S. Pat. No. 7,445,855
		US20070190359, US20080297033
		U.S. Pat. No. 7,338,722
		US20020134984
		Angew. Chem. Int. Ed. 47, 1 (2008)

TABLE 2-continued

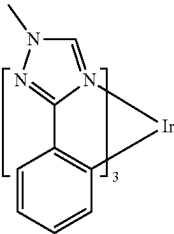
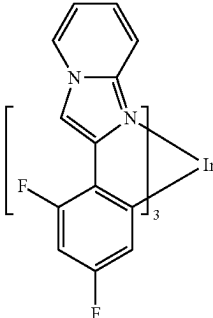
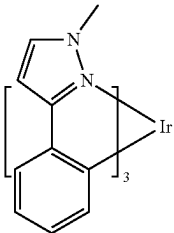
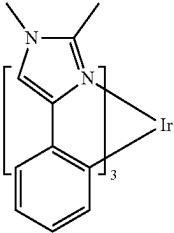
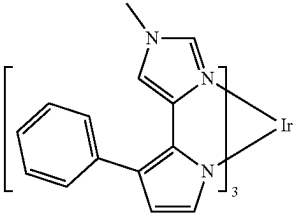
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
		Chem. Mater. 18, 5119 (2006)
		Inorg. Chem. 46, 4308 (2007)
		WO2005123873
		WO2005123873
		WO2007004380

TABLE 2-continued

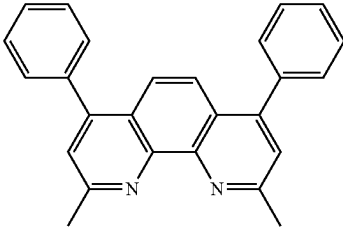
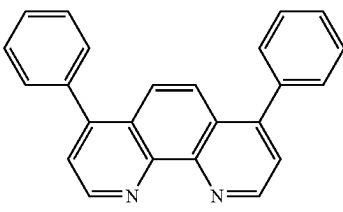
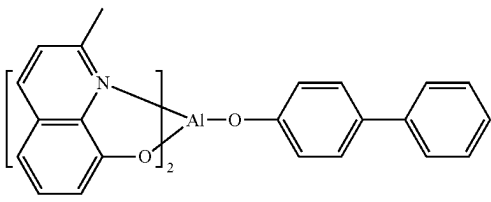
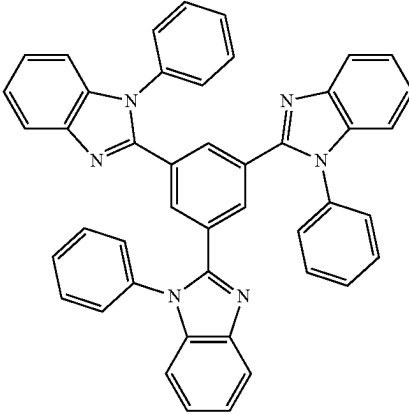
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Exciton/hole blocking layer materials		
Bathocuproine compounds (e.g., BCP, BPhen)		Appl. Phys. Lett. 75, 4 (1999)
		Appl. Phys. Lett. 79, 449 (2001)
Metal 8-hydroxyquinolates (e.g., BAlq)		Appl. Phys. Lett. 81, 162 (2002)
5-member ring electron deficient heterocycles such as triazole, oxadiazole, imidazole, benzoimidazole		Appl. Phys. Lett. 81, 162 (2002)

TABLE 2-continued

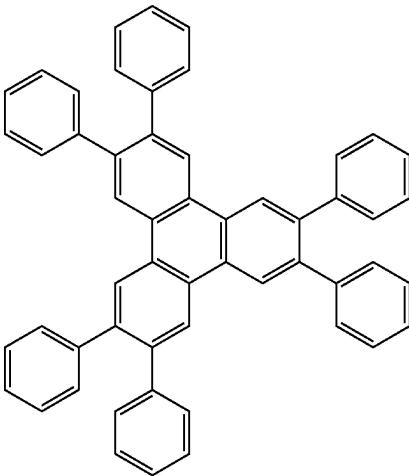
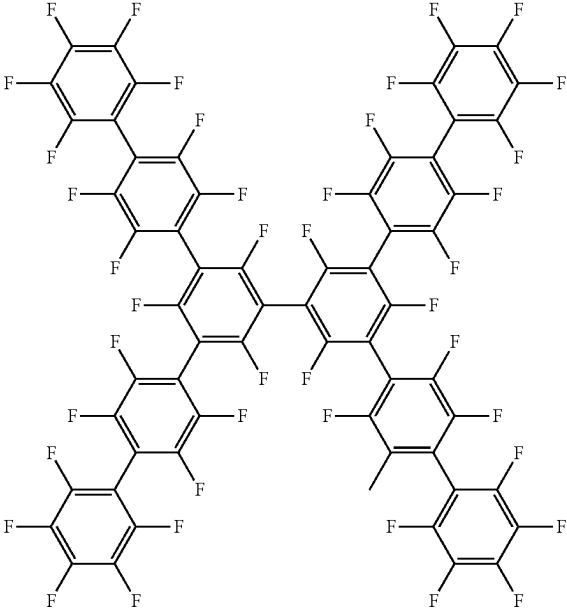
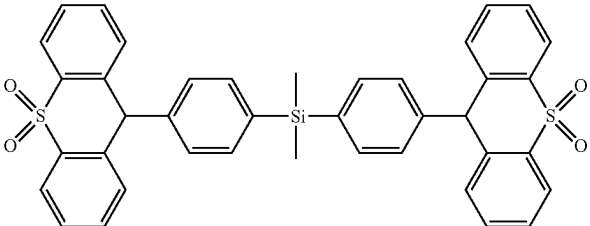
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Triphenylene compounds		US20050025993
Fluorinated aromatic compounds		Appl. Phys. Lett. 79, 156 (2001)
Phenothiazine-S-oxide		WO2008132085

TABLE 2-continued

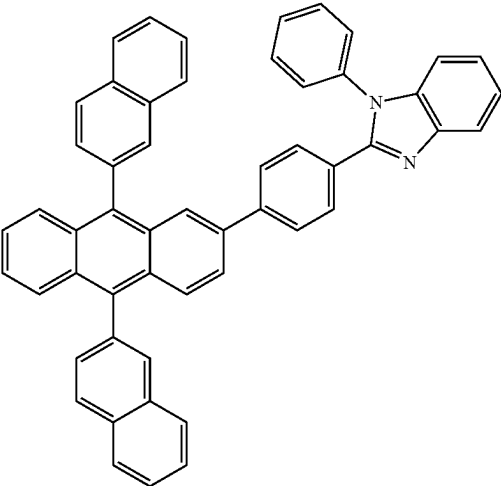
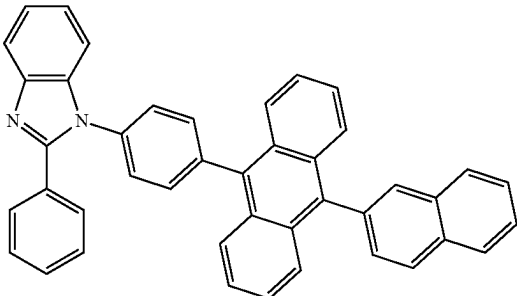
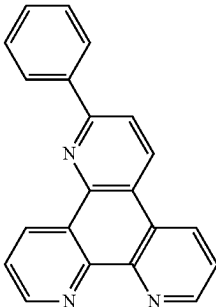
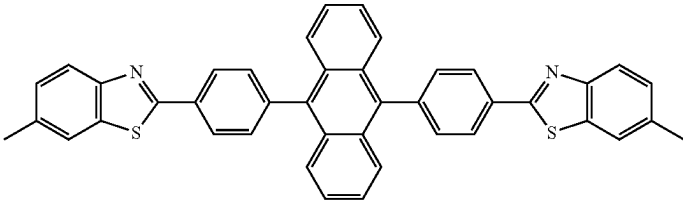
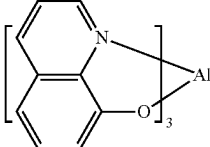
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Electron transporting materials		
Anthracene-benzimidazole compounds		WO2003060956
		US20090179554
Aza triphenylene derivatives		US20090115316
		Appl. Phys. Lett. 89, 063504 (2006)
Metal 8-hydroxyquinolates (e.g., Alq ₃ , Zr _q ₄)		Appl. Phys. Lett. 51, 913 (1987) U.S. Pat. No. 7,230,107

TABLE 2-continued

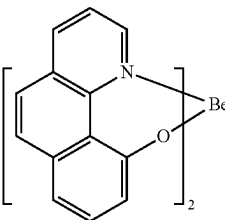
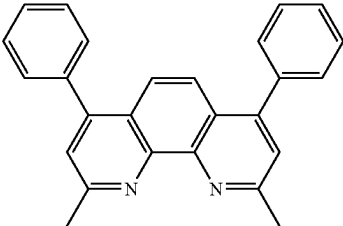
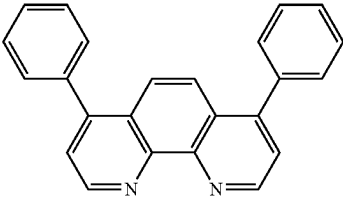
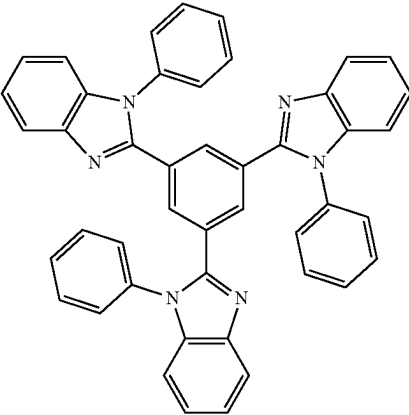
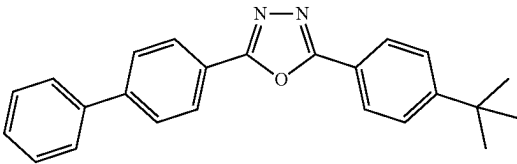
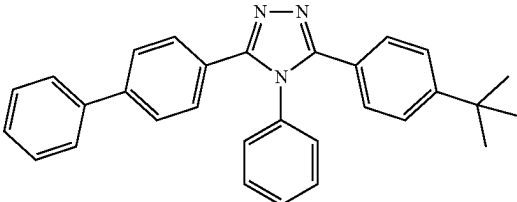
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Metal hydroxybenzoquinolates		Chem. Lett. 5, 905 (1993)
Bathocuprine compounds such as BCP, BPhen, etc		Appl. Phys. Lett. 91, 263503 (2007)
Bathocuprine compounds such as BCP, BPhen, etc		Appl. Phys. Lett. 79, 449 (2001)
5-member ring electron deficient heterocycles (e.g., triazole, oxadiazole, imidazole, benzimidazole)		Appl. Phys. Lett. 74, 865 (1999)
5-member ring electron deficient heterocycles (e.g., triazole, oxadiazole, imidazole, benzimidazole)		Appl. Phys. Lett. 55, 1489 (1989)
5-member ring electron deficient heterocycles (e.g., triazole, oxadiazole, imidazole, benzimidazole)		Jpn. J. Apply. Phys. 32, L917 (1993)

TABLE 2-continued

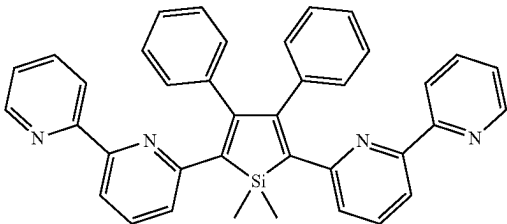
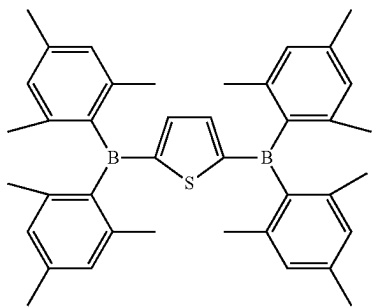
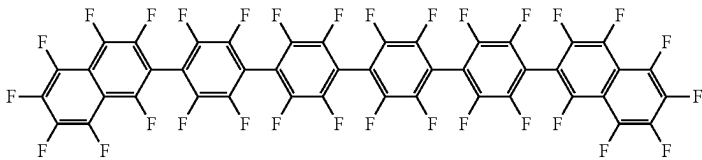
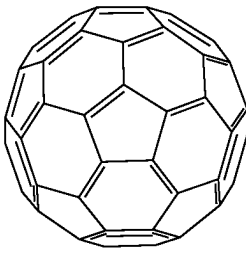
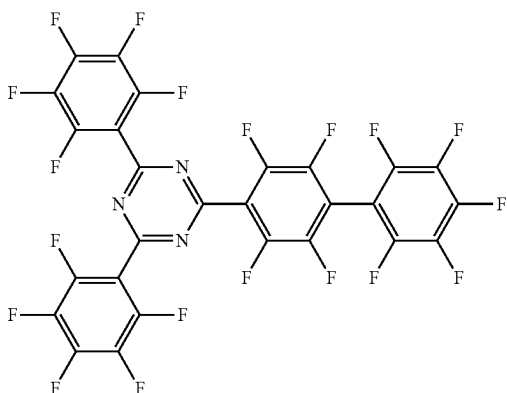
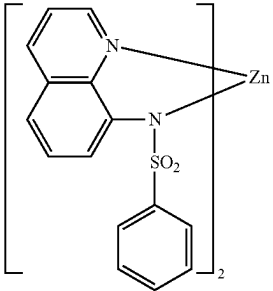
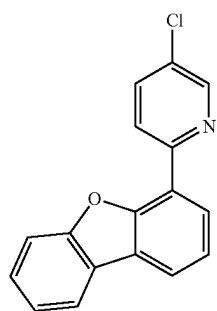
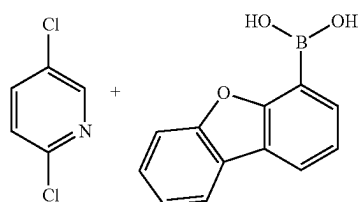
MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Silole compounds		Org. Electron. 4, 113 (2003)
Arylborane compounds		J. Am. Chem. Soc. 120, 9714 (1998)
Fluorinated aromatic compounds		J. Am. Chem. Soc. 122, 1832 (2000)
Fullerene (e.g., C60)		US20090101870
Triazine complexes		US20040036077

TABLE 2-continued

MATERIAL	EXAMPLES OF MATERIAL	PUBLICATIONS
Zn (N`N) complexes		U.S. Pat. No. 6,528,187

EXPERIMENTAL

Chemical abbreviations used throughout this document are as follows: Cy is cyclohexyl, dba is dibenzylideneacetone, EtOAc is ethyl acetate, DME is dimethoxyethane, dppe is 1,2-bis(diphenylphosphino)ethane, THE is tetrahydrofuran, DCM is dichloromethane, S-Phos is dicyclohexyl (2',6'-dimethoxy-[1,1'-biphenyl]-2-yl)phosphine.



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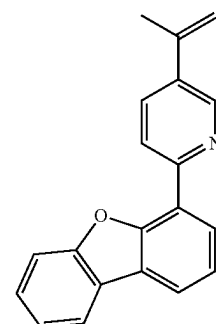
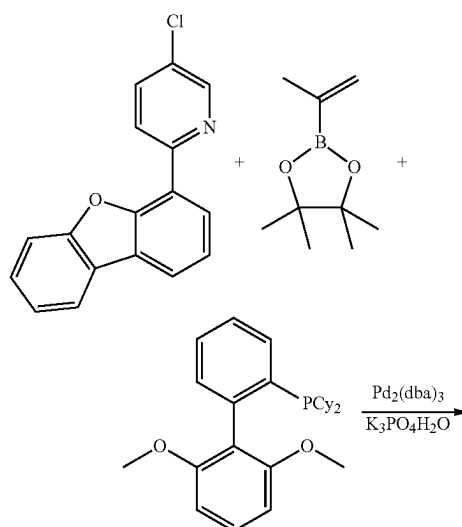
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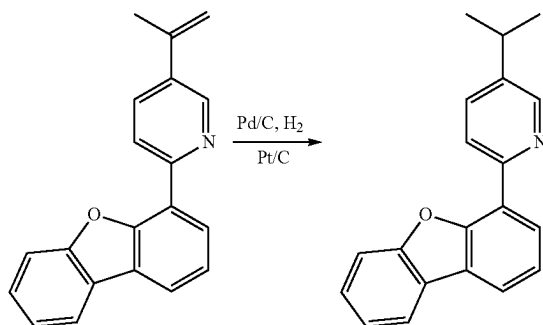
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Synthesis of 5-chloro-2-(dibenzo[b,d]furan-4-yl)pyridine: Dibenzo[b,d]furan-4-ylboronic acid (9.5 g, 44.8 mmol), 2,5-dichloropyridine (7.0 g, 47.0 mmol), Pd(PPh₃)₄ (2.6 g, 2.2 mmol) and potassium carbonate (18.6 g, 134 mmol) were added to dimethoxyethane (75 mL) and water (75 mL). The reaction mixture was degassed with nitrogen before being heated to reflux overnight. EtOAc and water were added, the organic layer separated and the aqueous layer was extracted with 3×50 mL dichloromethane and dried over sodium sulfate. After removing the solvent under reduced pressure, the crude product was chromatographed on silica gel with dichloromethane to yield 11.7 g of crude product. The product was crystallized from hexane to give 9.5 g (76%) of 5-chloro-2-(dibenzo[b,d]furan-4-yl)pyridine as white needles. The product was confirmed by GC/MS.

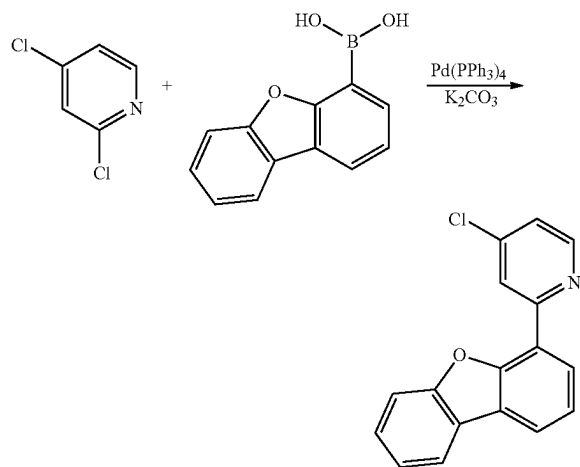
Synthesis of 2-(dibenzo[b,d]furan-4-yl)-5-(prop-1-en-2-yl)pyridine: 5-Chloro-2-(dibenzo[b,d]furan-4-yl)pyridine (9.5 g, 34.0 mmol), dicyclohexyl(2',6'-dimethoxy-[1,1'-biphenyl]-2-yl)phosphine (1.1 g, 2.7 mmol), and potassium phosphate tribasic monohydrate (23.5 g, 102 mmol) were added to toluene (200 mL) and water (20 mL) and the reaction mixture was degassed with nitrogen. Pd₂(dba)₃ (0.622 g, 0.679 mmol) and 4,4,5,5-tetramethyl-2-(prop-1-en-2-yl)-1,3,2-dioxaborolane (7.7 mL, 40.8 mmol) were added and the reaction mixture was heated to reflux overnight. EtOAc and water were added, the organic layer separated and the aqueous layer was extracted with 3×50 mL dichloromethane and dried over sodium sulfate. After removing the solvent under reduced pressure, 12.7 g of amber oil was obtained. The crude material was chromato-

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graphed on silica with 9/1 (v/v) hexane/EtOAc to give 7.5 g (77%) of 2-(dibenzo[b,d]furan-4-yl)-5-(prop-1-en-2-yl)pyridine as a white solid. The product was confirmed by GC/MS and used without further purification.

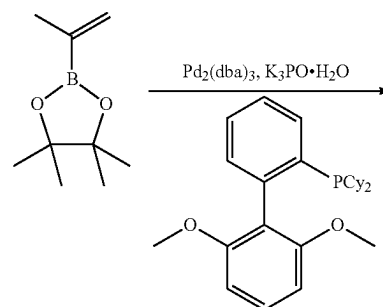
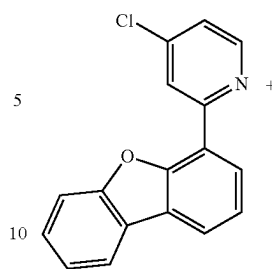


Synthesis of 2-(dibenzo[b,d]furan-4-yl)-5-isopropylpyridine: 2-(Dibenzo[b,d]furan-4-yl)-5-(prop-1-en-2-yl)pyridine (7.5 g, 26.3 mmol) was added to a hydrogenator bottle containing EtOH (150 mL). The reaction mixture was degassed by bubbling nitrogen for 10 min. Pd/C (0.28 g, 2.63 mmol) and Pt/C (0.26 g, 1.3 mmol) were added to the reaction mixture. The reaction mixture was placed on the Parr hydrogenator for 1 h. The reaction mixture was filtered on a tightly packed Celite® bed and washed with dichloromethane to yield 7.5 g (99%) of the desired product. The product was confirmed by GC/MS and NMR.



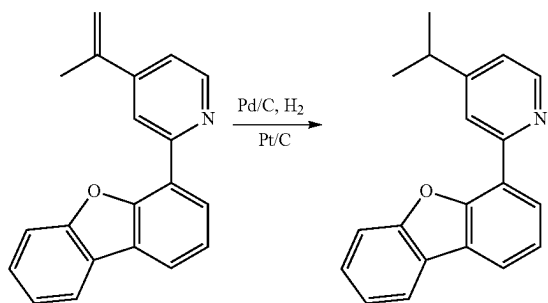
Synthesis of 5-chloro-2-(dibenzo[b,d]furan-4-yl)pyridine: Dibenzo[b,d]furan-4-ylboronic acid (25 g, 118 mmol), 2,4-dichloropyridine (19.2 g, 130 mmol), Pd(PPh₃)₄ (4.1 g, 3.5 mmol) and potassium carbonate (48.9 g, 354 mmol) were added to dimethoxyethane (200 mL) and water (200 mL). The reaction mixture was degassed with nitrogen before being heated to reflux overnight. EtOAc and water were added, the organic layer separated and the aqueous layer was extracted with 3×50 mL dichloromethane and dried over sodium sulfate. After removing the solvent under reduced pressure, the crude product was chromatographed on silica gel with dichloromethane to yield 33.4 g of crude product. The product was crystallized from hexane to give 27.0 g (82%) of 4-chloro-2-(dibenzo[b,d]furan-4-yl)pyridine as white needles. The product was confirmed by GC/MS and NMR.

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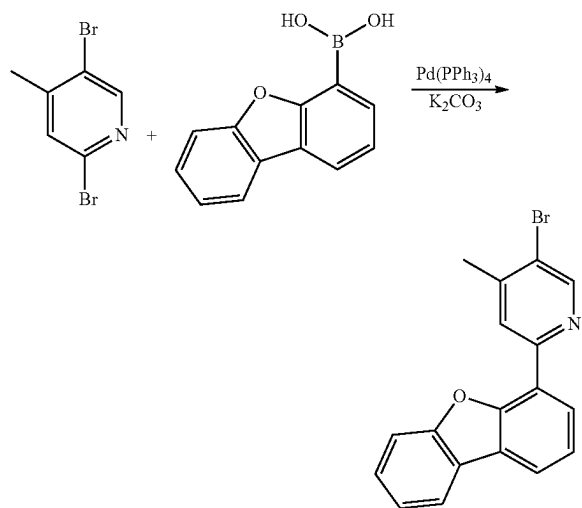


Synthesis of 2-(dibenzo[b,d]furan-4-yl)-4-(prop-1-en-2-yl)pyridine: 4-Chloro-2-(dibenzo[b,d]furan-4-yl)pyridine (24.0 g, 86.0 mmol), dicyclohexyl(2',6'-dimethoxy-[1,1'-biphenyl]-2-yl)phosphine (2.8 g, 6.9 mmol), and potassium phosphate tribasic monohydrate (59.3 g, 257 mmol) were added to toluene (400 mL) and water (40 mL) and the reaction mixture was degassed. Pd₂(dba)₃ (1.6 g, 1.7 mmol) and 4,4,5,5-tetramethyl-2-(prop-1-en-2-yl)-1,3,2-dioxaborolane (19.4 mL, 103 mmol) were added and the reaction mixture was heated to reflux overnight. EtOAc and water were added, the organic layer separated and the aqueous layer was extracted with 3×50 mL dichloromethane and dried over sodium sulfate. After removing the solvent under reduced pressure, 33.0 g of amber oil was obtained. The crude material was chromatographed on silica with 9/1 (v/v) DCM/EtOAc to give 23.5 g (96%) of 2-(dibenzo[b,d]furan-4-yl)-4-(prop-1-en-2-yl)pyridine as a white solid. The product was confirmed by GC/MS and used without further purification.

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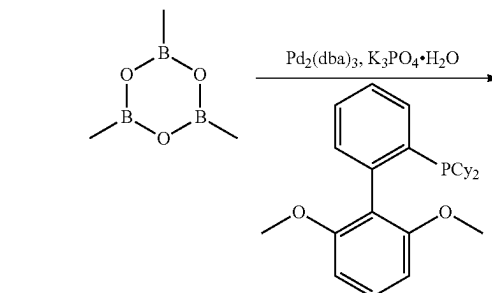
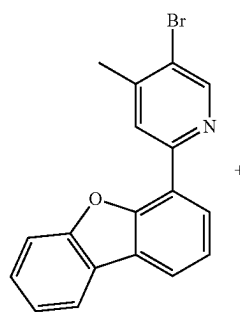


Synthesis of 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine: 2-(Dibenzo[b,d]furan-4-yl)-5-(prop-1-en-2-yl)pyridine (8.0 g, 28 mmol) was added to a hydrogenator bottle containing EtOH (150 mL). The reaction mixture was degassed by bubbling N₂ for 10 min. Pd/C (0.60 g, 5.6 mmol) and Pt/C (0.55 g, 2.8 mmol) were added to the reaction mixture. The reaction mixture was placed on the Parr hydrogenator for 1 h. The reaction mixture was filtered on a tightly packed Celite® bed and washed with dichloromethane. The crude product was chromatographed on silica gel with 9/1 (v/v) hexane/EtOAc to yield 7.2 g (96%) of 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine. The product was confirmed by GC/MS and NMR.



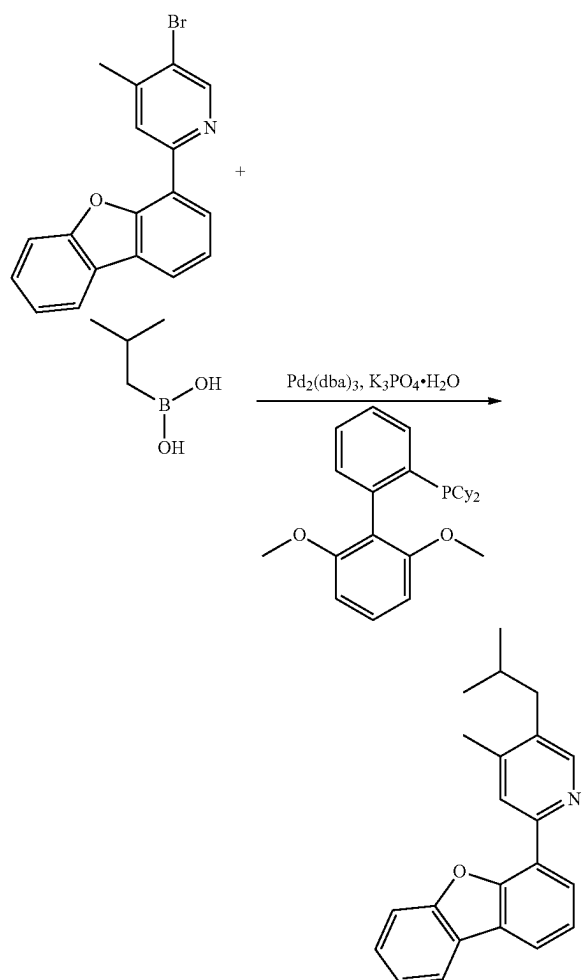
Synthesis of 5-bromo-2-(dibenzo[b,d]furan-4-yl)-4-methylpyridine: 2,5-Dibromo-4-methylpyridine (30 g, 118 mmol), dibenzo[b,d]furan-4-ylboronic acid (25 g, 118 mmol), Pd(PPh₃)₄ (1.4 g, 1.18 mmol), and K₂CO₃ (49 g, 354 mmol) were added to a flask with dimethoxyethane (450 mL) and water (100 mL) and degassed with nitrogen. The reaction mixture was heated to reflux for 15 h before cooling to room temperature. EtOAc and water were added, the organic layer separated and the aqueous layer was extracted with 3×50 mL dichloromethane and dried over sodium sulfate. After removing the solvent under reduced pressure, the crude product was chromatographed on silica gel with dichloromethane and 29.7 g of crude product was obtained. The product was crystallized from hexane to give 28.8 g (72%) of pure product. The product was confirmed by NMR and HPLC (99.3% pure)

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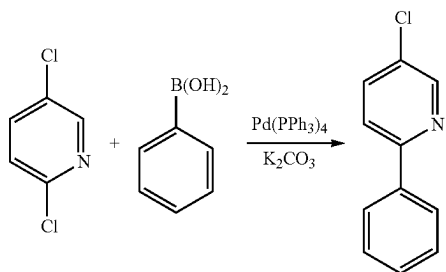


Synthesis of 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethylpyridine: 5-Bromo-2-(dibenzo[b,d]furan-4-yl)-4-methylpyridine (28.7 g, 85 mmol), dicyclohexyl(2',6'-dimethoxy[1,1'-biphenyl]-2-yl)phosphine (1.394 g, 3.39 mmol) and potassium phosphate monohydrate (58.6 g, 255 mmol) were added to toluene (500 mL) and water (50 mL) and degassed for 20 min. Trimethylboroxine (14.83 mL, 106 mmol) and Pd₂(dba)₃ (0.777 g, 0.849 mmol) were added and the reaction mixture heated to reflux overnight. After cooling, the organic layer was separated and the aqueous layer extracted 3×50 mL with EtOAc, dried over sodium sulfate and evaporated. The crude product was chromatographed on silica gel with 8/2 (v/v) dichloromethane/EtOAc in hexane to give 19.2 g of an off-white solid, which was recrystallized from hexane to give 16.8 g (83%) of the product as white needles. The product was confirmed by NMR and HPLC (99.97% pure).

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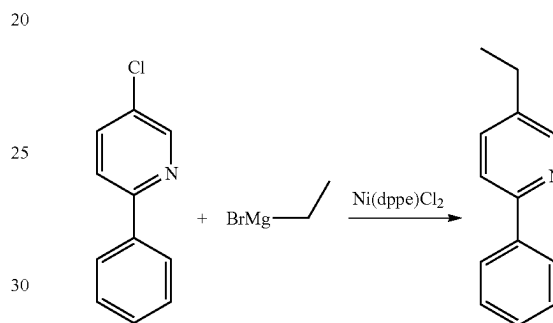
Synthesis of 2-(dibenzo[b,d]furan-4-yl)-5-isobutyl-4-methylpyridine: 5-Bromo-2-(dibenzo[b,d]furan-4-yl)-4-methylpyridine (13.0 g, 38.3 mmol), isobutylboronic acid (11.7 g, 115 mmol), dicyclohexyl(2',6'-dimethoxy-[1,1'-biphenyl]-2-yl)phosphine (0.63 g, 1.53 mmol) and potassium phosphate monohydrate (22.1 g, 96 mmol) were mixed in water (10 mL) and toluene (210 mL). The system was degassed for 20 min. with nitrogen and $\text{Pd}_2(\text{dba})_3$ (0.35 g, 0.38 mmol) was then added and the system was refluxed overnight. After cooling to room temperature, the reaction mixture was filtered through a small plug of silica gel and eluted with dichloromethane. The filtrate was concentrated and then crystallized from hexane to give 2-(dibenzo[b,d]furan-4-yl)-5-isobutyl-4-methylpyridine (9.0 g, 74%).



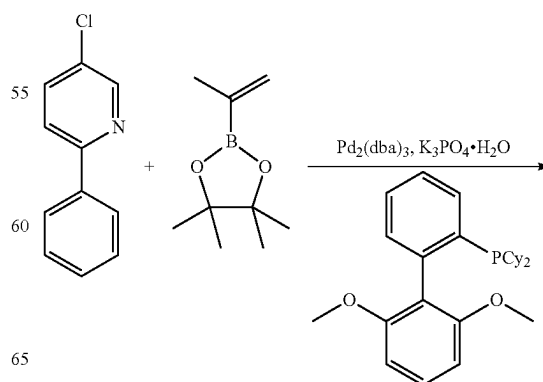
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Synthesis of 5-chloro-2-phenylpyridine: 2,5-Dichloropyridine (30 g, 203 mmol), phenylboronic acid (24.72 g, 203 mmol) and potassium carbonate (84 g, 608 mmol) were added to dimethoxyethane (500 mL) and water (100 mL).

The reaction mixture was degassed with nitrogen for 20 min, and $\text{Pd}(\text{PPh}_3)_4$ (2.3 g, 2.0 mmol) was added and reaction mixture was allowed to reflux for 18 h. The reaction was cooled to room temperature, the aqueous layer removed and dimethoxyethane was concentrated to dryness by rotary evaporation under vacuum. The residue was dissolved in DCM and passed through a silica gel plug, eluting with DCM. The solvent was removed and the crude product was chromatographed on silica with 40/60 (v/v) DCM/hexane to 50/50 (v/v) DCM/hexane to yield 28 g (73%) of the product as a white solid (HPLC purity: 99.7%).

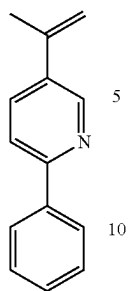


Synthesis of 5-ethyl-2-phenylpyridine: 5-Chloro-2-phenylpyridine (16 g, 84 mmol) and $\text{Ni}(\text{dppe})\text{Cl}_2$ (0.891 g, 1.687 mmol) were added to 300 mL of THF and the reaction mixture was degassed with nitrogen for 20 min. before being cooled to 0° C. Ethylmagnesium bromide (169 mL, 169 mmol) was added dropwise over a period of 60 min. and the reaction mixture stirred for an additional 3 h at before warming to room temperature overnight. The reaction mixture was recooled to 0° C. and quenched with 250 mL of water, extracted with EtOAc and the organic layer dried over sodium sulfate and filtered. The crude material was chromatographed on silica with 95/5 hexane/EtOAc to give 2.9 g (19%) of 5-ethyl-2-phenylpyridine as a white solid.

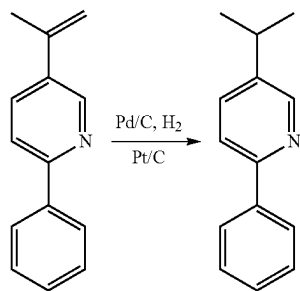


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-continued

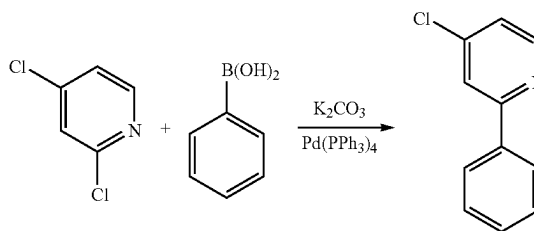


Synthesis of 2-phenyl-5-(prop-1-en-2-yl)pyridine: To a 1 L round bottom flask was added 5-chloro-2-phenylpyridine (10.15 g, 53.5 mmol), dicyclohexyl(2',6'-dimethoxy-[1,1'-biphenyl]-2-yl)phosphine (1.8 g, 4.3 mmol), potassium phosphate tribasic monohydrate (37.0 g, 161 mmol) with toluene (200 mL) and water (20 mL). The reaction mixture was degassed with nitrogen for 20 mins. 4,4,5,5-tetramethyl-2-(prop-1-en-2-yl)-1,3,2-dioxaborolane (12.07 mL, 64.2 mmol) and $\text{Pd}_2(\text{dba})_3$ (0.980 g, 1.070 mmol) were added and the reaction mixture was refluxed for 18 h. The aqueous layer was removed and the organic layer was concentrated to dryness. The crude product was chromatographed on silica gel with 0-20% EtOAc in hexane to yield 11 g of the desired product (HPLC purity: 95%). The product was confirmed by GC/MS.

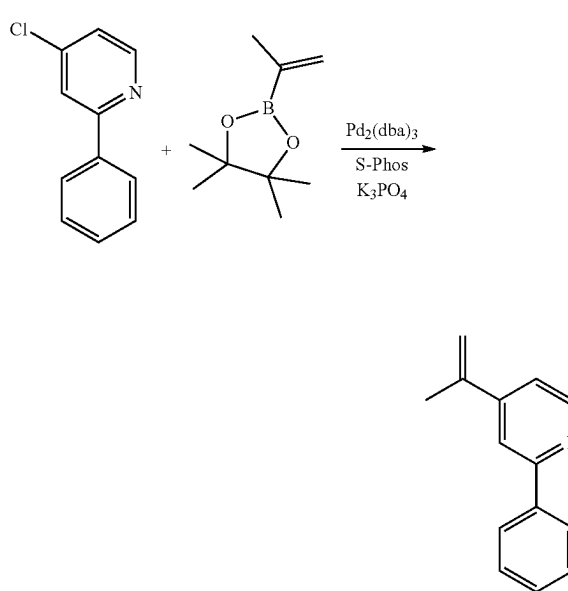


Synthesis of 2-phenyl-5-isopropylpyridine: 2-Phenyl-5-(prop-1-en-2-yl)pyridine (11 g, 56.3 mmol) was added to a hydrogenator bottle containing EtOH (150 mL). The reaction mixture was degassed by bubbling N_2 for 10 min. Pd/C (0.60 g, 5.63 mmol) and Pt/C (0.55 g, 2.82 mmol) were added to the reaction mixture. The reaction mixture was placed on the Parr hydrogenator for 1.5 h. The reaction mixture was filtered on a tightly packed Celite® bed and washed with dichloromethane. The solvent was removed on rotoevaporator and GC/MS confirmed complete conversion. The crude product was adsorbed on Celite® for column chromatography. The crude product was chromatographed on silica gel with 10% EtOAc in hexane to yield 6 g (54%) of 2-phenyl-5-isopropylpyridine (HPLC purity: 100%). The product was confirmed by GC/MS.

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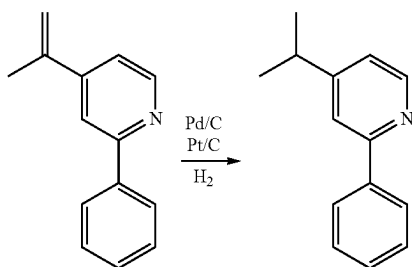


Synthesis of 4-chloro-2-phenylpyridine: A 1 L round bottom flask was charged with 2,4-dichloropyridine (30 g, 203 mmol), phenylboronic acid (24.7 g, 203 mmol), potassium carbonate (84 g, 608 mmol), $\text{Pd}(\text{PPh}_3)_4$ (2.3 g, 2.0 mmol), dimethoxyethane (500 mL) and water (150 mL). The mixture was degassed and heated to reflux for 20 h. After cooling, the aqueous layer was extracted with EtOAc; the organic portion was combined and subjected to column chromatography (SiO_2 , 5% EtOAc in hexane to 10% EtOAc in hexane) to give 34 g (88%) of 4-chloro-2-phenylpyridine. The product was confirmed by GC/MS and NMR.

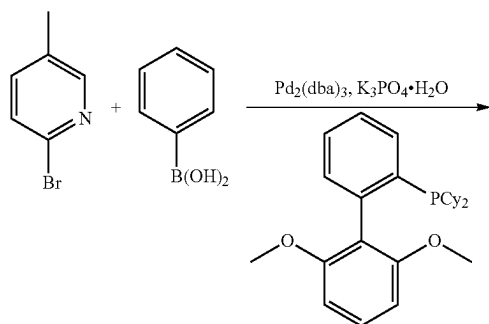


Synthesis of 2-phenyl-4-(prop-1-en-2-yl)pyridine: 4-Chloro-2-phenylpyridine (14 g, 73.8 mmol) and potassium phosphate (51.0 g, 221 mmol) were dissolved in 300 mL of toluene and 30 mL of water. The reaction was purged with nitrogen for 20 minutes and then 4,4,5,5-tetramethyl-2-(prop-1-en-2-yl)-1,3,2-dioxaborolane (16.65 mL, 89 mmol), $\text{Pd}_2(\text{dba})_3$ (1.35 g, 1.48 mmol) and S-Phos (2.42 g, 5.91 mmol) were added. The reaction was brought to reflux for 18 h. After cooling, 100 mL of water was added, separated and the aqueous layer extracted twice with 100 mL of ethyl acetate. The organic layers were passed through a plug of silica gel, eluting with DCM. After evaporation of the solvent, the crude product was subjected to column chromatography (SiO_2 , 5% EtOAc in hexane to 10% EtOAc in hexane) to get 13.5 g (90%) of 2-phenyl-4-(prop-1-en-2-yl)pyridine.

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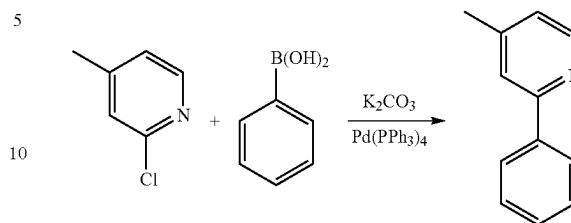
Synthesis of 2-phenyl-4-isopropylpyridine: 2-Phenyl-4-(prop-1-en-2-yl) pyridine (13.5 g, 69.1 mmol) was added to a hydrogenator bottle containing EtOH (150 mL). The reaction mixture was degassed by bubbling with nitrogen for 10 min. Pd/C (0.736 g, 6.9 mmol) and Pt/C (0.674 g, 3.5 mmol) was added to the reaction mixture. The reaction mixture was placed on the Parr hydrogenator for 2 h. The reaction mixture was filtered on a tightly packed Celite® bed and washed with dichloromethane. The solvent was removed on a rotary evaporator and GC/MS confirmed complete conversion. The crude product was adsorbed on Celite® for column chromatography. The crude product was chromatographed on silica gel with 10% EtOAc in hexane to yield 10 g (75%) of 2-phenyl-4-isopropylpyridine (HPLC purity: 99.8%). The product was confirmed by GC/MS.



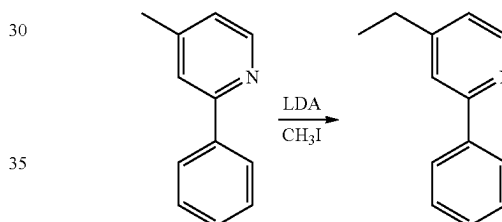
Synthesis of 5-methyl-2-phenylpyridine: 2-Bromo-5-methylpyridine (30 g, 174 mmol), phenylboronic acid (25.5 g, 209 mmol), dicyclohexyl(2',6'-dimethoxy-[1,1'-biphenyl]-2-yl)phosphine (2.86 g, 6.98 mmol) and potassium phosphate tribasic monohydrate (120 g, 523 mmol) were added to toluene (600 mL) and water (60 mL). The reaction mixture was degassed with nitrogen for 20 min. Pd₂(dba)₃ (3.19 g, 3.49 mmol) was added and the reaction mixture was refluxed for 18 h. After cooling, the organic layer was separated and the aqueous layer extracted with 3×50 mL dichloromethane, dried over sodium sulfate and evaporated. The crude product was chromatographed on silica gel with 75/25 (v/v) hexane/EtOAc and then distilled on a Kugelrohr apparatus (150° C., 100 mbar) to give 26 g (88%) of

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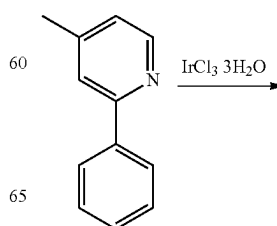
5-methyl-2-phenylpyridine as a white solid. The product was confirmed by NMR and GC/MS. HPLC purity: 99.2%.



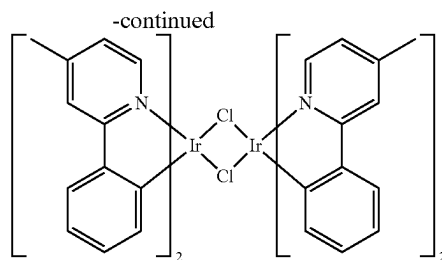
Synthesis of 4-methyl-2-phenylpyridine: A 1 L round bottom flask was charged with 2-chloro-4-methylpyridine (25 g, 196 mmol), phenylboronic acid (23.9 g, 196 mmol), potassium carbonate (81 g, 588 mmol), Pd(PPh₃)₄ (2.3 g, 1.9 mmol), dimethoxyethane (500 mL) and water (150 mL). The reaction mixture was degassed with nitrogen and heated to reflux for 22 h. After cooling, the aqueous layer was extracted with EtOAc; the organic portion was combined and subjected to column chromatography (SiO₂, 5% EtOAc in hexane to 10% EtOAc in hexane) to give 28 g (78%) of 4-methyl-2-phenylpyridine. The product was confirmed by NMR and GC/MS.



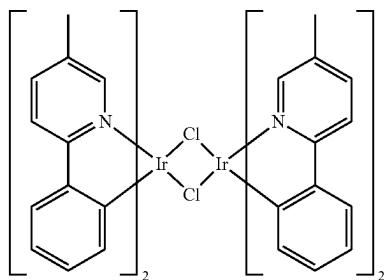
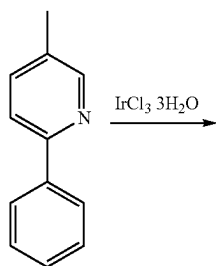
Synthesis of 4-ethyl-2-phenylpyridine: To 4-methyl-2-phenylpyridine (8 g, 47.3 mmol) in dry THF (150 mL) at -78° C. was added dropwise lithium diisopropylamide (LDA) (30.7 mL, 61.5 mmol). The dark solution was stirred for 3 h at -78° C. and then CH₃I was added (4.1 mL, 66.2 mmol) dropwise. The reaction mixture was allowed to slowly warm to room temperature overnight. Ammonium chloride solution and EtOAc were added and the reaction transferred to a separatory funnel. The layers were separated, washing the aqueous twice with EtOAc and combined organics once with water. After removal of the solvent, the crude product was chromatographed on silica gel with 9/1 (v/v) hexane/EtOAc to give 5.5 g (63.5%) of 4-ethyl-2-phenylpyridine. HPLC purity: 99.0%.



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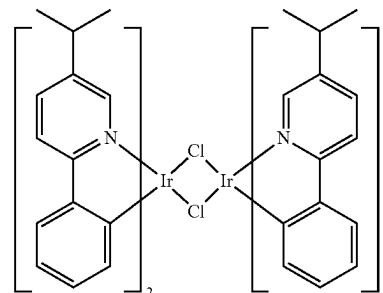
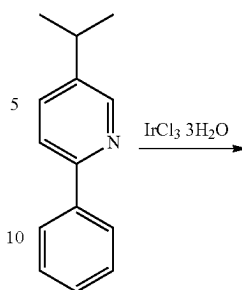


Synthesis of 4-methyl-2-phenylpyridine chloro-bridged dimer: To a 500 mL round-bottom flask was added 4-methyl-2-phenylpyridine (7 g, 41 mmol) and iridium(III) chloride hydrate (4.86 g, 13.79 mmol) with 2-ethoxyethanol (90 mL) and water (30 mL) under a nitrogen atmosphere. The resulting reaction mixture was refluxed at 130° C. for 18 h. The resulting precipitate was filtered and washed with methanol (3-4 times) and hexane (3-4 times). The product obtained was dried to give 7.5 g (90%) of the desired product. The product was used without further purification.

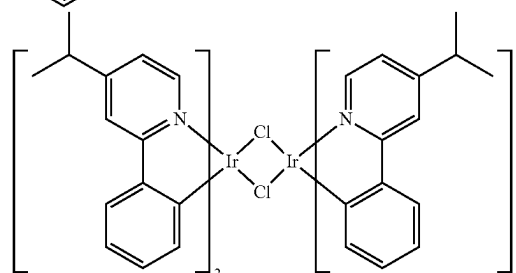
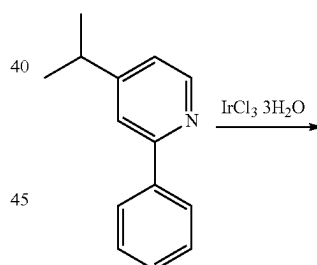


Synthesis of 5-methyl-2-phenylpyridine chloro-bridged dimer: To a 500 mL round bottom flask was added 5-methyl-2-phenylpyridine (12 g, 70.9 mmol) and iridium(III) chloride hydrate (7.1 g, 20.3 mmol) with 2-ethoxyethanol (100 mL) and water (33.3 mL) under a nitrogen atmosphere. The resulting reaction mixture was refluxed at 130° C. for 18 h. The resulting precipitate was filtered and washed with methanol (3-4 times) and hexane (3-4 times). The product obtained was dried to give 11.0 g (96%) of the desired product. The product was used without further purification.

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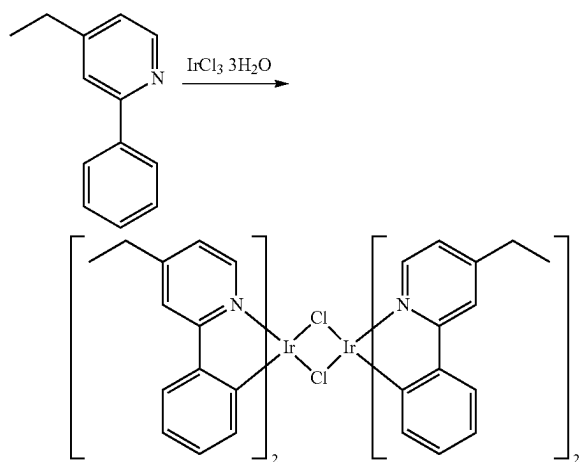
Synthesis of 2-phenyl-5-isopropylpyridine chloro-bridged dimer: To a 500 mL round-bottom flask was added 5-isopropyl-2-phenylpyridine (6.0 g, 30.4 mmol) and iridium(III) chloride hydrate (3.6 g, 10.1 mmol) with 2-ethoxyethanol (100 mL) and water (33.3 mL) under a nitrogen atmosphere. The resulting reaction mixture was refluxed at 130° C. for 18 h. The resulting precipitate was filtered and washed with methanol (3-4 times) and hexane (3-4 times). The product obtained was dried to give 7 g (100%) of the desired product. The product was used without further purification.



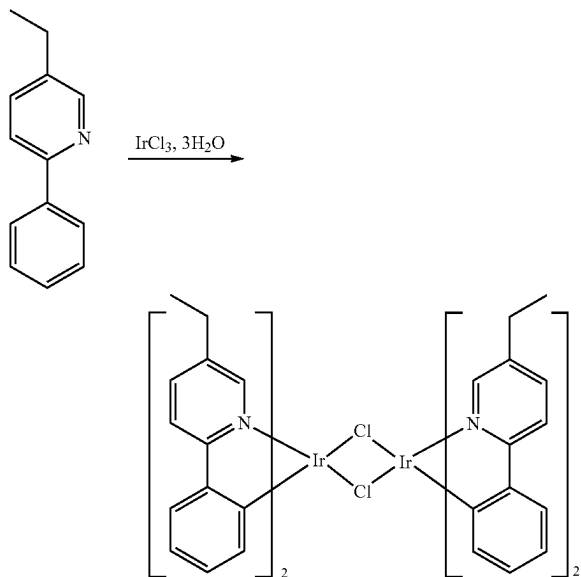
Synthesis of 2-phenyl-4-isopropylpyridine chloro-bridged dimer: To a 500 mL round-bottom flask was added 4-isopropyl-2-phenylpyridine (8.0 g, 40.6 mmol) and iridium(III) chloride hydrate (7.4 g, 20.3 mmol) with 2-ethoxyethanol (90 mL) and water (30 mL) under a nitrogen atmosphere. The resulting reaction mixture was refluxed at 130° C. for 18 h. The resulting precipitate was filtered and washed with methanol (3-4 times) and hexane (3-4 times).

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The product obtained was dried to give 6.1 g (95%) of the desired product. The product was used without further purification.



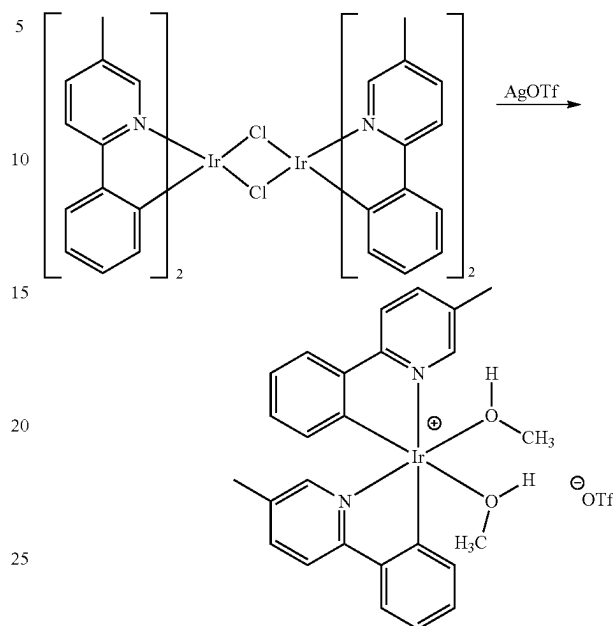
Synthesis of 4-ethyl-2-phenylpyridine chloro-bridged dimer: To a 500 mL round-bottom flask was added 4-isopropyl-2-phenylpyridine (5.5 g, 30.0 mmol) and iridium(III) chloride hydrate (5.8 g, 16.5 mmol) with 2-ethoxyethanol (90 mL) and water (30 mL) under a nitrogen atmosphere. The resulting reaction mixture was refluxed at 130° C. for 18 h. The resulting precipitate was filtered and washed with methanol (3-4 times) and hexane (3-4 times). The product obtained was dried to give 6.5 g (72%) of the desired product. The product was used without further purification.



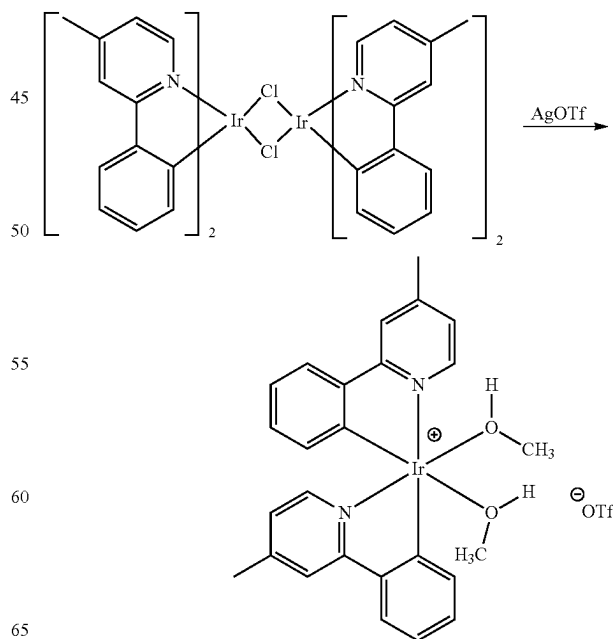
Synthesis of 5-ethyl-2-phenylpyridine chloro-bridged dimer: To a 500 mL round bottom flask was added 5-ethyl-2-phenylpyridine (2.9 g, 15.7 mmol) and iridium(III) chloride hydrate (1.8 g, 5.2 mmol) with 2-ethoxyethanol (60 mL) and water (20 mL) under a nitrogen atmosphere. The reaction mixture was refluxed at 130° C. for 18 h. The resulting precipitate was filtered and washed with methanol (3-4 times) and hexane (3-4 times). The product obtained was

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dried to give 2.45 g (89.3%) of the desired product. The product was used without further purification.

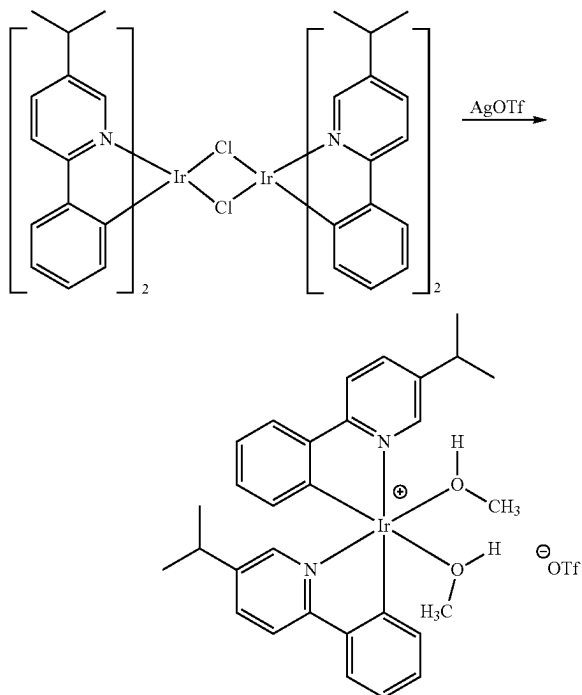


Synthesis of 5-methyl-2-phenylpyridine iridium trifluoromethanesulfonate salt: The iridium dimer (11 g, 9.8 mmol) was suspended in 600 mL of dichloromethane. In a separate flask, silver(I) trifluoromethanesulfonate (5.3 g, 20.5 mmol) was dissolved in MeOH (300 mL) and added slowly to the dichloromethane solution with continuous stirring at room temperature. The reaction mixture was stirred overnight in the dark. The reaction mixture was filtered through a tightly packed Celite® bed and the solvent was removed under vacuum to give 15 g (100%) of product as a brownish green solid. The product was used without further purification.

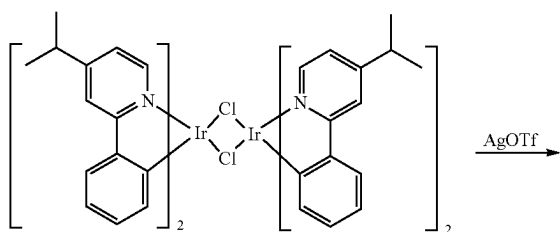


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Synthesis of 4-methyl-2-phenylpyridine iridium trifluoromethanesulfonate salt: The iridium dimer (7.5 g, 6.6 mmol) was dissolved in 600 mL of dichloromethane. In a separate flask, silver(I) trifluoromethanesulfonate (3.5 g, 13.8 mmol) was dissolved in MeOH (300 mL) and added slowly to the dichloromethane solution with continuous stirring at room temperature. The reaction mixture was stirred overnight in the dark. The reaction mixture was filtered through a tightly packed Celite® bed and the solvent was removed under vacuum to give 10 g (100%) of product as a brownish green solid. The product was used without further purification.

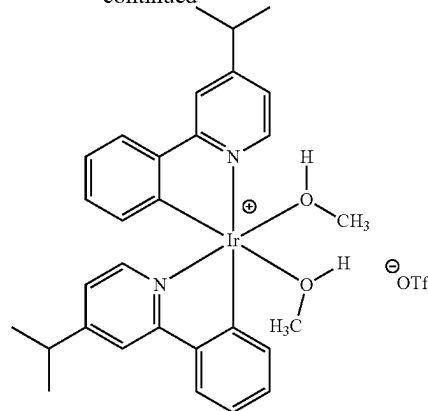


Synthesis of 2-phenyl-5-isopropylpyridine iridium trifluoromethanesulfonate salt: The iridium dimer (5.3 g, 4.3 mmol) was dissolved in 500 mL of dichloromethane. In a separate flask, silver(I) trifluoromethanesulfonate (2.3 g, 8.9 mmol) was dissolved in MeOH (250 mL) and added slowly to the dichloromethane solution with continuous stirring at room temperature. The reaction mixture was stirred overnight in the dark. The reaction mixture was filtered through a tightly packed Celite® bed and the solvent was removed under vacuum to give 6.9 g (100%) of product as a brownish solid. The product was used without further purification.

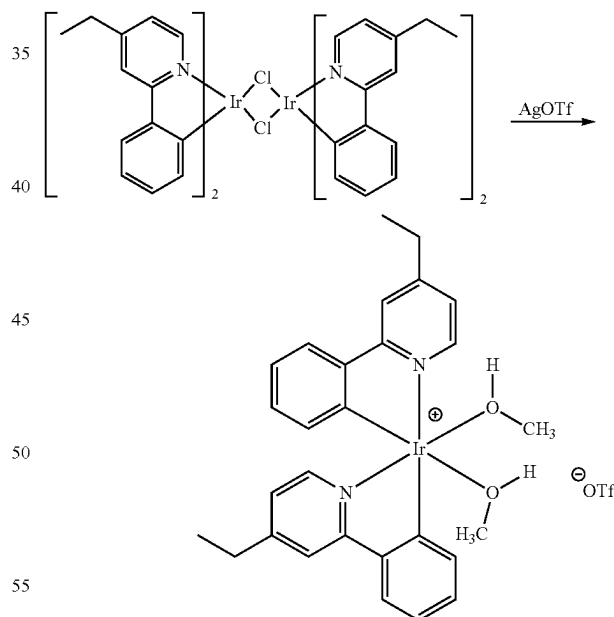


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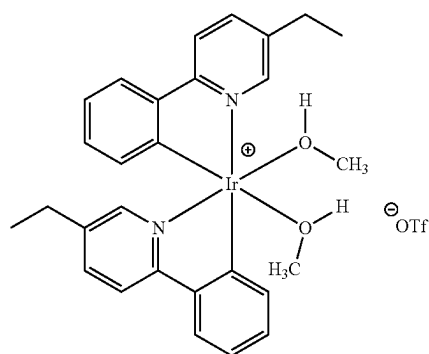
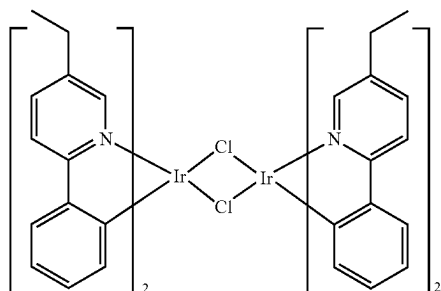
Synthesis of 2-phenyl-4-isopropylpyridine iridium trifluoromethanesulfonate salt: The iridium dimer (6.2 g, 4.94 mmol) was dissolved in 500 mL of dichloromethane. In a separate flask, silver(I) trifluoromethanesulfonate (2.7 g, 10.4 mmol) was dissolved in MeOH (250 mL) and added slowly to the dichloromethane solution with continuous stirring at room temperature. The reaction mixture was stirred overnight in the dark. The reaction mixture was filtered through a tightly packed Celite® bed and the solvent was removed under vacuum to give 7.8 g (100%) of product as a brownish green solid. The product was used without further purification.



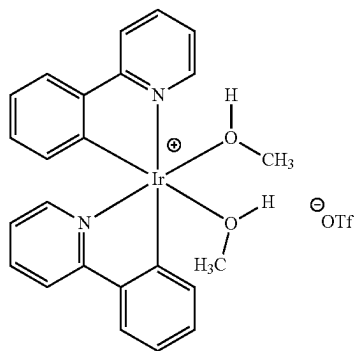
Synthesis of 4-ethyl-2-phenylpyridine iridium trifluoromethanesulfonate salt: The iridium dimer (6.8 g, 5.7 mmol) was dissolved in 500 mL of dichloromethane. In a separate flask, silver(I) trifluoromethanesulfonate (3.2 g, 12.5 mmol) was dissolved in MeOH (250 mL) and added slowly to the dichloromethane solution with continuous stirring at room temperature. The reaction mixture was stirred overnight in the dark. The reaction mixture was filtered through a tightly packed Celite® bed and the solvent

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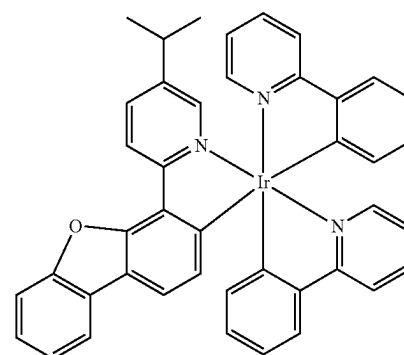
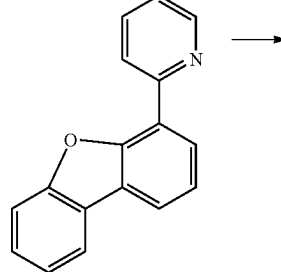
was removed under vacuum to give 5.5 g (63%) of product as a brownish green solid. The product was used without further purification.



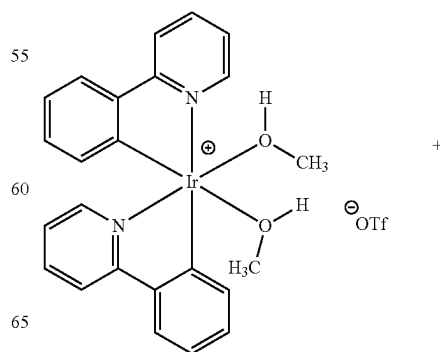
Synthesis of 5-ethyl-2-phenylpyridine iridium trifluoromethanesulfonate salt: The iridium dimer (2.8 g, 2.4 mmol) was suspended in 500 mL of dichloromethane. In a separate flask, silver(I) trifluoromethanesulfonate (1.3 g, 4.91 mmol) was dissolved in MeOH (250 mL) and added slowly to the dichloromethane solution with continuous stirring at room temperature. The reaction mixture was stirred overnight in the dark. The reaction mixture was filtered through a tightly packed Celite® bed and the solvent was removed under vacuum to give 3.6 g (100%) of product as a brownish green solid. The product was used without further purification.

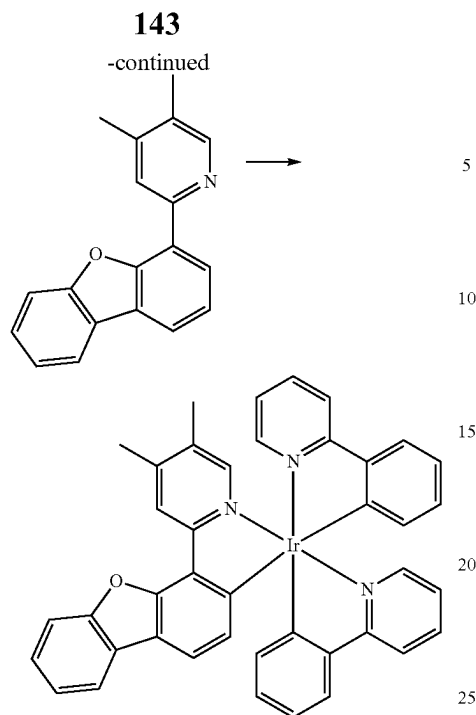
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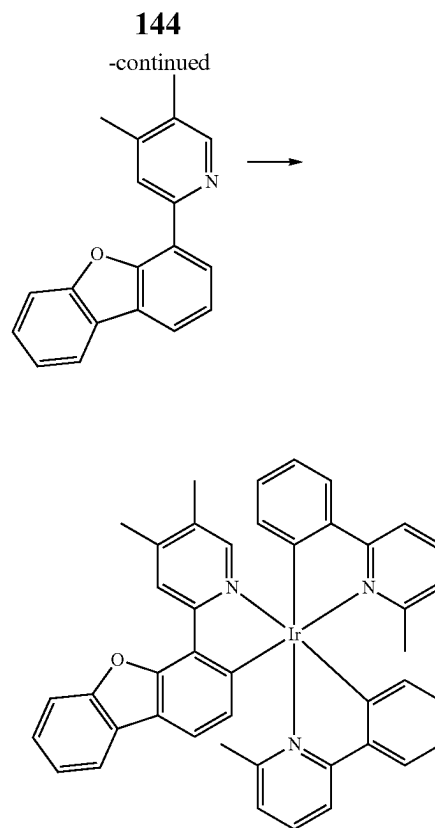
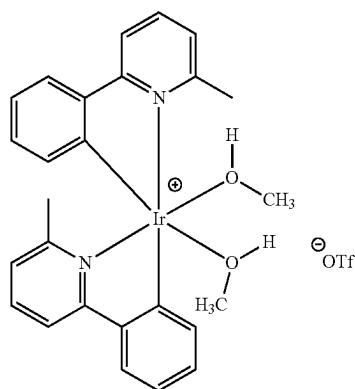


Synthesis of Compound 53: A mixture of the appropriate iridium trifluoromethanesulfonate complex (3.5 g, 4.9 mmol) and 2-(dibenzo[b,d]furan-4-yl)-5-isopropylpyridine (3.5 g, 12.18 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 1.3 g (33%) of Compound 53 as a yellow solid. The product was confirmed by HPLC (99.5% pure) and LC/MS.

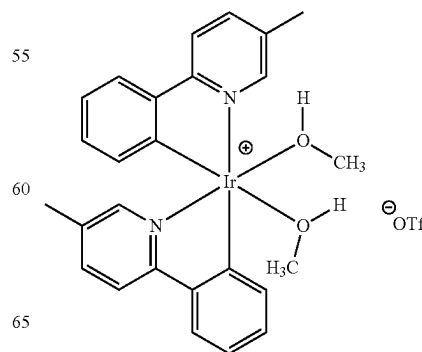


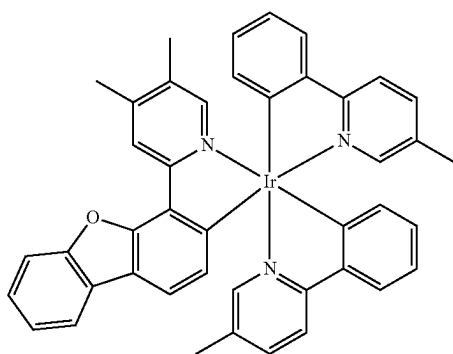
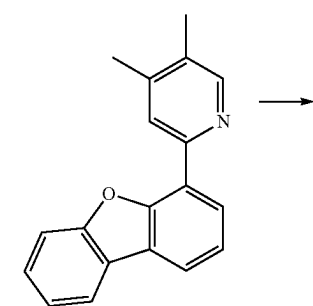
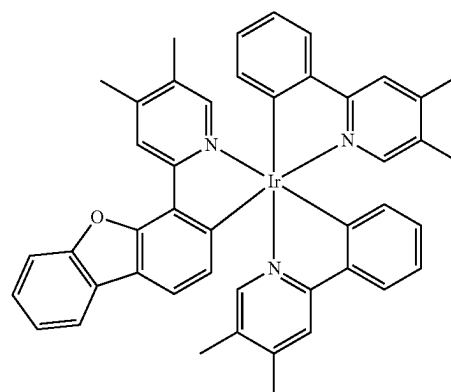
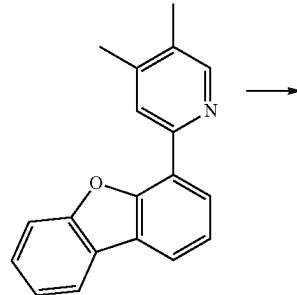


Synthesis of Compound 157—A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.5 g, 3.50 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethylpyridine (2.5 g, 9.15 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was refluxed for 24 h. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solvent was removed to half the volume and the product precipitated by addition of isopropanol and removing dichloromethane under reduced pressure. The filtered material was washed with isopropanol and hexane to yield a mixture of fac- and mer-isomers by LC/MS. The mixture was isomerized to the fac-isomer in a Rayonet at 350 nm in DMSO. The crude product was chromatographed on silica gel with 1/1 dichloromethane/hexane to yield 1.4 g (52%) of Compound 157 as a yellow solid. The product was confirmed by HPLC (98.7% pure) and LC/MS.

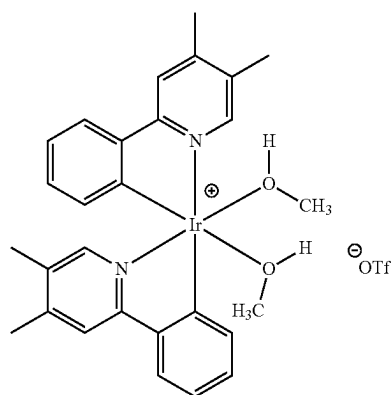


Synthesis of Compound 158: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.5 g, 3.37 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethylpyridine (2.5 g, 9.15 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solvent was removed to half the volume and the product precipitated by addition of isopropanol and removing dichloromethane under reduced pressure. The filtered material was washed with isopropanol and hexane and dried to yield 2.7 g (100%) of Compound 158 as a yellow solid. The product was confirmed by HPLC (99.4% pure) and LC/MS.

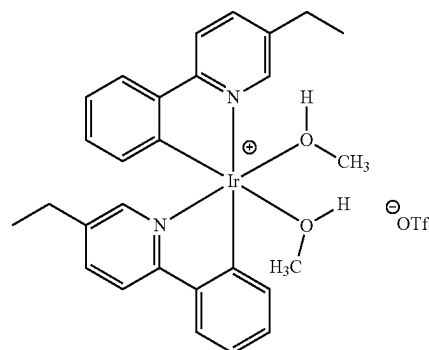


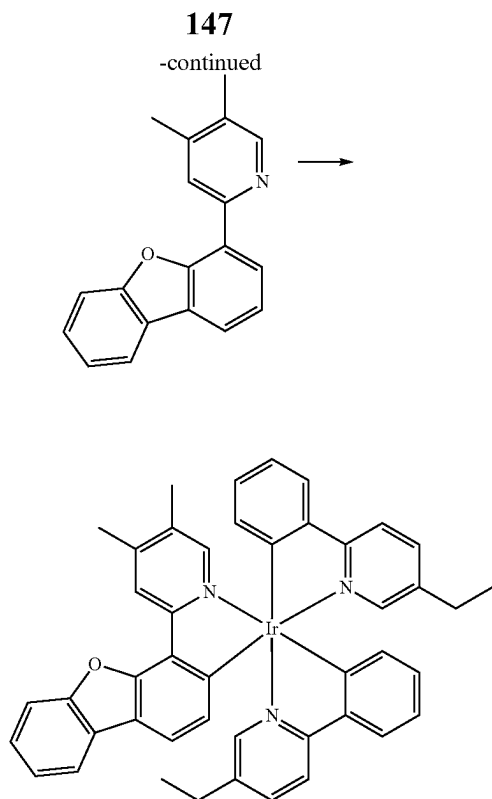
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Synthesis of Compound 159: A mixture of the appropriate iridium trifluoromethanesulfonate complex (3.0 g, 4.04 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethylpyridine (3.0 g, 10.98 mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solubility of the desired product was very poor. Large amount of solvent was used to elute the product. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane and later 4/1 dichloromethane/hexane to yield 0.3 g of the product as a yellow solid. The product was confirmed by HPLC (99.9% pure) and LC/MS.

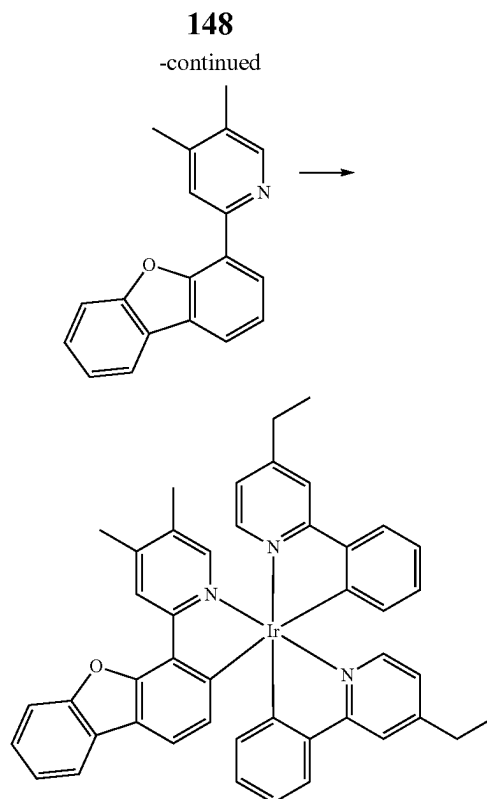
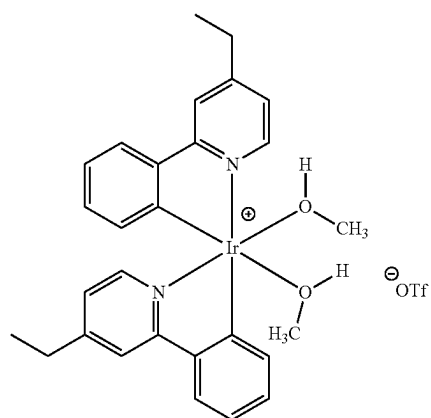


Synthesis of Compound 165: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.5 g, 3.25 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethylpyridine (2.5 g, 9.15 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solvent was removed to half the volume and the product precipitated by addition of isopropanol and removing dichloromethane under reduced pressure. The filtered material was washed with isopropanol and hexane and dried. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 1.2 g (43%) of Compound 165 as a yellow solid. The product was confirmed by HPLC (99.4% pure) and LC/MS.





Synthesis of Compound 174: A mixture of the appropriate iridium trifluoromethanesulfonate complex (3.6 g, 4.68 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethylpyridine (3.6 g, 13.17 mmol) in EtOH (50 mL) and MeOH (50 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 0.8 g of product as a yellow solid was confirmed by HPLC (98.6% pure) and LC/MS.



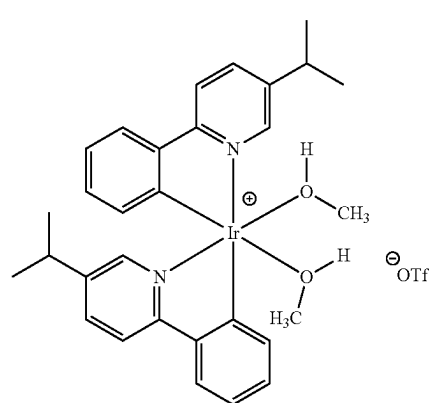
Synthesis of Compound 175: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.5 g, 3.25 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethylpyridine (2.66 g, 9.74 mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under nitrogen atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 2/3 (v/v) THF/hexane to yield 0.8 g of product by HPLC. The product was recrystallized by slow evaporation of DCM from a 1/3 DCM/hexane solution to yield 0.6 g (22%) as a yellow crystalline solid. The product was confirmed by HPLC (99.4% pure) and LC/MS.

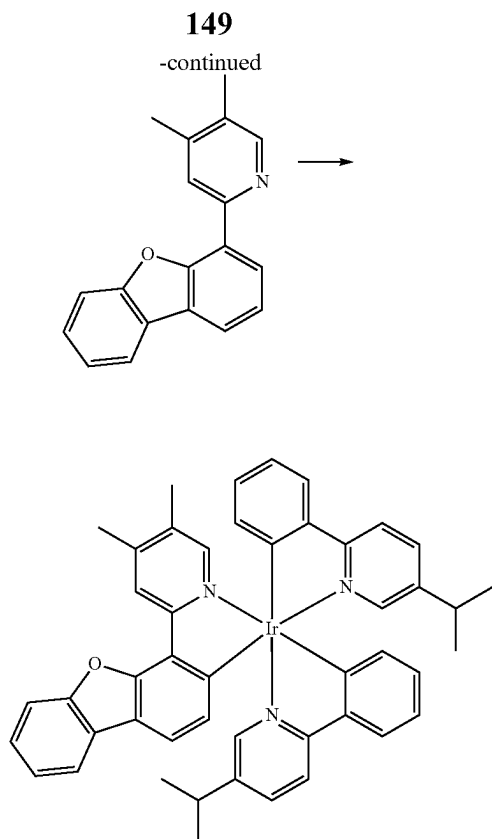
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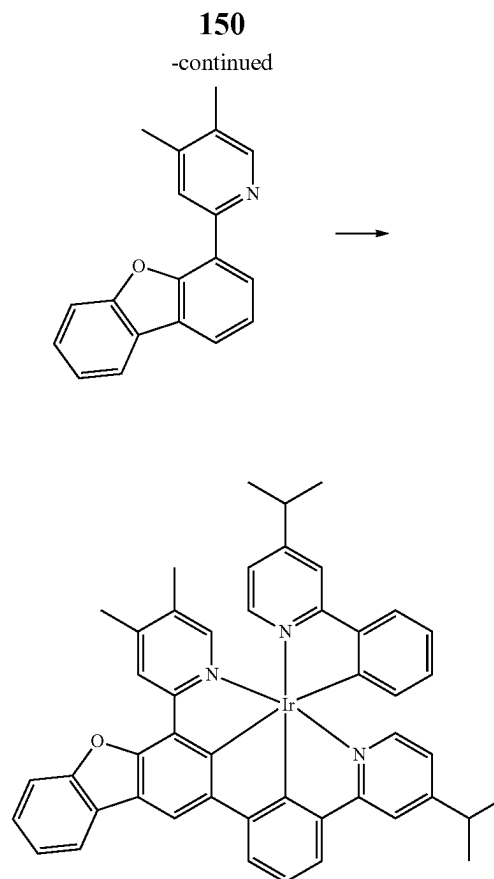
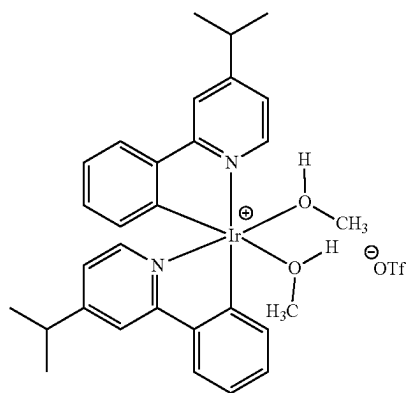
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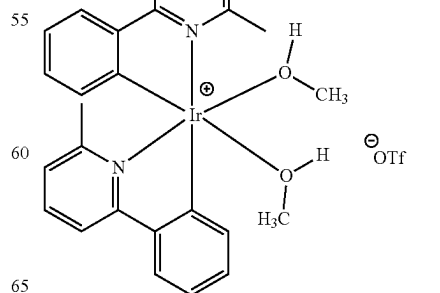


Synthesis of Compound 184: A mixture of the appropriate iridium trifluoromethanesulfonate complex (3.0 g, 3.76 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethylpyridine (3.0 g, 10.98 mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 2.1 g (65%) of product as a yellow solid. The product was confirmed by HPLC (99.8% pure) and LC/MS.

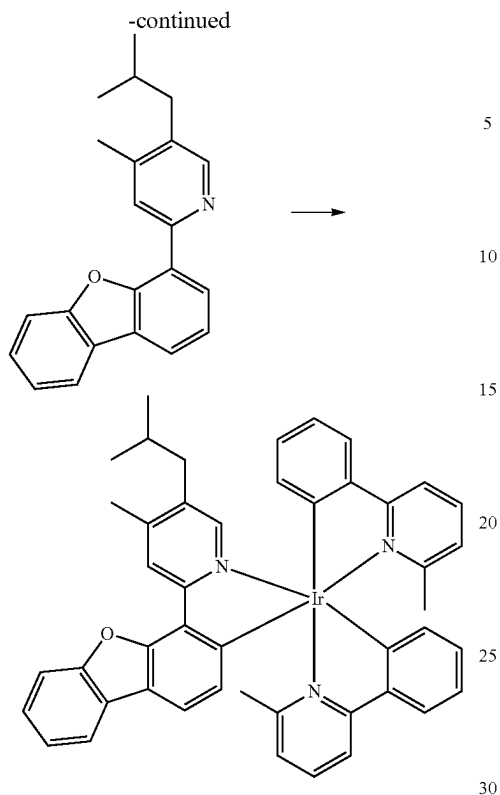


Synthesis of Compound 185: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.8 g, 3.51 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4,5-dimethylpyridine (2.88 g, 10.53 mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under N₂ atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 2/3 (v/v) dichloromethane/hexane to yield 2.1 g (69%) of product as a yellow solid. The product was confirmed by HPLC (99.9% pure) and LC/MS.

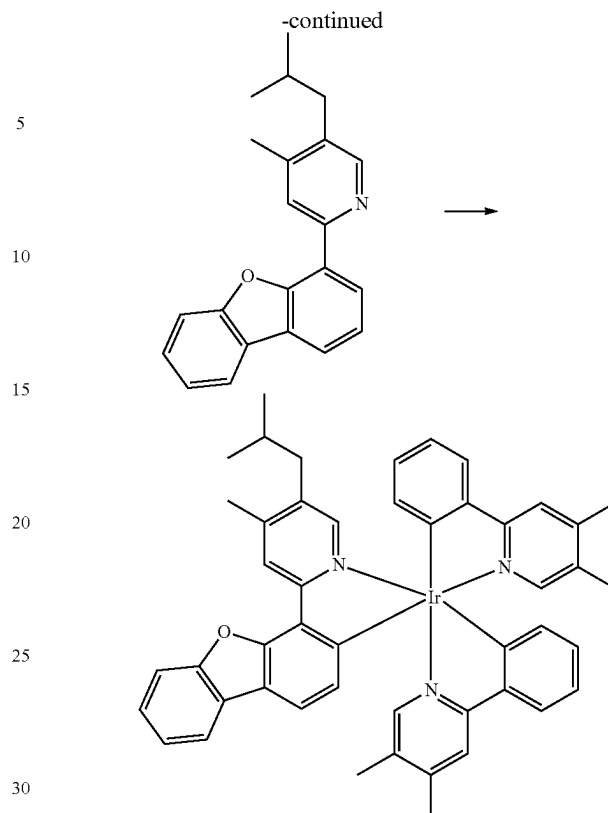
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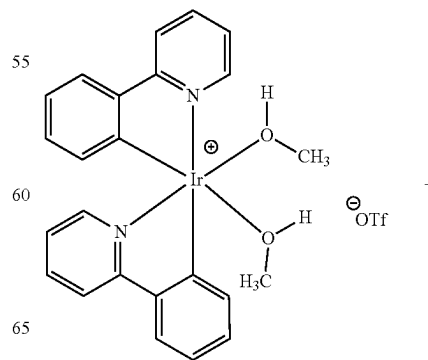
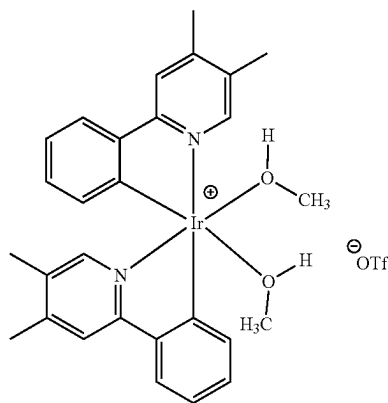


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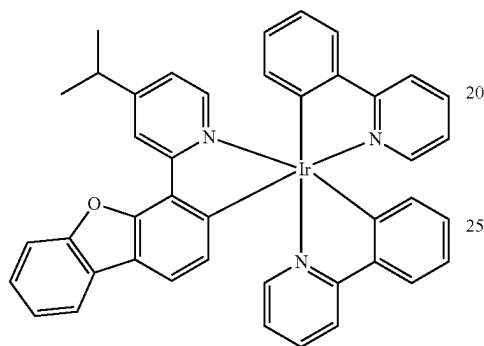
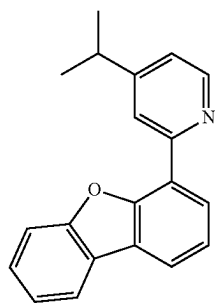
Synthesis of Compound 314: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.5 g, 3.37 mmol) and 2-(dibenzo[b,d]furan-4-yl)-5-isobutyl-4-methylpyridine (2.5 g, 7.93 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solvent was removed to half the volume and the product precipitated by addition of isopropanol and removing dichloromethane under reduced pressure. The filtered material was washed with isopropanol and hexane and dried to yield 3.0 g (100%) of Compound 314 as a yellow solid. The product was confirmed by HPLC (99.6% pure) and LC/MS.

Synthesis of Compound 321: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.2 g, 2.86 mmol) and 2-(dibenzo[b,d]furan-4-yl)-5-isobutyl-4-methylpyridine (2.2 g, 6.98 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solvent was removed to half the volume and the product precipitated by addition of isopropanol and removing dichloromethane under reduced pressure. The filtered material was washed with isopropanol and hexane. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 1.6 g (50%) of Compound 321 as a yellow solid. The product was confirmed by HPLC (99.0% pure) and LC/MS.

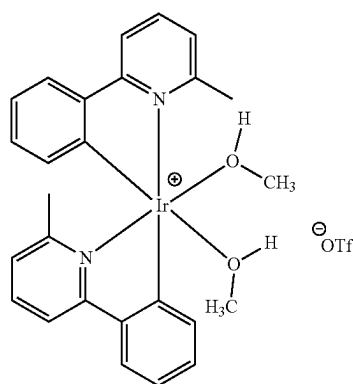


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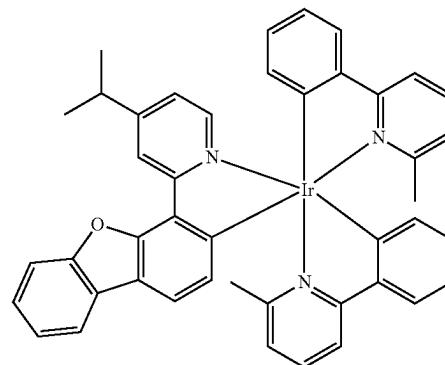
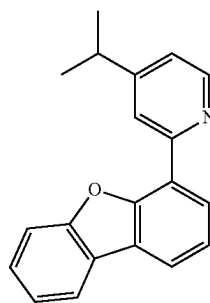
Synthesis of Compound 625: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.2 g, 3.08 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine (2.2 g, 7.66 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solvent was removed to half the volume and the product precipitated by addition of isopropanol and removing dichloromethane under reduced pressure. The filtered material was washed with isopropanol and hexane and dried to yield 1.7 g (67%) of Compound 625 as a yellow solid. The product was confirmed by HPLC (99.8% pure) and LC/MS.



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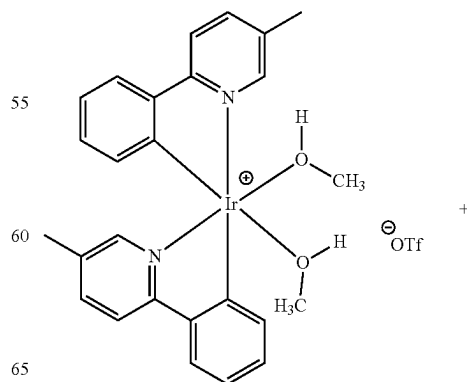
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Synthesis of Compound 626: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.5 g, 3.37 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine (2.5 g, 8.70 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solvent was removed to half the volume and the product precipitated by addition of isopropanol and removing dichloromethane under reduced pressure. The filtered material was washed with isopropanol and hexane and dried. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 2.5 g (89%) of Compound 626 as a yellow solid. The product was confirmed by HPLC (99.4% pure) and LC/MS.

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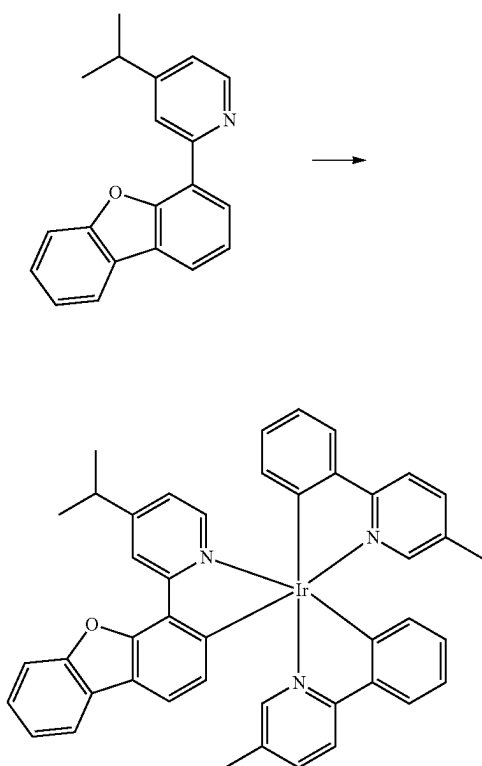
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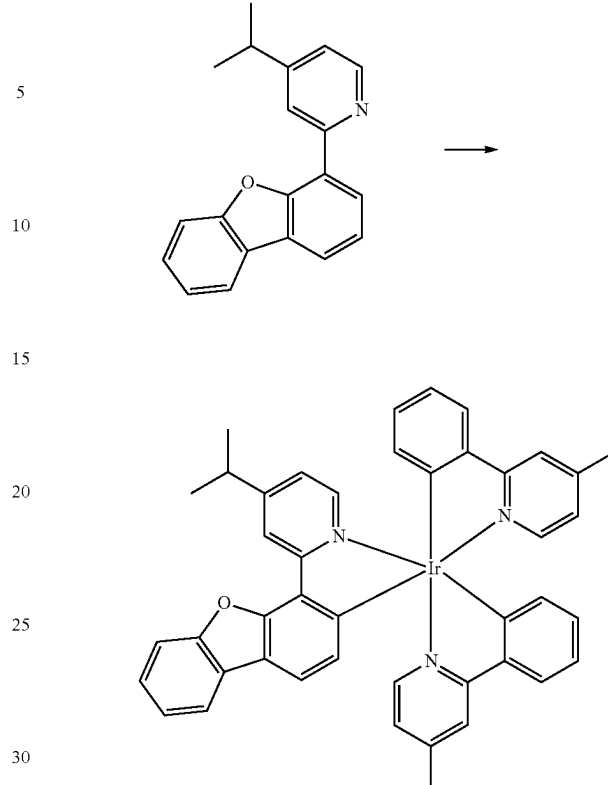
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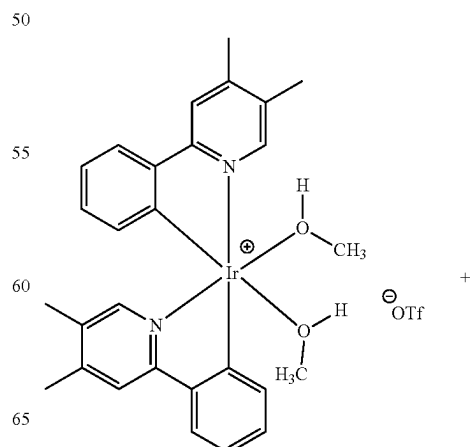
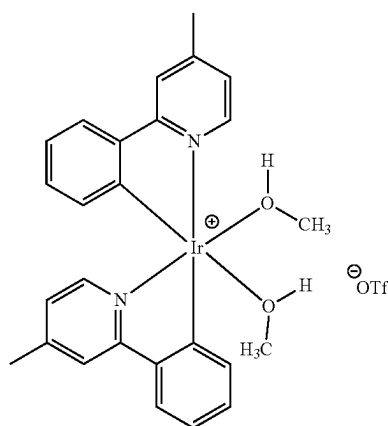
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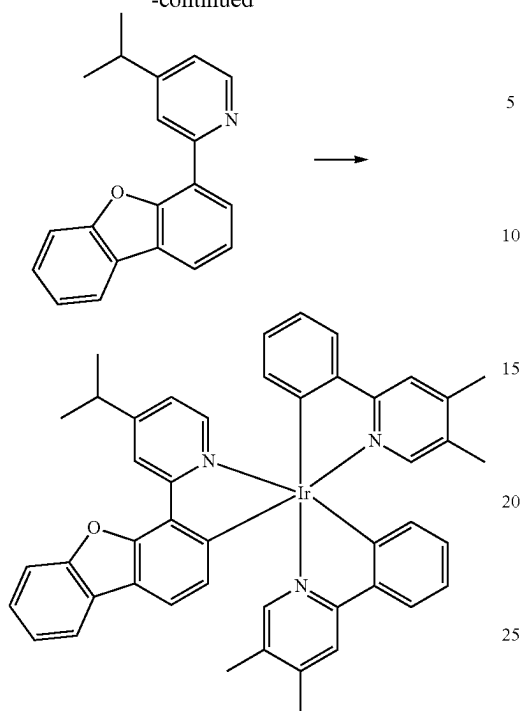
Synthesis of Compound 627: A mixture of the appropriate iridium trifluoromethanesulfonate complex (3.0 g, 4.0 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine (3.0 g, 10.4 mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 2.0 g (60%) of Compound 627 as a yellow solid. The product was confirmed by HPLC (99.9% pure) and LC/MS.

Synthesis of Compound 628: A mixture of the appropriate iridium trifluoromethanesulfonate complex (3.0 g, 4.0 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine (3.0 g, 10.5 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 24 h under nitrogen atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 2/3 (v/v) dichloromethane/hexane to yield 2.1 g (64%) of product as a yellow solid. The product was confirmed by HPLC (99.95% pure) and LC/MS.



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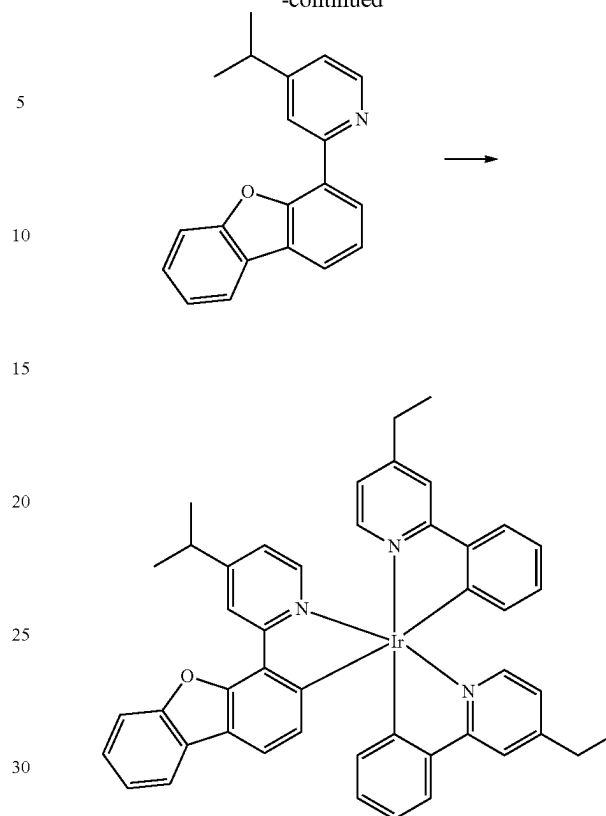
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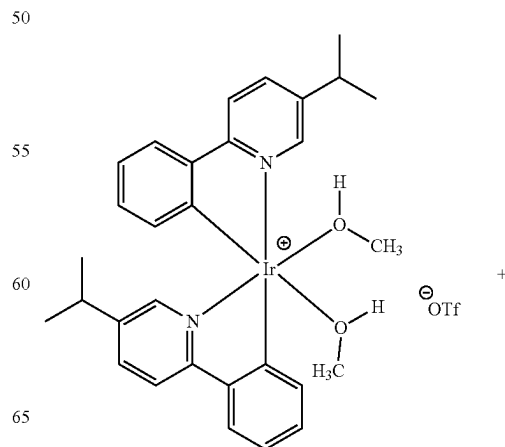
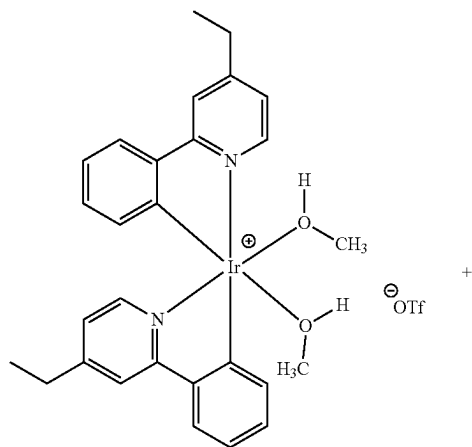
Synthesis of Compound 633: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.5 g, 3.25 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine (2.5 g, 8.70 mmol) in EtOH (25 mL) and MeOH (25 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The solvent was removed to half the volume and the product precipitated by addition of isopropanol and removing dichloromethane under reduced pressure. The filtered material was washed with isopropanol and hexane and dried. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 1.6 g (59%) of Compound 633 as a yellow solid. The product was confirmed by HPLC (99.7% pure) and LC/MS.

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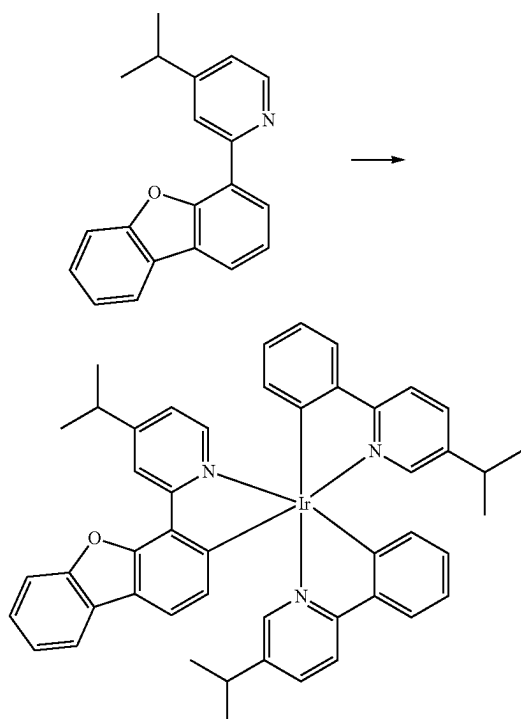


Synthesis of Compound 643: A mixture of the appropriate iridium trifluoromethanesulfonate complex (2.4 g, 3.12 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-ethylpyridine (2.69 g, 9.35 mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under nitrogen atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 2/3 (v/v) dichloromethane/hexane to yield 1.3 g (50%) of product as a yellow solid. The product was confirmed by HPLC (100% pure) and LC/MS.



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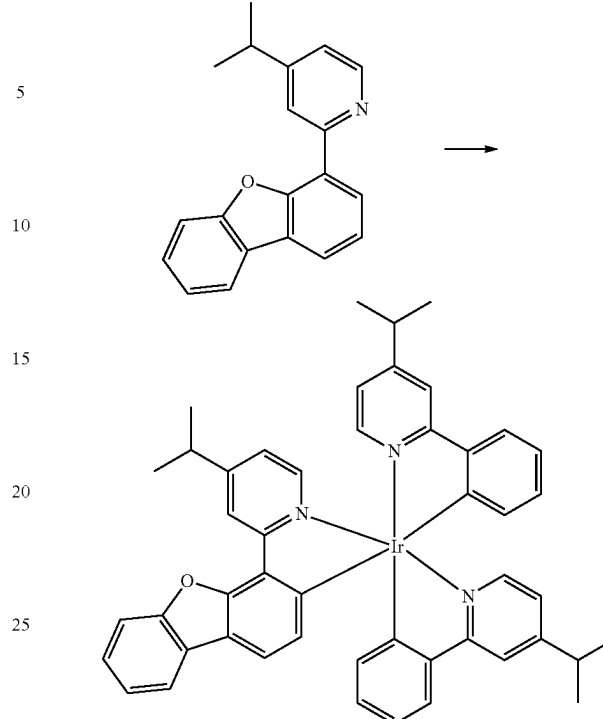
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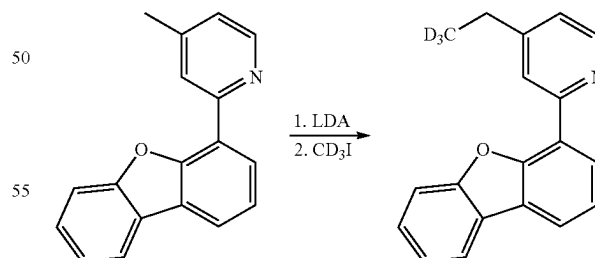
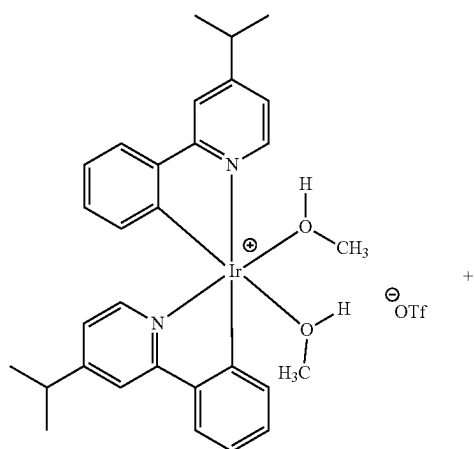
Synthesis of Compound 652: A mixture of the appropriate iridium trifluoromethanesulfonate complex (3.1 g, 3.9 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine (3.1 g, 10.9 mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 1/1 (v/v) dichloromethane/hexane to yield 2.1 g (62%) of Compound 652 as a yellow solid. The product was confirmed by HPLC (99.9% pure) and LC/MS

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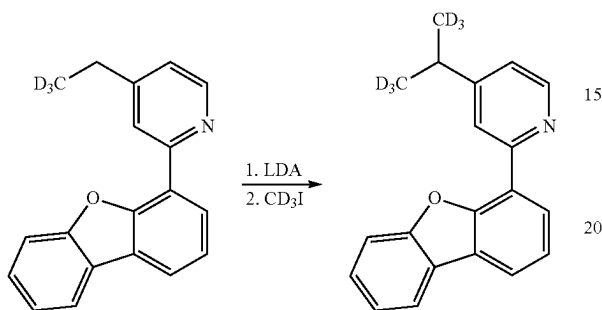
Synthesis of Compound 653: A mixture of iridium trifluoromethanesulfonate complex (2.4 g, 3.01 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-isopropylpyridine (3.0 g, 9.02 mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under nitrogen atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with 2/3 (v/v) dichloromethane/hexane to yield 0.96 g (45%) of product as a yellow solid. The product was confirmed by HPLC (99.8% pure) and LC/MS.



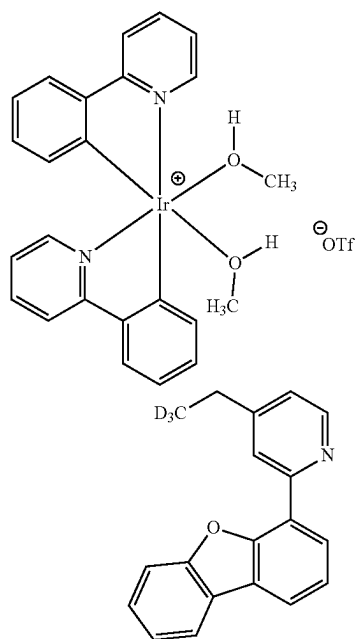
Synthesis of 2-(dibenzo[b,d]furan-4-yl)-4-ethyl-d₃-pyridine: To 2-(dibenzo[b,d]furan-4-yl)-4-methylpyridine (15.3 g, 59.0 mmol) in dry THE (250 mL) at -78° C. was added dropwise lithium diisopropylamide (35.4 mL, 70.8 mmol). The dark solution was stirred for 2 h at -78° C. and then CD₃I was added (4.41 mL, 70.8 mmol) dropwise. The reaction mixture was allowed to slowly warm to RT overnight. Ammonium chloride solution and EtOAc were added

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and the reaction transferred separatory funnel. The layers were separated, washing the aqueous twice with EtOAc and combined organics once with water. After removal of the solvent, the crude product was chromatographed on silica gel with 8/2 (v/v) hexane/EtOAc and then 7/3 hexane/EtOAc to give 14.5 g of product as a pale yellow solid. Recrystallization from hexane gave 12.9 g (79%) of 2-(dibenzo[b,d]furan-4-yl)-4-ethyl-d₃-pyridine. HPLC purity: 99.4%.

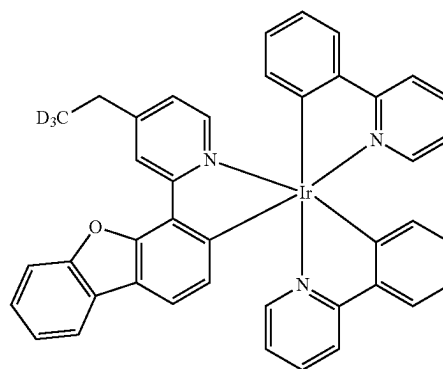


Synthesis of 2-(dibenzo[b,d]furan-4-yl)-4-isopropyl-d₆-pyridine: 2-(Dibenzo[b,d]furan-4-yl)-4-ethyl-d₃-pyridine was dissolved in dry THE (100 mL) and cooled to -78° C. Lithium diisopropylamide (19.0 mL, 38.0 mmol) was added dropwise and the reaction mixture stirred for 2 h at -78° C. CD₃I was added dropwise and the reaction mixture was allowed to slowly warm to room temperature overnight. The reaction was quenched with MeOH, NH₄Cl (aq.) and EtOAc were added and the biphasic mixture transferred to a separatory funnel. The layers were separated, washing the aqueous layer twice with EtOAc and the combined organics with water. After removal of the solvent, the crude product was chromatographed on silica gel with 8/2 (v/v) hexane/EtOAc to give 6.4 g (86%) of 2-(dibenzo[b,d]furan-4-yl)-4-isopropyl-d₆-pyridine. HPLC purity: 99.2%.

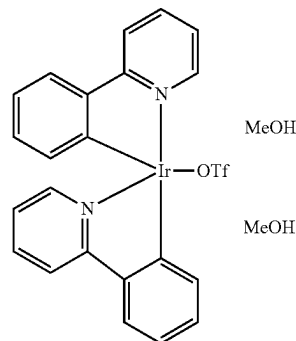
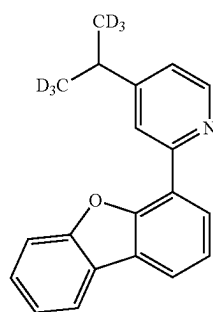


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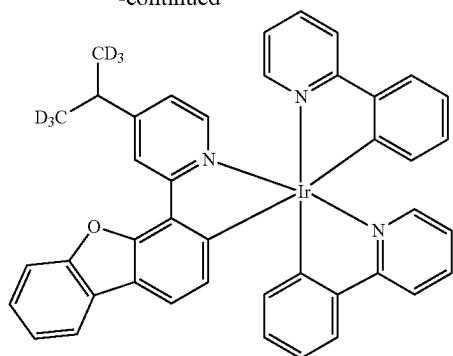


Synthesis of Compound 1145: A mixture of the appropriate iridium trifluoromethanesulfonate complex (3.5 g, 4.9 mmol) and 2-(dibenzo[b,d]furan-4-yl)-4-d³-ethylpyridine (3.5 g, 12.7 mmol) in EtOH (30 mL) and MeOH (30 mL) was refluxed for 20 h under inert atmosphere. The reaction mixture was cooled to room temperature, diluted with ethanol, celite was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The Celite®/silica plug was then washed with dichloromethane to elute the product. The crude product was chromatographed on silica gel with dichloromethane to yield 1.8 g (47%) of Compound 1145 as a yellow solid. The product was confirmed by HPLC (98.7% pure) and LC/MS.



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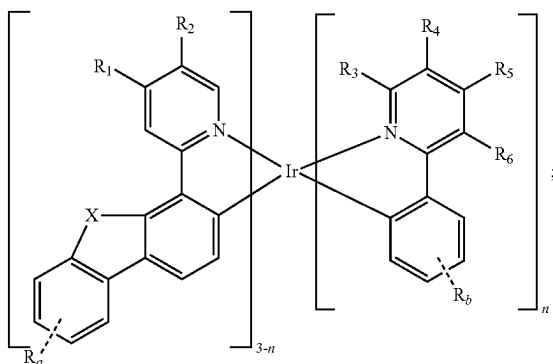
Synthesis of Compound 1146: 2-(Dibenzo[b,d]furan-4-yl)-4-isopropyl-d₆-pyridine and the appropriate iridium trifluoromethanesulfonate complex were combined in ethanol (25 mL) and methanol (25 mL) and heated to reflux for 16 h. The reaction mixture was cooled to room temperature, diluted with ethanol, Celite® was added and the mixture stirred for 10 min. The mixture was filtered on a small silica gel plug on a frit and washed with ethanol (3-4 times) and with hexane (3-4 times). The filtrate was discarded. The celite/silica plug was then washed with dichloromethane to dissolve the product. The crude product was chromatographed on silica gel with 50-70% dichloromethane in hexane to and then sublimed to yield 1.7 g (43%) of Compound 1146 as a yellow solid. The product was confirmed by HPLC (99.5% pure) and LC/MS.

It is understood that the various embodiments described herein are by way of example only, and are not intended to limit the scope of the invention. For example, many of the materials and structures described herein may be substituted with other materials and structures without deviating from the spirit of the invention. The present invention as claimed may therefore include variations from the particular examples and preferred embodiments described herein, as will be apparent to one of skill in the art. It is understood that various theories as to why the invention works are not intended to be limiting.

We claim:

1. A compound having the formula:

Formula I



wherein R_1 and R_2 are optionally linked;
 wherein the sum of the number of carbon atoms in R_1 and R_2 is at least 2;
 wherein R_3 , R_4 , R_5 , R_6 are optionally linked;

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wherein R_a and R_b represent mono-, di-, tri- or tetra-substitution;

wherein X is selected from the group consisting of BR, NR, PR, O, S, Se, S=O, SO₂, SiRR', and GeRR';

wherein R_a , R_b , R, R', R_1 , R_2 , R_3 , R_4 , R_5 , and R_6 are independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; wherein n is 1 or 2;

wherein at least one R_a is nitrile; and

wherein at least one of R_1 , R_2 , R_3 , R_4 , R_5 , and R_6 is selected from the group consisting of branched alkyl, cyclic alkyl, bicyclic alkyl, and multicyclic alkyl.

2. The compound of claim 1, wherein n is 2.

3. The compound of claim 1, wherein X is O.

4. The compound of claim 1, wherein R_1 is hydrogen and R_2 is selected from the group consisting of halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof.

5. The compound of claim 1, wherein R_1 is selected from the group consisting of halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof and R_2 is hydrogen.

6. The compound of claim 1, wherein R_1 and R_2 are selected from the group consisting of halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof.

7. The compound of claim 1, wherein R_1 or R_2 are independently selected from the group consisting of branched alkyl, cyclic alkyl, bicyclic alkyl, and multicyclic alkyl.

8. The compound of claim 1, wherein R_1 or R_2 contain one or more deuterium atoms.

9. The compound of claim 1, wherein R_1 and R_2 contain one or more deuterium atoms.

10. The compound of claim 1, wherein R_3 , R_4 , R_5 and R_6 are independently selected from the group consisting of hydrogen, deuterium, alkyl, aryl, and combinations thereof.

11. The compound of claim 1, wherein at least one of R_3 , R_4 , R_5 and R_6 comprises a branched alkyl, cyclic alkyl, bicyclic alkyl, or a multicyclic alkyl.

12. The compound of claim 1, wherein R_3 , R_4 , R_5 or R_6 contain one or more deuterium atoms.

13. The compound of claim 1, wherein R_4 and R_5 are selected from the group consisting of halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof.

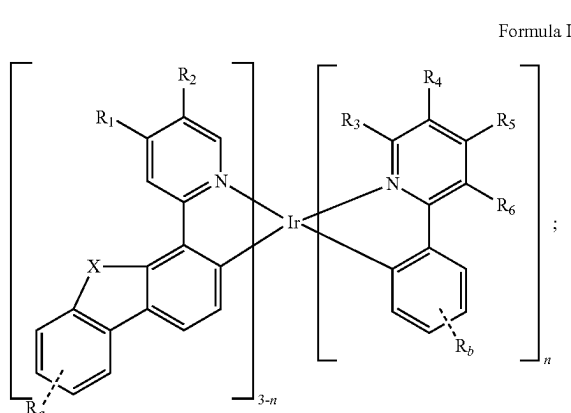
14. The compound of claim 1, wherein R_4 and R_5 are selected from the group consisting of halide, alkyl, cycloalkyl, silyl, aryl, heteroaryl, nitrile, and combinations thereof.

15. The compound of claim 1, wherein R_4 is alkyl.

16. The compound of claim 1, wherein R_5 is alkyl.

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17. The compound of claim 1, wherein R_4 is silyl.
 18. The compound of claim 1, wherein R_5 is silyl.
 19. A first device comprising a first organic light emitting device, comprising:
 an anode;
 a cathode; and
 an organic layer, disposed between the anode and the cathode, comprising a compound having the formula:

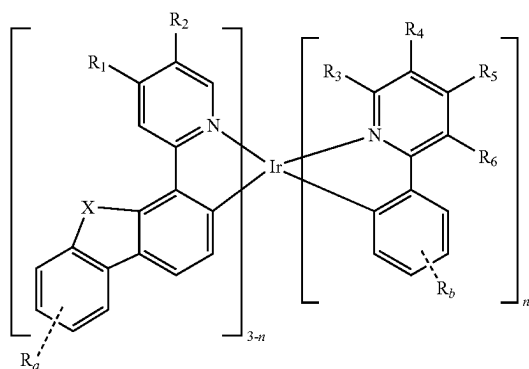


- wherein R_1 and R_2 are optionally linked;
 wherein the sum of the number of carbon atoms in R_1 and R_2 is at least 2;
 wherein R_3 , R_4 , R_5 , R_6 are optionally linked;
 wherein R_a and R_b represent mono-, di-, tri- or tetra-substitution;
 wherein X is selected from the group consisting of BR, NR, PR, O, S, Se, S=O, SO₂, SiRR', and GeRR';
 wherein R_a , R_b , R, R', R_1 , R_2 , R_3 , R_4 , R_5 , and R_6 are independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; wherein n is 1 or 2;
 wherein at least one R_a is nitrile; and
 wherein at least one of R_1 , R_2 , R_3 , R_4 , R_5 , and R_6 is selected from the group consisting of branched alkyl, cyclic alkyl, bicyclic alkyl, and multicyclic alkyl.

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20. A consumer product comprising a first organic light emitting device, comprising:
 an anode;
 a cathode; and
 an organic layer, disposed between the anode and the cathode, comprising a compound having the formula:

Formula I



- wherein R_1 and R_2 are optionally linked;
 wherein the sum of the number of carbon atoms in R_1 and R_2 is at least 2;
 wherein R_3 , R_4 , R_5 , R_6 are optionally linked;
 wherein R_a and R_b represent mono-, di-, tri- or tetra-substitution;
 wherein X is selected from the group consisting of BR, NR, PR, O, S, Se, S=O, SO₂, SiRR', and GeRR';
 wherein R_a , R_b , R, R', R_1 , R_2 , R_3 , R_4 , R_5 , and R_6 are independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; wherein n is 1 or 2;
 wherein at least one R_a is nitrile; and
 wherein at least one of R_1 , R_2 , R_3 , R_4 , R_5 , and R_6 is selected from the group consisting of branched alkyl, cyclic alkyl, bicyclic alkyl, and multicyclic alkyl.

* * * * *